Multivariable Calculus

Mihir Rao

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Lecture 1: Dot product

Multivariable calculus studies functions of several variables. Its basic language is that of vectors, viewed both geometrically (arrows) and algebraically (lists of components).

1.1 Vectors

A vector $\mathbf{A}$ has a direction and a length $|\mathbf{A}|$, also called its magnitude or norm. It is represented by a directed segment, an arrow from a tail to a head; two arrows with the same length and direction represent the same vector. The vector from a point P to a point Q is written $\overrightarrow{PQ}$. In coordinates a vector in space is given by its components,

$$\mathbf{A} = \langle a_1, a_2, a_3 \rangle = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}, \qquad \mathbf{i} = \langle 1,0,0 \rangle,\ \mathbf{j} = \langle 0,1,0 \rangle,\ \mathbf{k} = \langle 0,0,1 \rangle.$$

Placing the tail at the origin O, the head is the point (a_1, a_2, a_3). For a point P the position vector is $\overrightarrow{OP}$, and $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$ is the displacement from P to Q. Real numbers are called scalars.

Proposition 1.1 [Length]. $|\mathbf{A}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$.

Proof. The projection of $\mathbf{A}$ to the xy-plane is $\langle a_1, a_2, 0 \rangle$, of length $r = \sqrt{a_1^2 + a_2^2}$ by the Pythagorean theorem in the plane. This projection and the vertical segment of length $|a_3|$ are the legs of a right triangle with hypotenuse $\mathbf{A}$, so $|\mathbf{A}|^2 = r^2 + a_3^2 = a_1^2 + a_2^2 + a_3^2$. □

For example, $|\langle 3,2,1 \rangle| = \sqrt{14}$.

The operations are defined geometrically and computed componentwise:

- Scalar multiplication: $c\mathbf{A}$ has length $|c||\mathbf{A}|$ and the direction of $\mathbf{A}$ (reversed if $c < 0$); $c\mathbf{A} = \langle ca_1, ca_2, ca_3 \rangle$.
- Addition: place $\mathbf{A}$ and $\mathbf{B}$ head to tail; $\mathbf{A} + \mathbf{B}$ is the displacement $\mathbf{A}$ followed by $\mathbf{B}$, and $\mathbf{A} + \mathbf{B} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$. In the parallelogram spanned by $\mathbf{A}$ and $\mathbf{B}$ the two diagonals are $\mathbf{A} + \mathbf{B}$ and $\mathbf{B} - \mathbf{A}$.
- Subtraction: $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$ is the displacement from the head of $\mathbf{B}$ to the head of $\mathbf{A}$ when both start at the same point.

A unit vector is a vector of length one, written with a hat, as in $\hat{\mathbf{u}}$. Every nonzero vector can be rescaled to one: since $|\langle 3,4 \rangle| = 5$, the vector $\frac{1}{5}\langle 3,4 \rangle = \langle \frac{3}{5}, \frac{4}{5} \rangle$ is a unit vector parallel to $\langle 3,4 \rangle$.

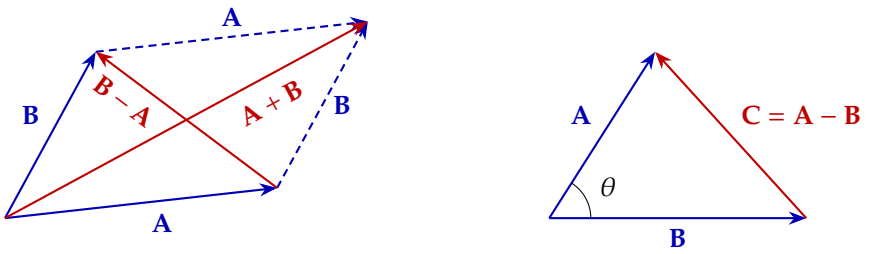


Figure 1: Left: the parallelogram spanned by $\mathbf{A}$ and $\mathbf{B}$, with diagonals $\mathbf{A}+\mathbf{B}$ and $\mathbf{B}-\mathbf{A}$. Right: the triangle with sides $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}=\mathbf{A}-\mathbf{B}$ used to prove Theorem 1.3.

1.2 The dot product

Definition 1.2 [Dot product]. The DOT PRODUCT (or SCALAR PRODUCT) of $\mathbf{A} = \langle a_1, a_2, a_3\rangle$ and $\mathbf{B} = \langle b_1, b_2, b_3\rangle$ is the scalar

$$\mathbf{A}\cdot\mathbf{B} = a_1b_1 + a_2b_2 + a_3b_3.$$

In the plane, $\mathbf{A}\cdot\mathbf{B} = a_1b_1 + a_2b_2$.

The definition gives directly $\mathbf{A}\cdot\mathbf{B} = \mathbf{B}\cdot\mathbf{A}$, $\mathbf{A}\cdot(\mathbf{B}+\mathbf{C}) = \mathbf{A}\cdot\mathbf{B}+\mathbf{A}\cdot\mathbf{C}$, and $\mathbf{A}\cdot\mathbf{A} = a_1^2+a_2^2+a_3^2 = |\mathbf{A}|^2$.

Theorem 1.3 [Geometric form]. *If θ is the angle between $\mathbf{A}$ and $\mathbf{B}$, then*

$$\boxed{\mathbf{A}\cdot\mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta}.$$

Proof. For $\mathbf{B}=\mathbf{A}$ both sides equal $|\mathbf{A}|^2$, since $\cos 0 = 1$. In general, consider the triangle with sides $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}=\mathbf{A}-\mathbf{B}$. The law of cosines gives $|\mathbf{C}|^2 = |\mathbf{A}|^2+|\mathbf{B}|^2-2|\mathbf{A}||\mathbf{B}|\cos\theta$. Expanding with the algebraic rules instead,

$$|\mathbf{C}|^2 = \mathbf{C}\cdot\mathbf{C} = (\mathbf{A}-\mathbf{B})\cdot(\mathbf{A}-\mathbf{B}) = |\mathbf{A}|^2+|\mathbf{B}|^2-2\,\mathbf{A}\cdot\mathbf{B}.$$

Comparing the two expressions gives the claim. So the theorem is the law of cosines in vector form. □

For instance, if $|\mathbf{A}|=2$, $|\mathbf{B}|=5$, and $\theta=\pi/4$, then $\mathbf{A}\cdot\mathbf{B} = 10\cdot\frac{\sqrt{2}}{2} = 5\sqrt{2}$; and $(\mathbf{i}+2\mathbf{j})\cdot(3\mathbf{i}+4\mathbf{j}) = 3+8 = 11$.

1.3 Applications: angles and orthogonality

Lengths and angles. By Theorem 1.3,

$$\cos\theta = \frac{\mathbf{A}\cdot\mathbf{B}}{|\mathbf{A}||\mathbf{B}|}.$$

The sign of $\mathbf{A} \cdot \mathbf{B}$ tells the type of angle: positive if $\theta < \pi/2$, negative if $\theta > \pi/2$, and zero if the vectors are perpendicular.

Example: *An angle of a triangle in space.* For $P = (1,0,0)$, $Q = (0,1,0)$, $R = (0,0,2)$, the angle at P satisfies

$$\cos\theta = \frac{\overrightarrow{PQ} \cdot \overrightarrow{PR}}{|\overrightarrow{PQ}||\overrightarrow{PR}|} = \frac{\langle -1,1,0\rangle \cdot \langle -1,0,2\rangle}{\sqrt{2}\,\sqrt{5}} = \frac{1}{\sqrt{10}}, \qquad \theta \approx 71.6^\circ.$$

Orthogonality. Vectors are ORTHOGONAL when they are perpendicular. Since $\cos(\pi/2) = 0$, for nonzero vectors

$$\mathbf{A} \perp \mathbf{B} \iff \mathbf{A} \cdot \mathbf{B} = 0.$$

Example: *A right triangle.* The points $A = (4,3,6)$, $B = (-2,0,8)$, $C = (1,5,0)$ form a right triangle: $\overrightarrow{AC} = \langle -3,2,-6\rangle$ and $\overrightarrow{AB} = \langle -6,-3,2\rangle$ give $\overrightarrow{AC} \cdot \overrightarrow{AB} = 18 - 6 - 12 = 0$, so the angle at A is $\pi/2$.

Example: *The set $x + 2y + 3z = 0$.* Let $\mathbf{A} = \langle 1,2,3\rangle$ and $P = (x,y,z)$. Then

$$x + 2y + 3z = 0 \iff \mathbf{A} \cdot \overrightarrow{OP} = 0 \iff \mathbf{A} \perp \overrightarrow{OP}$$

(the point $P = O$ is included). So the set is the plane through the origin perpendicular to $\mathbf{A}$, called its NORMAL VECTOR.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 1, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 2: Determinants; cross product

2.1 Components and projection

A third use of the dot product, after angles and orthogonality, is extracting components.

Definition 2.1 [Component]. If $\hat{\mathbf{u}}$ is a unit vector, the component of $\mathbf{A}$ in the direction of $\hat{\mathbf{u}}$ is the scalar

$$\mathbf{A} \cdot \hat{\mathbf{u}} = |\mathbf{A}| \cos\theta,$$

where θ is the angle between $\mathbf{A}$ and $\hat{\mathbf{u}}$. The component of $\mathbf{A}$ in the direction of a nonzero vector $\mathbf{B}$ is its component along $\hat{\mathbf{u}} = \mathbf{B}/|\mathbf{B}|$, namely $\mathbf{A} \cdot \mathbf{B}/|\mathbf{B}|$.

Geometrically, $|\mathbf{A}| \cos\theta$ is the signed length of the leg parallel to $\hat{\mathbf{u}}$ of the right triangle with hypotenuse $\mathbf{A}$; that leg is the orthogonal projection of $\mathbf{A}$ on $\hat{\mathbf{u}}$. For instance $\mathbf{A} \cdot \mathbf{i}$ is the x-component of $\mathbf{A}$. Only the direction of $\mathbf{B}$ matters, and a negative component means that the projection points opposite to $\hat{\mathbf{u}}$.

Example: *Three components.* (i) If $|\mathbf{A}| = 2$ and the angle with $\mathbf{B}$ is $\pi/4$, the component is $2\cos(\pi/4) = \sqrt{2}$, whatever $|\mathbf{B}|$ is. (ii) For $\mathbf{A} = \mathbf{i} + 2\mathbf{j}$ and $\mathbf{B} = 3\mathbf{i} + 4\mathbf{j}$, $\mathbf{B}/|\mathbf{B}| = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$ and the component is $\frac{3}{5} + \frac{8}{5} = \frac{11}{5}$. (iii) For $\mathbf{A} = \langle 2, 2\rangle$ and $\hat{\mathbf{u}} = \langle -1, 0\rangle$ the component is -2.

In physics one decomposes a force this way. For a pendulum, the weight $\mathbf{F}$ points straight down; its component along the direction of motion (tangent to the swing) drives the motion, and its component along the string determines the tension.

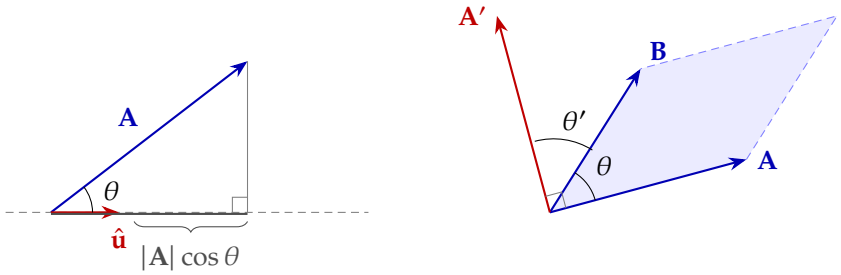


Figure 2: Left: the component $|\mathbf{A}| \cos\theta$ of $\mathbf{A}$ along $\hat{\mathbf{u}}$. Right: $\mathbf{A}'$ is $\mathbf{A}$ rotated by 90° counterclockwise, so $\theta' = \pi/2 - \theta$ and the shaded parallelogram has area $|\mathbf{A}||\mathbf{B}| \sin\theta = \mathbf{A}' \cdot \mathbf{B}$. Schematic.

2.2 Area and the 2×2 determinant

A plane polygon can be cut into triangles, and a triangle with sides $\mathbf{A}$, $\mathbf{B}$ at angle θ has area $\frac{1}{2}|\mathbf{A}||\mathbf{B}| \sin\theta$, half the area of the parallelogram they span. Computing $\sin\theta$ from $\cos\theta$ and

$\sin^2\theta + \cos^2\theta = 1$ gives an unpleasant formula. Instead, let $\mathbf{A}'$ be $\mathbf{A}$ rotated by 90° counterclockwise. If $\mathbf{A} = \langle a_1, a_2\rangle$ then $\mathbf{A}' = \langle -a_2, a_1\rangle$, and $|\mathbf{A}'| = |\mathbf{A}|$. When $\mathbf{B}$ is counterclockwise from $\mathbf{A}$ by $\theta \in (0, \pi)$, the angle from $\mathbf{B}$ to $\mathbf{A}'$ is $\theta' = \pi/2 - \theta$, so

$$\text{area of parallelogram} = |\mathbf{A}||\mathbf{B}| \sin\theta = |\mathbf{A}'||\mathbf{B}| \cos\theta' = \mathbf{A}' \cdot \mathbf{B} = a_1 b_2 - a_2 b_1.$$

If $\mathbf{B}$ is clockwise from $\mathbf{A}$, the same computation gives the negative of the area.

Definition 2.2 [Determinant of order two**].** The determinant of $\mathbf{A} = \langle a_1, a_2\rangle$ and $\mathbf{B} = \langle b_1, b_2\rangle$ is

$$\det(\mathbf{A}, \mathbf{B}) = \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} = a_1 b_2 - a_2 b_1.$$

Proposition 2.3 [Area**].** $\det(\mathbf{A}, \mathbf{B}) = \pm$ *the area of the parallelogram with sides* $\mathbf{A}$ *and* $\mathbf{B}$*. The sign is* $+$ *when* $\mathbf{B}$ *is counterclockwise from* $\mathbf{A}$ *and* $-$ *when it is clockwise.*

2.3 Determinants of order three and higher

Definition 2.4 [Minors, cofactors, determinant**].** For a square array $A = (a_{ij})$, the minor $|A_{ij}|$ is the determinant left after deleting row i and column j, and the cofactor is $A_{ij} = (-1)^{i+j}|A_{ij}|$, that is, the minor with the sign of the checkerboard pattern $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$. The determinant of order three is the expansion along the first row,

$$\det(\mathbf{A}, \mathbf{B}, \mathbf{C}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}.$$

For $n \times n$ arrays the same first-row expansion $|A| = a_{11}A_{11} + \cdots + a_{1n}A_{1n}$ defines the determinant inductively from determinants of order $n - 1$.

Expanding gives the six-term formula

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg) = aei + bfg + cdh - ceg - afh - bdi.$$

Proposition 2.5 [Laplace expansion and row operations**].** *Let* A *be a square array.*

(a) Multiplying each entry of any one row (or column) by its cofactor and adding gives $|A|$*:* $|A| = a_{i1}A_{i1} + \cdots + a_{in}A_{in} = a_{1j}A_{1j} + \cdots + a_{nj}A_{nj}$*.*

(b) Interchanging two rows or two columns multiplies $|A|$ *by* -1*.*

(c) $|A| = 0$ *if a row or column is zero, or if two rows or two columns are equal.*

(d) Multiplying every entry of one row or column by c multiplies $|A|$ by c.

(e) Adding a multiple of one row (column) to another leaves $|A|$ unchanged.

Facts (b) to (e) are immediate for 2×2 determinants from the formula and follow for 3×3 through the Laplace expansion.

Example: *Two expansions.* For

$$|A| = \begin{vmatrix} 1 & 0 & 3 \\ 1 & 2 & -1 \\ 2 & 1 & -1 \end{vmatrix},$$

the minors and cofactors in positions $(1,2)$ and $(2,2)$ are $|A_{12}| = 1$, $A_{12} = -1$, $|A_{22}| = -7$, $A_{22} = -7$. Expanding along the first row and along the second column,

$$|A| = 1 \cdot A_{11} + 0 \cdot A_{12} + 3A_{13} = 1 \cdot (-1) + 0 + 3 \cdot (-3) = -10,$$
$$|A| = 0 \cdot A_{12} + 2A_{22} + 1 \cdot A_{32} = 0 + 2 \cdot (-7) + 1 \cdot 4 = -10.$$

For an order-four example, expanding along the first row (the term with $a_{12} = 0$ drops out) gives

$$\begin{vmatrix} 1 & 0 & 2 & 3 \\ 2 & -1 & 1 & 4 \\ -1 & 4 & 1 & 0 \\ 0 & 4 & 2 & -1 \end{vmatrix} = 1 \cdot \begin{vmatrix} -1 & 1 & 4 \\ 4 & 1 & 0 \\ 4 & 2 & -1 \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 & 4 \\ -1 & 4 & 0 \\ 0 & 4 & -1 \end{vmatrix} - 3 \begin{vmatrix} 2 & -1 & 1 \\ -1 & 4 & 1 \\ 0 & 4 & 2 \end{vmatrix} = 21 + 2 \cdot (-23) - 3 \cdot 2 = -31.$$

Proposition 2.6 [Volume]. $\det(\mathbf{A}, \mathbf{B}, \mathbf{C}) = \pm$ *the volume of the parallelepiped with edges* $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$.

The proof is through the triple product at the end of this lecture. In general an $n \times n$ determinant is $\pm$ the n-dimensional volume of the parallelotope spanned by its rows.

2.4 The cross product

Definition 2.7 [Cross product]. For two vectors in space, $\mathbf{A} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{B} = \langle b_1, b_2, b_3 \rangle$, the cross product (or vector product) is the vector

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \mathbf{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \mathbf{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \mathbf{k} = \langle a_2 b_3 - a_3 b_2,\ a_3 b_1 - a_1 b_3,\ a_1 b_2 - a_2 b_1 \rangle.$$

The 3×3 determinant with vector entries is only a mnemonic; the expansion is the definition.

Unlike the dot product, the result is a vector. For example $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, and similarly $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$.

From the determinant properties,

$$\mathbf{A}\times\mathbf{A}=\mathbf{0},\qquad \mathbf{A}\times\mathbf{B}=-\mathbf{B}\times\mathbf{A},\qquad \mathbf{A}\times(\mathbf{B}+\mathbf{C})=\mathbf{A}\times\mathbf{B}+\mathbf{A}\times\mathbf{C}.$$

The product is not associative: $(\mathbf{i}\times\mathbf{j})\times\mathbf{j}=\mathbf{k}\times\mathbf{j}=-\mathbf{i}$, while $\mathbf{i}\times(\mathbf{j}\times\mathbf{j})=\mathbf{0}$. The rules also give products without the determinant: $(2\mathbf{i}+3\mathbf{j})\times(3\mathbf{i}-2\mathbf{j})=-4\,\mathbf{i}\times\mathbf{j}+9\,\mathbf{j}\times\mathbf{i}=-13\mathbf{k}$.

Theorem 2.8 [Geometric description]. *Let $\theta\in[0,\pi]$ be the angle between $\mathbf{A}$ and $\mathbf{B}$.*

(a) $|\mathbf{A}\times\mathbf{B}|=|\mathbf{A}||\mathbf{B}|\sin\theta$*, the area of the parallelogram in space with sides $\mathbf{A}$ and $\mathbf{B}$.*

(b) $\mathbf{A}\times\mathbf{B}$ *is perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$, and of the two normal directions it is the one given by the right-hand rule: point the right hand along $\mathbf{A}$, curl the fingers toward $\mathbf{B}$; the thumb points along $\mathbf{A}\times\mathbf{B}$.*

Proof. (a) Expanding both sides in components proves the Lagrange identity $|\mathbf{A}\times\mathbf{B}|^2=|\mathbf{A}|^2|\mathbf{B}|^2-(\mathbf{A}\cdot\mathbf{B})^2$. By the geometric form of the dot product (Lecture 1) the right side is $|\mathbf{A}|^2|\mathbf{B}|^2(1-\cos^2\theta)=|\mathbf{A}|^2|\mathbf{B}|^2\sin^2\theta$, and $\sin\theta\ge 0$ for $\theta\in[0,\pi]$.
(b) Perpendicularity: dotting with $\mathbf{A}$ replaces the row $\mathbf{i},\mathbf{j},\mathbf{k}$ of the mnemonic by $\mathbf{A}$, so $\mathbf{A}\cdot(\mathbf{A}\times\mathbf{B})=\det(\mathbf{A},\mathbf{A},\mathbf{B})=0$, a determinant with two equal rows; likewise for $\mathbf{B}$. The orientation can be checked on $\mathbf{i}\times\mathbf{j}=\mathbf{k}$, which agrees with the right-hand rule. □

Example: *Area of a triangle in space.* The triangle with vertices O, $(1,2,4)$, $(0,5,1)$ has half the area of the parallelogram spanned by its two edges from O:

$$\langle 1,2,4\rangle\times\langle 0,5,1\rangle=\begin{vmatrix}\mathbf{i}&\mathbf{j}&\mathbf{k}\\1&2&4\\0&5&1\end{vmatrix}=-18\mathbf{i}-\mathbf{j}+5\mathbf{k},\qquad \text{area}=\tfrac{1}{2}\sqrt{18^2+1^2+5^2}=\tfrac{1}{2}\sqrt{350}.$$

2.5 The triple product and volume

Proposition 2.9 [Triple product]. $\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})=\det(\mathbf{A},\mathbf{B},\mathbf{C})$*, and its absolute value is the volume of the parallelepiped with edges $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$.*

Proof. The identity is the first-row expansion: the components of $\mathbf{B}\times\mathbf{C}$ are exactly the cofactors of a_1,a_2,a_3. For the volume, take the face spanned by $\mathbf{B}$ and $\mathbf{C}$ as base. Its area is $|\mathbf{B}\times\mathbf{C}|$ by Theorem 2.8, and the height is the component of $\mathbf{A}$ along the unit normal $\hat{\mathbf{n}}=\mathbf{B}\times\mathbf{C}/|\mathbf{B}\times\mathbf{C}|$, up to sign. Hence volume $=|\mathbf{B}\times\mathbf{C}|\,|\mathbf{A}\cdot\hat{\mathbf{n}}|=|\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})|$. This proves Proposition 2.6. □

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 2, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 3: Matrices; inverse matrices

3.1 The plane through three points

Recall $\mathbf{A}\times\mathbf{B} = -\mathbf{B}\times\mathbf{A}$ and $\mathbf{A}\times\mathbf{A} = \mathbf{0}$. The cross product gives the plane through three points P_1, P_2, P_3 not on a line. A point $P = (x, y, z)$ lies in it exactly when $\overrightarrow{P_1P}$, $\overrightarrow{P_1P_2}$, $\overrightarrow{P_1P_3}$ span a parallelepiped of zero volume:

$$\det\big(\overrightarrow{P_1P}, \overrightarrow{P_1P_2}, \overrightarrow{P_1P_3}\big) = 0 \iff \overrightarrow{P_1P}\cdot\mathbf{N} = 0, \qquad \mathbf{N} = \overrightarrow{P_1P_2}\times\overrightarrow{P_1P_3}.$$

The two conditions agree because the determinant is the triple product $\overrightarrow{P_1P}\cdot(\overrightarrow{P_1P_2}\times\overrightarrow{P_1P_3})$. Geometrically, $\mathbf{N}$ is perpendicular to the plane (a normal vector), and P is in the plane exactly when $\overrightarrow{P_1P}\perp\mathbf{N}$.

3.2 Matrices and matrix multiplication

Quantities are often related by linear transformations, for instance when changing from coordinates (x_1, x_2, x_3) to coordinates (u_1, u_2, u_3) better adapted to a problem:

$$\begin{cases} u_1 = 2x_1 + 3x_2 + 3x_3 \\ u_2 = 2x_1 + 4x_2 + 5x_3 \\ u_3 = x_1 + x_2 + 2x_3 \end{cases} \iff \begin{pmatrix} 2 & 3 & 3 \\ 2 & 4 & 5 \\ 1 & 1 & 2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}, \quad \text{i.e. } AX = U.$$

Definition 3.1 [Matrix]. An $m\times n$ matrix $A = (a_{ij})$ is a rectangular array with m rows and n columns; a_{ij} is its ij-entry. A $1\times n$ matrix is a row vector and an $m\times 1$ matrix a column vector. Matrices of the same size are added entrywise, $cA = (ca_{ij})$, and the transpose $A^T = (a_{ji})$ is the $n\times m$ matrix whose rows are the columns of A.

Definition 3.2 [Matrix product]. If A is $m\times n$ and B is $n\times p$, the product $C = AB$ is the $m\times p$ matrix with

$$c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj},$$

the dot product of row i of A with column j of B. The product is defined only when the width of A equals the height of B.

For example, if the first row of A is $\begin{pmatrix}1 & 2 & 3 & 4\end{pmatrix}$ and the second column of B is $(0, 3, 0, 2)^T$, then the entry of AB in position $(1, 2)$ is $0 + 6 + 0 + 8 = 14$. Writing A to the left and B above, each entry of AB sits where the row to its left meets the column above it. Multiplying by a standard basis vector

picks out a column, and a standard row vector picks out a row:

$$\begin{pmatrix}1&2&3\\4&5&6\\7&8&9\end{pmatrix}\begin{pmatrix}0\\1\\0\end{pmatrix}=\begin{pmatrix}2\\5\\8\end{pmatrix},\qquad \begin{pmatrix}1&0&0\end{pmatrix}\begin{pmatrix}1&2&3\\4&5&6\\7&8&9\end{pmatrix}=\begin{pmatrix}1&2&3\end{pmatrix}.$$

Meaning of the product. BX applies the transformation B to X, so $(AB)X = A(BX)$ applies first B, then A. Matrix multiplication is composition of transformations, read from right to left.

Proposition 3.3 [Laws of matrix multiplication]. *For matrices of compatible sizes:*

(a) $A(B+C) = AB + AC$ *and* $(A+B)C = AC + BC$;

(b) $(AB)C = A(BC)$ *and* $(cA)B = c(AB)$;

(c) in general $AB \neq BA$, *and one of the two may not even be defined;*

(d) for square A *and* B *of the same size,* $\det(AB) = \det(A)\det(B)$.

Associativity reflects composition; the determinant law is checked directly for 2×2 matrices and is assumed in general.

The identity matrix I_n has ones on the diagonal and zeros elsewhere; it is the identity transformation, $I_n X = X$, and $AI = IA = A$.

Example: *Rotation.* $R = \left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$ is the rotation of the plane by 90° counterclockwise: its columns are $R\mathbf{i} = \mathbf{j}$ and $R\mathbf{j} = -\mathbf{i}$, and rotating twice gives $R^2 = -I$.

Example: *The image of the unit square.* The matrix $A = \left(\begin{smallmatrix}4&1\\1&3\end{smallmatrix}\right)$ sends $\mathbf{i}$ and $\mathbf{j}$ to its columns $A\mathbf{i} = \langle 4,1\rangle$ and $A\mathbf{j} = \langle 1,3\rangle$. By linearity it maps the unit square onto the parallelogram they span, whose area is $\det A = 12 - 1 = 11$.

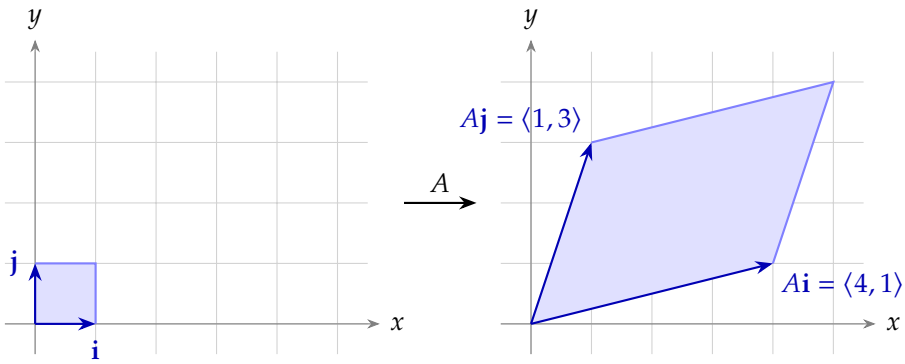


Figure 3: The matrix $A = \left(\begin{smallmatrix}4&1\\1&3\end{smallmatrix}\right)$ maps the unit square spanned by $\mathbf{i}, \mathbf{j}$ onto the parallelogram spanned by its columns, of area $\det A = 11$.

3.3 Inverse matrices

Definition 3.4 [Inverse]. The inverse of an $n \times n$ matrix A is a matrix $M = A^{-1}$ with $AM = MA = I_n$. Only square matrices have inverses.

The inverse expresses the reciprocal linear relation. If $AX = U$, then multiplying by A^{-1} and using associativity, $X = A^{-1}(AX) = A^{-1}U$, so X is determined by U. In particular the solution of $AX = U$ is unique when A^{-1} exists.

Example: *Solving with an inverse.* For $A = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right)$ and $M = \left(\begin{smallmatrix} -3 & 2 \\ 2 & -1 \end{smallmatrix}\right)$, one checks $MA = I$. The system $x_1 + 2x_2 = -1$, $2x_1 + 3x_2 = 4$ then has the solution $\left(\begin{smallmatrix} x_1 \\ x_2 \end{smallmatrix}\right) = M \left(\begin{smallmatrix} -1 \\ 4 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 11 \\ -6 \end{smallmatrix}\right)$. Likewise the change of coordinates $x_1 = y_1 + 2y_2$, $x_2 = 2y_1 + 3y_2$ inverts to $y_1 = -3x_1 + 2x_2$, $y_2 = 2x_1 - x_2$.

Theorem 3.5 [Inverse by cofactors]. *A square matrix A is invertible if and only if* $\det A \neq 0$*, and then*

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A), \qquad \operatorname{adj}(A) = (A_{ij})^T,$$

where A_{ij} is the cofactor of a_{ij} (Lecture 2). The matrix $\operatorname{adj}(A)$ *is the* adjoint *(or* adjugate*) of A.*

Proof. If $MA = I$, the determinant law gives $\det M \det A = \det I = 1$, so $\det A \neq 0$. Conversely, it suffices to show $A \operatorname{adj}(A) = (\det A)\, I$ (the same argument with columns gives $\operatorname{adj}(A)A = (\det A)I$). The (i, k) entry of $A \operatorname{adj}(A)$ is $a_{i1}A_{k1} + a_{i2}A_{k2} + \cdots + a_{in}A_{kn}$. For $k = i$ this is the Laplace expansion of $\det A$ along row i. For $k \neq i$ it is the Laplace expansion along row k of the matrix obtained from A by replacing row k with a copy of row i; the cofactors of row k do not depend on the entries of row k, so they are unchanged. That matrix has two equal rows, so the entry is 0. □

So for small matrices the inverse is computed in four steps:

1. the matrix of minors (delete one row and one column, take the determinant);
2. the cofactors: flip signs by the checkerboard pattern;
3. transpose, giving $\operatorname{adj}(A)$;
4. divide by $\det A$.

For large matrices other algorithms are used.

Example: *A* 3×3 *inverse.* For the matrix A of the change of coordinates above, the four steps give

$$A = \begin{pmatrix} 2 & 3 & 3 \\ 2 & 4 & 5 \\ 1 & 1 & 2 \end{pmatrix}: \qquad \begin{pmatrix} 3 & -1 & -2 \\ 3 & 1 & -1 \\ 3 & 4 & 2 \end{pmatrix} \to \begin{pmatrix} 3 & 1 & -2 \\ -3 & 1 & 1 \\ 3 & -4 & 2 \end{pmatrix}$$

$$\to \operatorname{adj}(A) = \begin{pmatrix} 3 & -3 & 3 \\ 1 & 1 & -4 \\ -2 & 1 & 2 \end{pmatrix} \to A^{-1} = \frac{1}{3}\begin{pmatrix} 3 & -3 & 3 \\ 1 & 1 & -4 \\ -2 & 1 & 2 \end{pmatrix}.$$

The top-left minor is $\left|\begin{smallmatrix} 4 & 5 \\ 1 & 2 \end{smallmatrix}\right| = 3$, and $\det A = 2 \cdot 3 + 3 \cdot 1 + 3 \cdot (-2) = 3$ by expansion along the first row. For a 2×2 matrix the same steps give

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

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Lecture 4: Square systems; equations of planes

4.1 Equations of planes

An equation $ax + by + cz = d$ defines a plane, and its coefficients are the components of a normal vector. A plane is determined by a point $P_0 = (x_0, y_0, z_0)$ in it and a normal vector $\mathbf{N} = \langle a, b, c\rangle$: a point $P = (x, y, z)$ lies in the plane exactly when $\overrightarrow{P_0P} \perp \mathbf{N}$, that is,

$$\mathbf{N} \cdot \overrightarrow{P_0P} = 0 \iff a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

This is the POINT-NORMAL FORM. Multiplying the equation by a nonzero constant gives the same plane.

Example: *Two parallel planes.* The plane through the origin with normal $\mathbf{N} = \langle 1, 5, 10\rangle$ is $\mathbf{N} \cdot \overrightarrow{OP} = 0$, i.e. $x + 5y + 10z = 0$. The plane through $P_0 = (2, 1, -1)$ with the same normal is $(x - 2) + 5(y - 1) + 10(z + 1) = 0$, i.e. $x + 5y + 10z = -3$; it is parallel to the first.

So finding a plane amounts to finding a normal vector, for instance as the cross product of two vectors in the plane.

Example: *The plane through three points.* For $P_1 = (1, 2, 3)$, $P_2 = (0, 0, 3)$, $P_3 = (2, 5, 5)$, the vectors $\overrightarrow{P_1P_2} = \langle -1, -2, 0\rangle$ and $\overrightarrow{P_1P_3} = \langle 1, 3, 2\rangle$ lie in the plane, so

$$\mathbf{N} = \overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -2 & 0 \\ 1 & 3 & 2 \end{vmatrix} = \langle -4, 2, -1\rangle,$$

and the plane is $-4(x - 1) + 2(y - 2) - (z - 3) = 0$, or $-4x + 2y - z = -3$.

Example: *Intercept form.* The plane with intercepts $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ contains $\langle -a, b, 0\rangle$ and $\langle -a, 0, c\rangle$, so $\mathbf{N} = \langle -a, b, 0\rangle \times \langle -a, 0, c\rangle = \langle bc, ac, ab\rangle$. The point-normal form $bc(x-a)+acy+abz = 0$ simplifies to

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1.$$

Likewise the plane with normal $\mathbf{k}$ through $(0, 0, 3)$ is $z = 3$, and in the xy-plane the line through (x_0, y_0) with normal $\langle a, b\rangle$ is $a(x - x_0) + b(y - y_0) = 0$.

Example: *A vector and a plane.* Is $\mathbf{v} = \langle 1, 2, -1\rangle$ parallel or perpendicular to the plane $x + y + 3z = 5$? A perpendicular vector would be proportional to $\mathbf{N} = \langle 1, 1, 3\rangle$, which $\mathbf{v}$ is not. Being parallel to the plane means being perpendicular to $\mathbf{N}$, and $\mathbf{v} \cdot \mathbf{N} = 1 + 2 - 3 = 0$; so $\mathbf{v}$ is parallel to the plane.

4.2 Distances to planes and lines

Proposition 4.1 [Distance formulas]. *Let P be a point.*

(a) The distance from P to the plane through Q with normal $\mathbf{N}$ is $d = \left|\overrightarrow{PQ} \cdot \mathbf{N}\right|/|\mathbf{N}|$.

(b) The distance from P to the line through Q with direction $\mathbf{v}$ is $d = \left|\overrightarrow{QP} \times \mathbf{v}\right|/|\mathbf{v}|$.

Proof. Let R be the point of the plane or line closest to P; it need not be found. (a) The segment PR is parallel to $\mathbf{N}$, so with θ the angle between $\overrightarrow{PQ}$ and $\mathbf{N}$ (or $-\mathbf{N}$), $d = |PR| = |\overrightarrow{PQ}||\cos\theta|$, which is the absolute value of the component of $\overrightarrow{PQ}$ along $\mathbf{N}/|\mathbf{N}|$. (b) In the right triangle QRP with angle θ at Q, $d = |\overrightarrow{QP}|\sin\theta = |\overrightarrow{QP} \times \mathbf{v}|/|\mathbf{v}|$ by the magnitude of the cross product. □

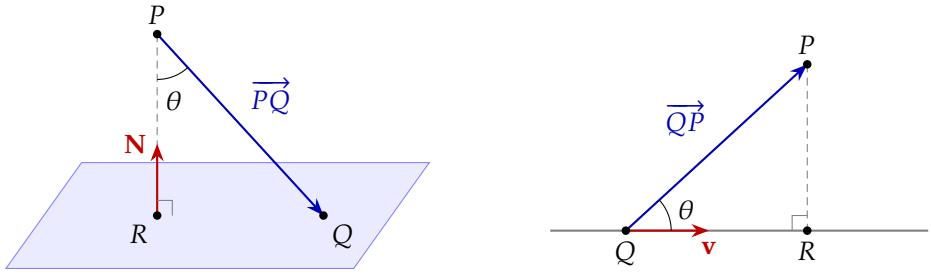


Figure 4: The distance $|PR|$ from P to a plane through Q with normal $\mathbf{N}$ (left) and to a line through Q with direction $\mathbf{v}$ (right); R is the closest point. Schematic.

Example: *Four distances.* (a) From $P = (1,3,2)$ to the plane $x + 2y = 3$: take $Q = (3,0,0)$ and $\mathbf{N} = \langle 1,2,0\rangle$; then $\overrightarrow{PQ} = \langle 2,-3,-2\rangle$ and $d = |2-6|/\sqrt{5} = 4/\sqrt{5}$. (b) From $P = (1,3,2)$ to the line through $(1,0,0)$ and $(1,2,0)$: take $Q = (1,0,0)$, $\mathbf{v} = 2\mathbf{j}$, so $\mathbf{v}/|\mathbf{v}| = \mathbf{j}$ and $\overrightarrow{QP} = 3\mathbf{j} + 2\mathbf{k}$; then $d = |(3\mathbf{j} + 2\mathbf{k}) \times \mathbf{j}| = |-2\mathbf{i}| = 2$. (c) The planes $x + 2y - z = 4$ and $x + 2y - z = 3$ share the normal $\mathbf{N} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, so they are parallel; their distance is the distance from $P = (4,0,0)$ on the first to the second, through $Q = (1,0,0)$: $d = |3\mathbf{i} \cdot \mathbf{N}|/\sqrt{6} = 3/\sqrt{6} = \sqrt{6}/2$. (d) Two skew lines with directions $\mathbf{v}_1, \mathbf{v}_2$ lie in parallel planes with normal $\mathbf{N} = \mathbf{v}_1 \times \mathbf{v}_2$, each through a point of its line; their distance is the distance between these planes.

4.3 Geometry of square systems

Each linear equation in two unknowns is a line, so a 2×2 system asks where two lines meet: in one point (one solution), nowhere if they are parallel and distinct (no solution), or everywhere along them if they are the same line (infinitely many solutions).

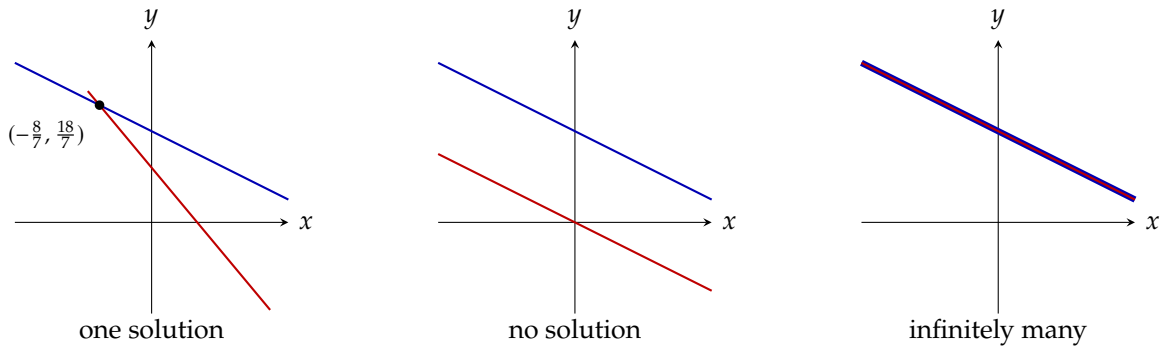


Figure 5: The systems $\{6x+5y=6,\ x+2y=4\}$ with the unique solution $(-8/7, 18/7)$; $\{x+2y=4,\ x+2y=0\}$ with no solution; and $\{x+2y=4,\ x+2y=4\}$ with infinitely many.

A 3×3 system $AX=B$ asks for the intersection of three planes. Two planes usually meet in a line, and the third plane usually meets that line in a single point, the unique solution $X=A^{-1}B$. The exceptions occur when the third plane is parallel to the line of intersection of the first two: if the line lies in the third plane there are infinitely many solutions (the line); otherwise there are none. Listing all cases:

1. one point (one solution);
2. a line (infinitely many): (a) three different planes, the third containing the line of intersection of the first two; (b) two equal planes crossed by the third;
3. a plane (infinitely many): (a) all three planes equal;
4. no common point (no solution): (a) three different parallel planes; (b) two parallel planes crossed by the third; (c) no two planes parallel, but the three lines of intersection of pairs parallel; (d) two equal planes parallel to the third.

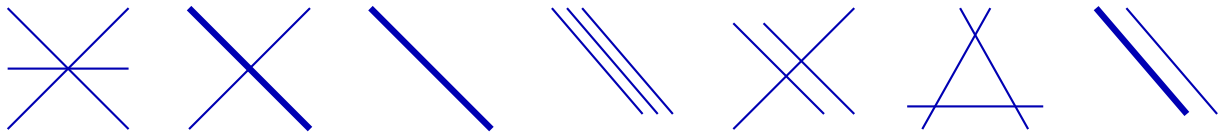


Figure 6: The degenerate cases, drawn with every plane perpendicular to the page and seen edge-on as a segment. Thick segments are two or three coincident planes. Schematic.

4.4 Homogeneous and inhomogeneous systems

The system $AX=B$ is HOMOGENEOUS if $B=0$ and INHOMOGENEOUS otherwise. A square matrix A is SINGULAR if $\det A=0$ and NONSINGULAR (or invertible) if $\det A\neq 0$; by Lecture 3, A is invertible exactly when $\det A\neq 0$. The special cases above are those with $\det A=0$.

Theorem 4.2 [SQUARE SYSTEMS]. *Let A be an $n \times n$ matrix.*

(a) If $\det A \neq 0$, then $AX = B$ has the unique solution $X = A^{-1}B$; in particular $AX = 0$ has only the TRIVIAL SOLUTION *$X = 0$.*

(b) If $\det A = 0$, then $AX = 0$ has nontrivial solutions (infinitely many, forming a line or a plane when $n = 3$).

(c) If $\det A = 0$, then $AX = B$ has either no solution or infinitely many, and for most B it has none.

Equivalently, $AX = 0$ has a nonzero solution if and only if $\det A = 0$.

Proof for $n = 3$. (a) Multiplying $AX = B$ by A^{-1} shows that any solution equals $A^{-1}B$, and $A^{-1}B$ is a solution.
(b) Let $\mathbf{N}_1, \mathbf{N}_2, \mathbf{N}_3$ be the rows of A, the normals of the three planes. The homogeneous system says $\mathbf{N}_i \cdot X = 0$ for each i: all three planes pass through the origin, and X must be orthogonal to all three normals. Now $\det A = \mathbf{N}_1 \cdot (\mathbf{N}_2 \times \mathbf{N}_3)$ is $\pm$ the volume of the parallelepiped they span, so $\det A = 0$ means the normals are coplanar. Any nonzero X perpendicular to a plane containing them solves the system. If $\mathbf{N}_1, \mathbf{N}_2$ are not parallel, the solutions form the line through O in the direction $\mathbf{N}_1 \times \mathbf{N}_2$.
(c) Compared with $AX = 0$, the planes of $AX = B$ are moved to parallel positions. Since the rows are coplanar, one of them, say $\mathbf{N}_3$, is a combination $\mathbf{N}_3 = r\mathbf{N}_1 + s\mathbf{N}_2$ (if $\mathbf{N}_1$ and $\mathbf{N}_2$ are parallel, renumber). Then any solution satisfies $b_3 = \mathbf{N}_3 \cdot X = rb_1 + sb_2$. So unless B satisfies this relation there is no solution. If it does, the third equation follows from the first two and can be dropped, leaving two equations in three unknowns, which have a line (or more) of solutions unless the two planes are parallel and distinct. In either case a solution X_0, if any, is not unique, since $X_0 + X$ is also a solution for every solution X of $AX = 0$. □

Elimination decides, for a given singular system, which of the two alternatives in (c) occurs.

Even for nonsingular A, solving $AX = B$ is delicate if $\det A \approx 0$ (an ILL-CONDITIONED system): $\det A$ appears in the denominator of A^{-1}, so the solution is very sensitive to small changes in the coefficients. For a 2×2 system with entries that are not small, the rows then span a parallelogram of small area, so the two lines are nearly parallel and their intersection point moves a lot when they move slightly.

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Lecture 5: Parametric equations for lines and curves

5.1 Parametric equations of lines

A line in space was seen as the intersection of two planes. Another description is as the trajectory of a moving point. A PARAMETRIC CURVE gives the position as functions of a PARAMETER t,

$$x = x(t), \qquad y = y(t), \qquad z = z(t),$$

often thought of as time; these are its PARAMETRIC EQUATIONS. A line is determined by a point $P_0 = (x_0, y_0, z_0)$ on it and a direction vector $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ parallel to it: P is on the line exactly when $\overrightarrow{P_0P} = t\mathbf{v}$ for some scalar t, i.e.

$$\langle x, y, z \rangle = \langle x_0 + tv_1,\ y_0 + tv_2,\ z_0 + tv_3 \rangle.$$

The point moves at constant velocity $\mathbf{v}$. Replacing $\mathbf{v}$ by a multiple such as $2\mathbf{v}$ traces the same line at a different rate, so the parametrization is not unique.

Example: *The line through two points.* Let $Q(t)$ start at $Q_0 = (-1, 2, 2)$ at $t = 0$ and reach $Q_1 = (1, 3, -1)$ at $t = 1$, moving at constant speed. Its velocity is $\mathbf{v} = \overrightarrow{Q_0Q_1} = \langle 2, 1, -3 \rangle$ and $\overrightarrow{Q_0Q(t)} = t\,\overrightarrow{Q_0Q_1}$, so $\langle x + 1, y - 2, z - 2 \rangle = t\langle 2, 1, -3 \rangle$:

$$x(t) = -1 + 2t, \qquad y(t) = 2 + t, \qquad z(t) = 2 - 3t.$$

5.2 Lines and planes

Example: *Which side of a plane?* Relative to the plane $x + 2y + 4z = 7$, plugging in the coordinates gives $x + 2y + 4z = 11 > 7$ at Q_0 and $3 < 7$ at Q_1, so Q_0 and Q_1 lie on opposite sides; neither is in the plane.

Example: *Where the line meets the plane.* The moving point $Q(t)$ is in the plane when

$$x + 2y + 4z = (-1 + 2t) + 2(2 + t) + 4(2 - 3t) = 11 - 8t = 7,$$

that is, at $t = \frac{1}{2}$. The intersection point is $Q(\frac{1}{2}) = (0, \frac{5}{2}, \frac{1}{2})$.

In general, substituting the parametric equations of a line into the equation of a plane gives a linear equation in t. If the line is parallel to the plane, the t-terms cancel and the left side is a constant: equal to the right side if the line lies in the plane (every t is a solution), and different from it otherwise (no solution).

5.3 General parametric curves

The POSITION VECTOR of a moving point is $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} = \langle x(t), y(t), z(t)\rangle$, the vector from the origin to the point. For example a projectile fired from the origin with initial velocity components $v_{0,x}, v_{0,y}$ follows $x(t) = v_{0,x}t$, $y(t) = -\frac{1}{2}gt^2 + v_{0,y}t$.

Example: *Circles, ellipses, and a helix.* The curve $x = a\cos t$, $y = a\sin t$ satisfies $x^2 + y^2 = a^2$: it traverses the circle of radius a about O. We call these equations its PARAMETRIC FORM and $x^2 + y^2 = a^2$ its SYMMETRIC FORM. The parametrization $x = a\cos 3t$, $y = a\sin 3t$ traces the same circle three times as fast. Similarly $x = a\cos t$, $y = b\sin t$ traces the ellipse $x^2/a^2 + y^2/b^2 = 1$, and $\mathbf{r}(t) = \cos t\,\mathbf{i} + \sin t\,\mathbf{j} + at\,\mathbf{k}$ is a helix winding around the z-axis.

Example: *The symmetric form can lose information.* For $x = 3\cos^2 t$, $y = 3\sin^2 t$ we get $x + y = 3$ with $x, y \geq 0$. The symmetric form is a line, but the point only moves back and forth along the segment of it in the first quadrant.

5.4 The cycloid

The CYCLOID is the trajectory of a point P on the rim of a wheel of radius a rolling without slipping along the floor (the x-axis), with P initially at the origin. A natural parameter is the angle θ the wheel has turned. Rolling without slipping means that the distance travelled equals the arc length $a\theta$ unrolled from the rim.

Let A be the point of contact with the floor and B the center of the wheel, and decompose $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BP}$. Then $\overrightarrow{OA} = \langle a\theta, 0\rangle$ and $\overrightarrow{AB} = \langle 0, a\rangle$. The vector $\overrightarrow{BP}$ has length a and makes angle θ with the downward vertical (turning clockwise from it), so $\overrightarrow{BP} = \langle -a\sin\theta, -a\cos\theta\rangle$. Hence

$$\boxed{x(\theta) = a\theta - a\sin\theta, \qquad y(\theta) = a - a\cos\theta}.$$

The cycloid has no simple symmetric form, so we work with these equations.

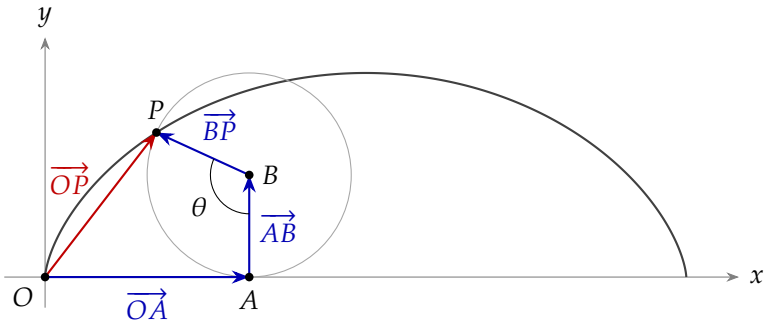


Figure 7: One arch of the cycloid with $a = 1$, drawn exactly, and the wheel after turning $\theta = 2$ radians: $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AB} + \overrightarrow{BP}$ with contact point A and center B.

Behavior at the bottom point. Near $\theta = 0$, use the Taylor approximations $\sin\theta \approx \theta - \theta^3/6$ and

$\cos\theta \approx 1 - \theta^2/2$. Then

$$x(\theta) \approx \frac{a\theta^3}{6}, \qquad y(\theta) \approx \frac{a\theta^2}{2}, \qquad \frac{y}{x} \approx \frac{3}{\theta} \to \pm\infty \quad (\theta \to 0^{\pm}).$$

So y is much larger than x near the origin: the curve has a vertical tangent there, a CUSP, rather than a corner, a loop, or a horizontal tangent. Equivalently, for $a = 1$ the slope is

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\sin\theta}{1 - \cos\theta},$$

an indeterminate form $0/0$ at $\theta = 0$; by l'Hôpital's rule it behaves like $\cos\theta/\sin\theta$, which tends to $-\infty$ as $\theta \to 0^-$ and to $+\infty$ as $\theta \to 0^+$. At the cusp both $dx/d\theta$ and $dy/d\theta$ vanish: the point stops before switching from moving down to moving up.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 5, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 6: Velocity, acceleration; Kepler's second law

6.1 Velocity and acceleration

Let $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}\,[+z(t)\mathbf{k}]$ be the position vector of a moving point. Over a short time Δt the position changes by $\Delta\mathbf{r} = \Delta x\,\mathbf{i} + \Delta y\,\mathbf{j}$, and the average velocity is $\Delta\mathbf{r}/\Delta t$. Letting $\Delta t \to 0$:

Definition 6.1 [Velocity, speed, acceleration]. The velocity vector is

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \left\langle \frac{dx}{dt}, \frac{dy}{dt} \right\rangle,$$

the speed is the scalar $|\mathbf{v}|$, and the acceleration is $\mathbf{a}(t) = d\mathbf{v}/dt = d^2\mathbf{r}/dt^2$.

As $\Delta t \to 0$ the vector $\Delta\mathbf{r}/\Delta t$ becomes tangent to the trajectory, so $\mathbf{v}$ is tangent to the curve. For a parameter other than time, $d\mathbf{r}/dt$ is still called the velocity, or more precisely the tangent vector. Note that the speed is $|d\mathbf{r}/dt|$, which is *not* the same as $d|\mathbf{r}|/dt$.

Example: *The cycloid.* For $\mathbf{r}(t) = \langle t - \sin t,\ 1 - \cos t\rangle$ (radius 1, parameter the angle turned),

$$\mathbf{v}(t) = \langle 1 - \cos t,\ \sin t\rangle, \qquad |\mathbf{v}| = \sqrt{(1-\cos t)^2 + \sin^2 t} = \sqrt{2 - 2\cos t}, \qquad \mathbf{a}(t) = \langle \sin t,\ \cos t\rangle.$$

At $t = 0$ the translation and rotation of the wheel cancel and $\mathbf{v} = \mathbf{0}$, while $\mathbf{a} = \langle 0, 1\rangle$ points straight up. At $t = \pi$ (top of the arch) they add, $\mathbf{v} = \langle 2, 0\rangle$. The speed is smallest at $t = 0, 2\pi, \ldots$ and largest at $t = \pi$. If the wheel turns at 3 revolutions per second, then $\theta = 6\pi t$ and $\mathbf{v}(t) = \langle 6\pi - 6\pi\cos 6\pi t,\ 6\pi \sin 6\pi t\rangle$.

Example: *A projectile.* For $\mathbf{r}(t) = v_{0,x}t\,\mathbf{i} + \left(-\frac{g}{2}t^2 + v_{0,y}t\right)\mathbf{j}$, we get $\mathbf{v} = v_{0,x}\mathbf{i} + (v_{0,y} - gt)\mathbf{j}$ and $\mathbf{a} = -g\mathbf{j}$.

6.2 Arc length and the unit tangent vector

Let s be the arc length, the distance travelled along the trajectory. Over a short time, $\Delta s \approx |\Delta\mathbf{r}| = \sqrt{(\Delta x)^2 + (\Delta y)^2}$; dividing by Δt and letting $\Delta t \to 0$,

$$\frac{ds}{dt} = \left|\frac{d\mathbf{r}}{dt}\right| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} = |\mathbf{v}|,$$

with a third term $(dz/dt)^2$ in space. So the speed is the rate of change of arc length, and the length of a trajectory is $\int |\mathbf{v}|\,dt$. Such integrals are often hard. For one arch of the cycloid the length is

$$\int_0^{2\pi} \sqrt{2 - 2\cos t}\,dt = \int_0^{2\pi} 2\sin\frac{t}{2}\,dt = 8,$$

where the half-angle identity $2 - 2\cos t = 4\sin^2(t/2)$ happens to remove the square root.

Average speed and average velocity differ: a point going once around a circle of radius 1 in 3 seconds has average speed $2\pi/3$ but average velocity $\mathbf{0}$, since its net displacement is zero.

Definition 6.2 [Unit tangent vector]. The unit tangent vector is $\hat{\mathbf{T}} = \mathbf{v}/|\mathbf{v}|$.

By the chain rule,

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d\mathbf{r}}{ds}\frac{ds}{dt} = \hat{\mathbf{T}}\,\frac{ds}{dt} = |\mathbf{v}|\,\hat{\mathbf{T}},$$

which splits the velocity into its magnitude ds/dt and its direction $\hat{\mathbf{T}}$. Intuitively, over an interval Δt, $\Delta\mathbf{r} \approx \hat{\mathbf{T}}\,\Delta s$; dividing by Δt and letting $\Delta t \to 0$ gives the same identity.

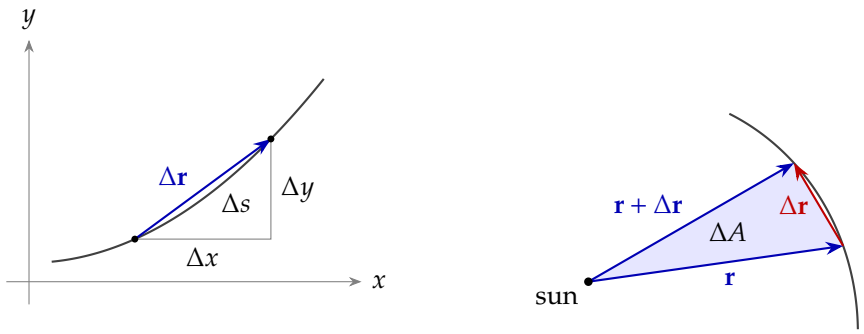


Figure 8: Left: over a short time the arc length Δs is close to $|\Delta\mathbf{r}| = \sqrt{(\Delta x)^2 + (\Delta y)^2}$. Right: the area ΔA swept by the radius vector is close to that of the triangle with sides $\mathbf{r}$ and $\Delta\mathbf{r}$. Schematic.

6.3 Differentiating products of vectors

Proposition 6.3 [Product rules]. *For differentiable vector functions $\mathbf{r}(t)$ and $\mathbf{s}(t)$ in the plane or in space,*

$$\frac{d}{dt}(\mathbf{r}\cdot\mathbf{s}) = \frac{d\mathbf{r}}{dt}\cdot\mathbf{s} + \mathbf{r}\cdot\frac{d\mathbf{s}}{dt}, \qquad \frac{d}{dt}(\mathbf{r}\times\mathbf{s}) = \frac{d\mathbf{r}}{dt}\times\mathbf{s} + \mathbf{r}\times\frac{d\mathbf{s}}{dt}.$$

In the second rule the order of the factors matters, since $\mathbf{a}\times\mathbf{b} = -\mathbf{b}\times\mathbf{a}$.

Proof. Both rules follow by writing out components. Directly: when t increases by Δt,

$$\Delta(\mathbf{r}\cdot\mathbf{s}) = (\mathbf{r}+\Delta\mathbf{r})\cdot(\mathbf{s}+\Delta\mathbf{s}) - \mathbf{r}\cdot\mathbf{s}, \qquad \frac{\Delta(\mathbf{r}\cdot\mathbf{s})}{\Delta t} = \frac{\Delta\mathbf{r}}{\Delta t}\cdot\mathbf{s} + \mathbf{r}\cdot\frac{\Delta\mathbf{s}}{\Delta t} + \frac{\Delta\mathbf{r}}{\Delta t}\cdot\Delta\mathbf{s}.$$

As $\Delta t \to 0$, $\Delta\mathbf{s} \to \mathbf{0}$ by continuity, so the last term vanishes. Replacing $\cdot$ by $\times$ throughout, keeping the order of factors, gives the second rule. □

6.4 Kepler's second law

Kepler's second law (1609) says that a planet moves in a plane, and the line from the sun to the planet sweeps out equal areas in equal times. Vector methods show that this is equivalent to the force on the planet being CENTRAL, that is, directed along the line to the sun. Put the sun at the origin, so $\mathbf{r}$ is the radius vector.

The area rate. From t to $t + \Delta t$ the swept area is approximately the triangle with sides $\mathbf{r}$ and $\Delta\mathbf{r}$, half the parallelogram they span:

$$\Delta A \approx \tfrac{1}{2}|\mathbf{r} \times \Delta\mathbf{r}| \approx \tfrac{1}{2}|\mathbf{r} \times \mathbf{v}|\,\Delta t, \qquad \text{so} \qquad \frac{dA}{dt} = \tfrac{1}{2}|\mathbf{r} \times \mathbf{v}|.$$

Theorem 6.4 [KEPLER'S SECOND LAW AND CENTRAL FORCES]. *For motion $\mathbf{r}(t)$ with $\mathbf{r} \times \mathbf{v} \neq \mathbf{0}$, the following are equivalent: (i) the motion is planar and the radius vector sweeps out area at a constant rate; (ii) $\mathbf{r} \times \mathbf{v}$ is a constant vector; (iii) $\mathbf{r} \times \mathbf{a} = \mathbf{0}$, i.e. the acceleration, and hence the force $\mathbf{F} = m\mathbf{a}$, is parallel to $\mathbf{r}$.*

Proof. (i)⇒(ii): by the area rate, $|\mathbf{r} \times \mathbf{v}|$ is constant. Since $\mathbf{r}$ and $\mathbf{v}$ lie in the plane of motion, $\mathbf{r} \times \mathbf{v}$ is perpendicular to it, so its direction is constant too.
(ii)⇔(iii): by the product rule for cross products,

$$\frac{d}{dt}(\mathbf{r} \times \mathbf{v}) = \frac{d\mathbf{r}}{dt} \times \mathbf{v} + \mathbf{r} \times \frac{d\mathbf{v}}{dt} = \mathbf{v} \times \mathbf{v} + \mathbf{r} \times \mathbf{a} = \mathbf{r} \times \mathbf{a},$$

so $\mathbf{r} \times \mathbf{v}$ is constant exactly when $\mathbf{r} \times \mathbf{a} = \mathbf{0}$, which means $\mathbf{a} \parallel \mathbf{r}$.
(ii)⇒(i): if $\mathbf{r} \times \mathbf{v} = \mathbf{K} \neq \mathbf{0}$ is constant, then $\mathbf{r} \cdot \mathbf{K} = 0$ for all t, so the motion stays in the plane through the sun perpendicular to $\mathbf{K}$, and $dA/dt = \tfrac{1}{2}|\mathbf{K}|$ is constant. □

So Kepler's second law really says that the force is directed along $\mathbf{r}$; for planets, which orbit rather than fly off, it points toward the sun. The argument uses nothing about gravity: the same conclusion holds for any central force, for instance the attraction between electric charges.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 6, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 7: Level curves; partial derivatives

A function of two variables is visualized by its graph, a surface in space, or by its contour plot, a family of curves in the plane. Partial derivatives measure how the function changes when one variable moves and the others are held fixed.

7.1 Functions of several variables and their graphs

A function of several variables assigns a number to each point of its domain, for example

$$f(x,y) = x^2 + y^2, \qquad f(x,y) = xy^2e^{x+y}, \qquad f(x,y,z) = xy\log z,$$

or the ideal gas law $P = kT/V$. In $z = f(x,y)$ the variables x, y are independent: they may take any values in the domain. The variable z is dependent: it is determined by them.

The graph of $z = f(x,y)$ is the surface in xyz-space consisting of the points $(x, y, f(x,y))$: go to (x,y) in the plane, then up to height $f(x,y)$. To sketch it, slice by the coordinate planes $x =$ const and $y =$ const (vertical traces) and by horizontal planes $z =$ const.

Example: *Two graphs.* For $z = -y$, every slice $x =$ const is the line $z = -y$, so the graph is a plane containing the x-axis and sloping down as y increases. For $z = 1 - x^2 - y^2$, the slices $x = 0$ and $y = 0$ are the downward parabolas $z = 1 - y^2$ and $z = 1 - x^2$, and the slice at height $z = c < 1$ is the circle $x^2 + y^2 = 1 - c$. The graph is a paraboloid with top at $(0,0,1)$. Similarly $z = x^2 + y^2$ is an upward paraboloid through $f(0,0) = 0$ and $f(1,1) = f(0,\sqrt{2}) = 2$; its trace $x = 0$ is the parabola $z = y^2$ and its slice $z = 3$ is the circle of radius $\sqrt{3}$.

The standard quadric surfaces, with $a, b, c > 0$, are

ellipsoid $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$,

hyperboloid of one sheet $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 1$,

hyperboloid of two sheets $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = -1$,

elliptic cone $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = \dfrac{z^2}{c^2}$,

elliptic paraboloid $z = ax^2 + by^2$,

hyperbolic paraboloid $z = by^2 - ax^2$.

The last two, and the cone's upper half, are graphs of functions of (x,y).

7.2 Level curves and contour plots

Definition 7.1 [Level curves]. The level curve of f at height c is the plane curve $f(x,y) = c$. A contour plot draws several level curves in the xy-plane, each labeled with its value c.

The level curve at height c is the slice of the graph by the horizontal plane $z = c$, projected to the xy-plane; the graph is the contour plot with each curve lifted to its height, as on a topographic map or a temperature map. For $z = -y$ the level curves are the horizontal lines $y = -c$. For $z = x^2 + y^2$ they are the circles of radius $\sqrt{c}$, and for $z = 1 - x^2 - y^2$ the circles of radius $\sqrt{1-c}$. For the saddle $z = y^2 - x^2$ they are hyperbolas, and the level curve $z = 0$ consists of the two lines $y = \pm x$ crossing at the origin. A level curve that crosses itself in this way marks a MOUNTAIN PASS: moving away from the crossing in one direction the values go up, in the other they go down.

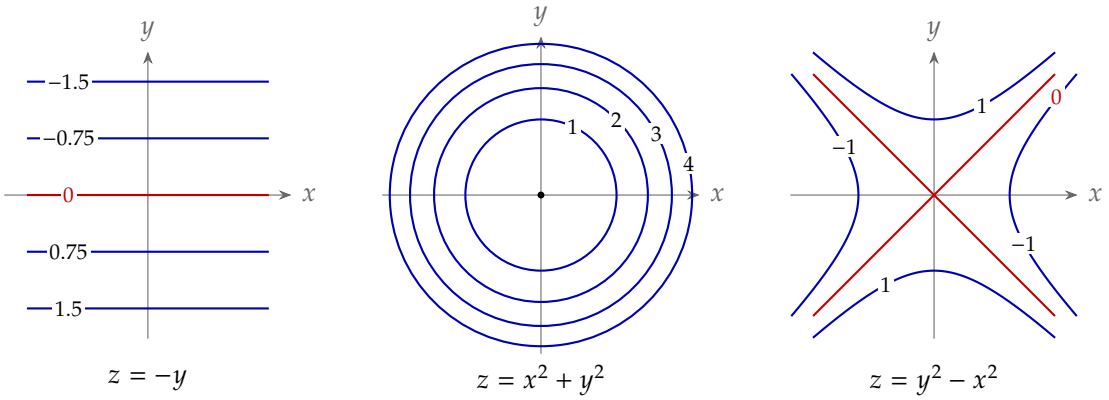


Figure 9: Contour plots of $z = -y$, $z = x^2 + y^2$, and $z = y^2 - x^2$. Each curve is labeled with its value of z. In the right panel the level curve $z = 0$ is the pair of lines $y = \pm x$.

A contour plot shows qualitatively how f varies. Moving across the level curves toward higher labels increases f; moving along a level curve leaves f unchanged; closely spaced curves mean f changes quickly. In the left panel, increasing y decreases z, and changing x has no effect.

7.3 Partial derivatives

Let $w = f(x, y)$. Fixing $y = y_0$ gives the PARTIAL FUNCTION $x \mapsto f(x, y_0)$ of one variable.

Definition 7.2 [PARTIAL DERIVATIVE]. The PARTIAL DERIVATIVE of f with respect to x at (x_0, y_0) is the derivative of this partial function:

$$f_x(x_0, y_0) = \frac{\partial f}{\partial x}(x_0, y_0) = \lim_{\Delta x \to 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}.$$

Similarly, $f_y(x_0, y_0)$ is the derivative of $y \mapsto f(x_0, y)$ at y_0.

Other notations are $\partial w/\partial x|_{(x_0,y_0)}$ and $(\partial w/\partial x)_0$, where the subscript 0 means evaluation at the point. As (x_0, y_0) varies, f_x and f_y are again functions of x and y.

Geometrically, the graph of $x \mapsto f(x, y_0)$ is the curve cut from the graph of f by the vertical plane $y = y_0$, and $f_x(x_0, y_0)$ is the slope of its tangent line at $P = (x_0, y_0, f(x_0, y_0))$. Likewise $f_y(x_0, y_0)$ is

the slope of the slice by the plane $x = x_0$. As a rate, $\partial w/\partial x$ says how fast w increases with x when y is held constant.

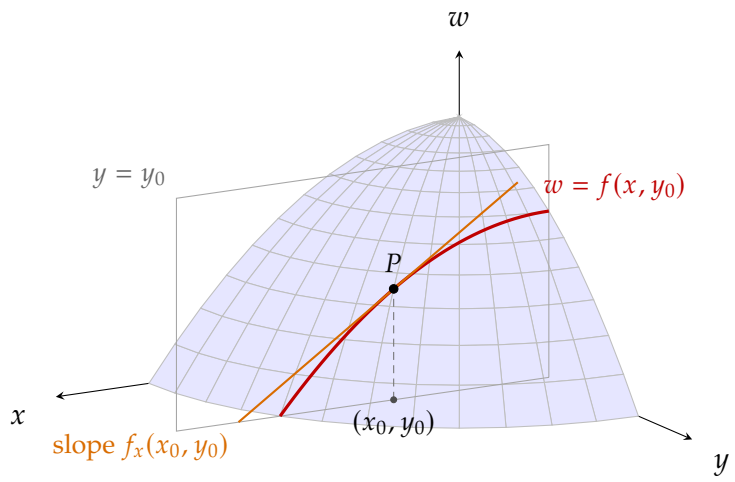


Figure 10: The graph of $w = 1 - x^2 - y^2$ cut by the vertical plane $y = y_0$ (here $y_0 = \frac{1}{2}$). The slice is the graph of the partial function $w = f(x, y_0)$, and $f_x(x_0, y_0)$ is the slope of its tangent line at P (here $x_0 = \frac{1}{2}$, slope -1).

To compute f_x, treat x as the variable and every other variable as a constant. For $f(x, y) = x^3y + y^2$,

$$f_x = 3x^2y, \qquad f_y = x^3 + 2y.$$

For $w = f(x, y, z, \ldots)$ the graph can no longer be drawn, but the definition is the same: hold all variables but one fixed and take the ordinary derivative in the remaining one.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 7, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 8: Max-min problems; least squares

The tangent plane to a graph gives the basic approximation formula for functions of several variables. At a local maximum or minimum the tangent plane is horizontal, which locates candidates for extrema. Fitting a line to data is a minimization problem of this kind.

8.1 The tangent plane and the approximation formula

Let $w = f(x, y)$ and $w_0 = f(x_0, y_0)$. The slices $y = y_0$ and $x = x_0$ of the graph have tangent lines at (x_0, y_0, w_0):

$$y = y_0,\ w = w_0 + f_x(x_0, y_0)(x - x_0) \qquad \text{and} \qquad x = x_0,\ w = w_0 + f_y(x_0, y_0)(y - y_0).$$

The tangent plane must contain both lines. A non-vertical plane through (x_0, y_0, w_0) has the form $w - w_0 = a(x - x_0) + b(y - y_0)$; setting $y = y_0$ shows that a is its slope in the **i** direction, so $a = f_x(x_0, y_0)$, and similarly $b = f_y(x_0, y_0)$.

Definition 8.1 [Tangent plane and linearization]. The tangent plane to the graph of $w = f(x, y)$ at (x_0, y_0, w_0) is

$$w - w_0 = \left(\frac{\partial w}{\partial x}\right)_0 (x - x_0) + \left(\frac{\partial w}{\partial y}\right)_0 (y - y_0).$$

The function on the right-hand side plus w_0, whose graph is this plane, is the linearization of f at (x_0, y_0).

Near (x_0, y_0) the graph is close to its tangent plane. With $\Delta x = x - x_0$, $\Delta y = y - y_0$, $\Delta w = w - w_0$, this is the approximation formula

$$\Delta w \approx \left(\frac{\partial w}{\partial x}\right)_0 \Delta x + \left(\frac{\partial w}{\partial y}\right)_0 \Delta y, \qquad \Delta x, \Delta y \approx 0, \tag{8.1}$$

and for $w = f(x, y, z)$ one adds the term $(\partial w/\partial z)_0 \Delta z$.

Example: *A square of tolerance.* Find a square centered at $(1, 1)$ on which $w = x^3 y^4$ varies by at most ± 0.1. Here $w_x = 3x^2 y^4$ and $w_y = 4x^3 y^3$, so at $(1, 1)$, $\Delta w \approx 3\Delta x + 4\Delta y$. If $|\Delta x|, |\Delta y| \le 0.01$ then $|\Delta w| \le 3|\Delta x| + 4|\Delta y| \le 0.07$, within the bound. So $|x - 1| \le 0.01, |y - 1| \le 0.01$ works.

Example: *Sensitivity of a volume.* A box has sides measured as $a = 1$, $b = 2$, $c = 3$. Since $V = abc$,

$$\Delta V \approx bc\,\Delta a + ac\,\Delta b + ab\,\Delta c = 6\,\Delta a + 3\,\Delta b + 2\,\Delta c \quad \text{at } (1, 2, 3).$$

The volume is most sensitive to the shortest side: an error of 0.01 in a changes V by about 0.06. In general, the value of $w = f(x, y, \ldots)$ is most sensitive to the variable whose partial derivative has

the largest absolute value at the point (the SENSITIVITY PRINCIPLE).

8.2 When the approximation is valid

The tangent plane was built to match the surface's slopes in the **i** and **j** directions only, so (8.1) needs a hypothesis guaranteeing that it matches in the other directions too. We call f SMOOTH at (x_0, y_0) if f_x and f_y exist and are continuous in some rectangle centered at (x_0, y_0). Under this hypothesis (8.1) holds. The usual way smoothness fails is that a partial derivative does not exist at the point.

Example: *A cone.* For $w = \sqrt{x^2 + y^2}$, $w_x = x/\sqrt{x^2 + y^2}$ and $w_y = y/\sqrt{x^2 + y^2}$, which are continuous except at the origin, where they are undefined. The graph is a right circular cone with vertex at the origin: it has a sharp point there, and no linear function approximates it.

The following argument avoids geometry, so it extends to any number of variables, and shows where smoothness enters. Go from $P = (x_0, y_0)$ to $R = (x_0 + \Delta x, y_0 + \Delta y)$ through $Q = (x_0 + \Delta x, y_0)$, and write $\Delta w = \Delta w_1 + \Delta w_2$ for the changes along PQ and QR.

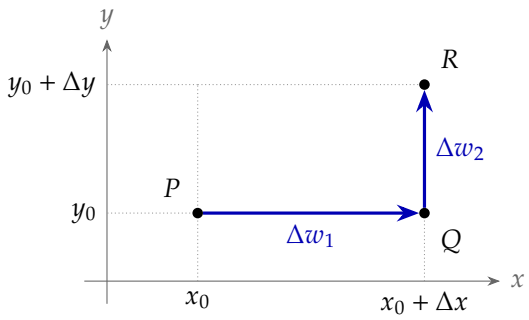


Figure 11: The change Δw from P to R split into the change Δw_1 along PQ, with $y = y_0$ fixed, and the change Δw_2 along QR, with $x = x_0 + \Delta x$ fixed.

Proposition 8.2 [APPROXIMATION FORMULA]. *If f is smooth at (x_0, y_0), then*

$$\Delta w = \big(f_x(x_0, y_0) + \varepsilon_1\big)\Delta x + \big(f_y(x_0, y_0) + \varepsilon_2\big)\Delta y,$$

where $\varepsilon_1, \varepsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0, 0)$.

Proof. Along PQ only x changes. By the mean value theorem for $x \mapsto f(x, y_0)$, $\Delta w_1 = f_x(\xi, y_0)\Delta x$ for some ξ between x_0 and $x_0 + \Delta x$. Along QR only y changes, so $\Delta w_2 = f_y(x_0 + \Delta x, \eta)\Delta y$ for some η between y_0 and $y_0 + \Delta y$. Both evaluation points tend to (x_0, y_0), so by continuity of f_x and f_y the differences $\varepsilon_1 = f_x(\xi, y_0) - f_x(x_0, y_0)$ and $\varepsilon_2 = f_y(x_0 + \Delta x, \eta) - f_y(x_0, y_0)$ tend to 0. □

The continuity of f_y is used essentially: the second step evaluates f_y at $(x_0 + \Delta x, y_0)$, not at (x_0, y_0). The same argument with one step per variable gives the approximation formula for $w = f(x, y, z)$ when w_x, w_y, w_z are continuous near the point.

8.3 Critical points

If f has a local maximum or minimum at (x_0, y_0), then x_0 is a local extremum of the slice $x \mapsto f(x, y_0)$, so $f_x(x_0, y_0) = 0$, and likewise $f_y(x_0, y_0) = 0$. Since the two tangent lines determine the tangent plane, the tangent plane is horizontal: by (8.1), $|\Delta f|$ is much smaller than $|\Delta x|$ and $|\Delta y|$.

Definition 8.3 [Critical point]. A CRITICAL POINT of $f(x, y)$ is a point (x_0, y_0) where $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$.

Local extrema of a differentiable function occur at critical points, but a critical point may be a local minimum, a local maximum, or a saddle. For $z = x^2 + y^2 + 0.5$, $z = 1 - x^2 - y^2$ and $z = -x^2 + y^2$ the only critical point is the origin, and it is a minimum, a maximum, and a saddle, respectively.

Example: *Completing the square.* Let $f(x, y) = x^2 - 2xy + 3y^2 + 2x - 2y$. The equations

$$f_x = 2x - 2y + 2 = 0, \qquad f_y = -2x + 6y - 2 = 0$$

give the single critical point $(x_0, y_0) = (-1, 0)$. Completing the square,

$$f = (x - y)^2 + 2y^2 + 2x - 2y = (x - y + 1)^2 + 2y^2 - 1 \ \geq \ -1,$$

with equality only at $(-1, 0)$. So the critical point is the minimum. A systematic test is the subject of Lecture 9.

Example: *A box with the least material.* A box with no top has double-thick sides, a triple-thick bottom, and single-thick front and back. Its dimensions are x (front width), y (depth) and z (height), and its volume is $xyz = 3$. The cardboard used is

$$w = 4yz + 2xz + 3xy = \frac{12}{x} + \frac{6}{y} + 3xy,$$

after substituting $z = 3/(xy)$. Then

$$w_x = -\frac{12}{x^2} + 3y = 0, \qquad w_y = -\frac{6}{y^2} + 3x = 0.$$

The first gives $y = 4/x^2$; substituting into the second gives $-\frac{6}{16}x^4 + 3x = 0$, so $x^3 = 8$ (the root $x = 0$ is not allowed). Thus $x = 2$, $y = 1$, $z = 3/2$, and $w = 18$. A minimum must exist on physical grounds, and this is the only critical point.

8.4 Least squares

Given data points $(x_1, y_1), \ldots, (x_n, y_n)$, we seek the line $y = ax + b$ that best fits them. The unknowns are a and b, not x and y. The DEVIATIONS $y_i - (ax_i + b)$ measure how far each observed y_i lies from the predicted value.

Definition 8.4 [Least-squares line]. The method of least squares chooses a, b to minimize the total square deviation

$$D(a,b) = \sum_{i=1}^{n} \big(y_i - (a x_i + b)\big)^2 .$$

The minimizing line is the least-squares line, or regression line.

Squaring keeps deviations of opposite sign from canceling and weights large deviations more heavily; it is also the right choice when the measurement errors are Gaussian.

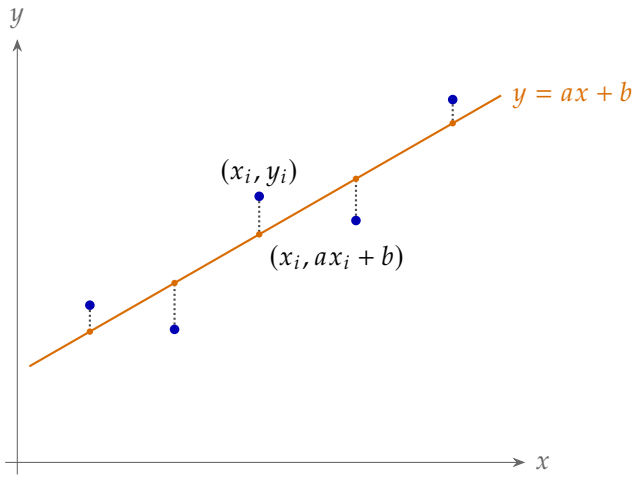


Figure 12: Schematic data points and a fitted line $y = ax + b$. The dotted segments are the deviations $y_i - (a x_i + b)$; least squares minimizes the sum of their squares.

Differentiate with the chain rule, without expanding the squares:

$$\frac{\partial D}{\partial a} = \sum_{i=1}^{n} 2(y_i - a x_i - b)(-x_i) = 0, \qquad \frac{\partial D}{\partial b} = \sum_{i=1}^{n} 2(y_i - a x_i - b)(-1) = 0.$$

Collecting terms gives a 2×2 linear system for a and b:

$$\Big(\sum x_i^2\Big) a + \Big(\sum x_i\Big) b = \sum x_i y_i, \qquad \Big(\sum x_i\Big) a + n b = \sum y_i. \tag{8.2}$$

Dividing by n, with averages $\bar{x}, \bar{y}$ and $\bar{s} = \frac{1}{n}\sum x_i^2$, the system reads $\bar{s}\, a + \bar{x}\, b = \frac{1}{n}\sum x_i y_i$ and $\bar{x}\, a + b = \bar{y}$; the second equation says the line passes through $(\bar{x}, \bar{y})$. The determinant of (8.2) is

$$n \sum x_i^2 - \Big(\sum x_i\Big)^2 = n \sum_{i=1}^{n} (x_i - \bar{x})^2 ,$$

so the system is solvable unless all x_i are equal, in which case the best line is vertical and not of the form $y = ax + b$.

The same setup fits other laws.

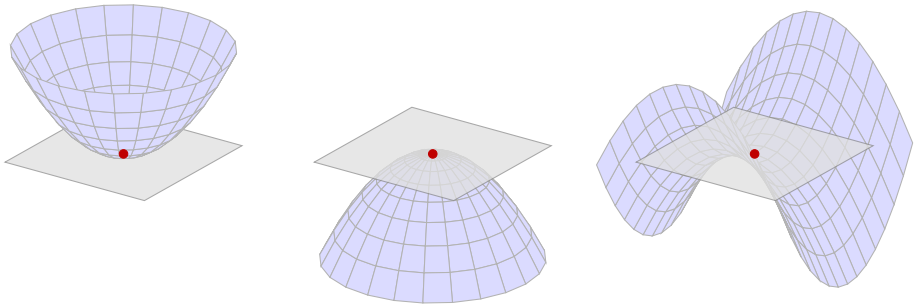
- An exponential law $y = ce^{ax}$ becomes linear after taking logarithms: $\ln y = \ln c + ax$, so with $b = \ln c$ one fits a line to the points $(x_i, \ln y_i)$. For example, Moore's law (the number of transistors on a chip grows exponentially with time) appears as a line on a log plot.
- A quadratic law $y = a_0 + a_1 x + a_2 x^2$ leads, by setting the three partials $\partial D/\partial a_j$ to zero, to a 3×3 linear system for a_0, a_1, a_2.
- More generally, for $y = a_0 f_0(x) + \cdots + a_r f_r(x)$ with given functions f_j (such as $1, x, x^2, 1/x, e^{kx}$), the method gives a square linear system in $a_0, \ldots, a_r$. A linear function $z = a_1 + a_2 x + a_3 y$ of two variables fit to points (x_i, y_i, z_i) gives a 3×3 system.

The essential point is that the unknown coefficients occur linearly in the trial function. For $y = ce^{kx}$ fitted directly the equations are not linear in k, which is why one takes logarithms first.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 8, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 9: Second derivative test; boundaries and infinity

A critical point can be a local minimum ($w = x^2 + y^2$), a local maximum ($w = -x^2 - y^2$), or a saddle ($w = y^2 - x^2$). The goal is to decide which from second derivatives, and then to find the global extrema, which may also lie on the boundary of the domain or at infinity.



Minimum: $w = x^2 + y^2$ Maximum: $w = -x^2 - y^2$ Saddle: $w = y^2 - x^2$

Figure 13: The three kinds of critical point at the origin: a minimum of $w = x^2 + y^2$, a maximum of $w = -x^2 - y^2$, and a saddle of $w = y^2 - x^2$. In each case the tangent plane at the origin (gray) is horizontal; the graph lies above it, below it, or on both sides.

9.1 Quadratic functions

Start with $w = ax^2 + bxy + cy^2$, whose only critical point (in general) is the origin. For instance $w = x^2 - 2xy + 3y^2 = (x - y)^2 + 2y^2$ has a minimum there. If $a \neq 0$, completing the square gives

$$w = a\Big(x^2 + \frac{b}{a}xy\Big) + cy^2 = a\Big(x + \frac{b}{2a}y\Big)^2 + \Big(c - \frac{b^2}{4a}\Big)y^2 = \frac{1}{4a}\Big(4a^2\Big(x + \frac{b}{2a}y\Big)^2 + (4ac - b^2)y^2\Big).$$

- If $4ac - b^2 > 0$, the two squares have the same sign: a minimum if $a > 0$, a maximum if $a < 0$.
- If $4ac - b^2 < 0$, they have opposite signs, so w takes both signs: a saddle.
- If $4ac - b^2 = 0$, the case is degenerate.

The same criterion comes from the quadratic formula. For $y \neq 0$, $w = y^2\big(at^2 + bt + c\big)$ with $t = x/y$. If $b^2 - 4ac < 0$, the polynomial $at^2 + bt + c$ has no real roots and keeps a constant sign, so w is always ≥ 0 or always ≤ 0. If $b^2 - 4ac > 0$, it changes sign, and so does w.

9.2 The second derivative test

Write $f_{xx} = \partial^2 f/\partial x^2$, f_{xy}, f_{yx}, f_{yy} for the second partial derivatives. For functions with continuous second derivatives, $f_{xy} = f_{yx}$.

Theorem 9.1 [Second derivative test]. *Let (x_0, y_0) be a critical point of f, where f has continuous second derivatives, and set*

$$A = f_{xx}(x_0, y_0), \qquad B = f_{xy}(x_0, y_0), \qquad C = f_{yy}(x_0, y_0).$$

- *If $AC - B^2 > 0$ and $A > 0$ (equivalently $C > 0$), then (x_0, y_0) is a local minimum.*
- *If $AC - B^2 > 0$ and $A < 0$ (equivalently $C < 0$), it is a local maximum.*
- *If $AC - B^2 < 0$, it is a saddle point.*
- *If $AC - B^2 = 0$, the test is inconclusive.*

If $AC - B^2 > 0$ then $AC > B^2 \geq 0$, so A and C have the same sign; this is why either may be used. In the degenerate case one needs higher derivatives or more information about the graph. For the quadratic $w = ax^2 + bxy + cy^2$ we have $A = 2a$, $B = b$, $C = 2c$, so $AC - B^2 = 4ac - b^2$, agreeing with the previous subsection.

Example: *Two critical points.* For $w = 12x^2 + y^3 - 12xy$,

$$w_x = 24x - 12y, \quad w_y = 3y^2 - 12x, \qquad A = 24, \quad B = -12, \quad C = 6y.$$

From $w_x = 0$, $y = 2x$; then $w_y = 0$ gives $12x^2 = 12x$, so $x = 0$ or $x = 1$. The critical points are $(0, 0)$ and $(1, 2)$, and $AC - B^2 = 144y - 144$. At $(0, 0)$ this is $-144 < 0$: a saddle. At $(1, 2)$ it is $144 > 0$ with $A > 0$: a local minimum. Near the origin $w \approx 12x^2 - 12xy = 12x(x - y)$, so the level curves there look like the hyperbolas $x(x - y) = c$, with asymptotes $x = 0$ and $y = x$ forming the crossing level curve.

9.3 Why the test works

Lemma 9.2 [Sign of a quadratic]. *For the quadratic $q(t) = At^2 + 2Bt + C$:*

- *if $AC - B^2 > 0$ and $A > 0$ or $C > 0$, then $q(t) > 0$ for all t;*
- *if $AC - B^2 > 0$ and $A < 0$ or $C < 0$, then $q(t) < 0$ for all t;*
- *if $AC - B^2 < 0$, then q takes both positive and negative values.*

Proof. The roots of q are $t = \left(-B \pm \sqrt{B^2 - AC}\right)/A$ when $A \neq 0$. If $AC - B^2 > 0$, then $A \neq 0$ and there are no real roots, so the graph of q lies on one side of the t-axis. The side is determined by A, since $q(t) \to +\infty$ as $t \to \infty$ when $A > 0$, or by C, since $q(0) = C$. If $AC - B^2 < 0$ and $A \neq 0$, there are two real roots and the parabola crosses the axis. If $A = 0$, then $B \neq 0$ and $q(t) = 2Bt + C$ is a nonconstant line, which takes both signs. □

Quadratic functions. Consider

$$w = w_0 + ax + by + \tfrac{1}{2}\big(Ax^2 + 2Bxy + Cy^2\big),$$

normalized so that $(w_x)_0 = a$, $(w_y)_0 = b$, $w_{xx} = A$, $w_{xy} = B$, $w_{yy} = C$; this explains the $\frac{1}{2}$ and the $2B$. If the origin is a critical point, then $a = b = 0$ and

$$w - w_0 = \tfrac{1}{2}\big(Ax^2 + 2Bxy + Cy^2\big) = \frac{y^2}{2}\left(A\left(\frac{x}{y}\right)^2 + 2B\frac{x}{y} + C\right) \qquad (y \neq 0).$$

The tangent plane at the origin is $w = w_0$, so the question is the sign of $w - w_0$ at points $(x, y) \neq (0, 0)$. When $y \neq 0$, apply Lemma 9.2 to the bracket. When $y = 0$, then $x \neq 0$ and $w - w_0 = \frac{1}{2}Ax^2$; when $AC - B^2 > 0$ this has the sign of A as well. Hence $AC - B^2 > 0$, $A > 0$ gives $w > w_0$ away from the origin, a minimum; $AC - B^2 > 0$, $A < 0$ gives a maximum; and $AC - B^2 < 0$ gives values on both sides of w_0, a saddle.

General functions. Suppose f has a critical point at (x_0, y_0) and continuous second derivatives nearby. With $u = x - x_0$, $v = y - y_0$, the quadratic approximation (Taylor's formula to second order) is

$$\Delta f \approx f_x\, u + f_y\, v + \tfrac{1}{2}f_{xx}u^2 + f_{xy}uv + \tfrac{1}{2}f_{yy}v^2,$$

and at a critical point the linear terms vanish:

$$f(x, y) \approx w_0 + \tfrac{1}{2}\big(Au^2 + 2Buv + Cv^2\big).$$

Since $u_x = 1$, $v_x = 0$ and $u_y = 0$, $v_y = 1$, the chain rule gives $w_x = w_u$, $w_y = w_v$, and likewise for the second derivatives, so A, B, C are the second partials of f at (x_0, y_0). Near the critical point the graph of f is essentially that of its quadratic approximation, so the classification of the quadratic case carries over. This is suggestive rather than rigorous; in the degenerate case the quadratic part does not determine the answer.

9.4 Boundaries and infinity

The global maximum or minimum of a function need not occur at a critical point. It may also occur on the boundary of the domain, or the function may approach its extreme values at infinity. To find global extrema, compare the values at critical points with the behavior on the boundary and at infinity.

Example: *A global minimum and no maximum.* Let $f(x, y) = x + y + \dfrac{1}{xy}$ for $x > 0$, $y > 0$. Then

$$f_x = 1 - \frac{1}{x^2 y} = 0, \qquad f_y = 1 - \frac{1}{xy^2} = 0,$$

so $x^2 y = 1 = xy^2$, and the only critical point is $(1, 1)$. The second derivatives are $f_{xx} = 2/(x^3 y)$,

$f_{xy} = 1/(x^2y^2)$, $f_{yy} = 2/(xy^3)$, so $A = 2$, $B = 1$, $C = 2$ and $AC - B^2 = 3 > 0$ with $A > 0$: a local minimum, where $f = 3$. As $x \to 0$ or $y \to 0$ (the boundary) and as $x \to \infty$ or $y \to \infty$, $f \to \infty$. So there is no maximum, and the local minimum is the global minimum.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 9, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 10: Differentials; chain rule

The total differential records how variations of a function are related to variations of its variables. It packages the approximation formula and, divided by dt, gives the chain rule.

10.1 Differentials

In one variable, $y = f(x)$ gives $dy = f'(x)\,dx$. Differentials can be manipulated directly: for $y = \sin^{-1} x$ we have $x = \sin y$, so $dx = \cos y\,dy$ and

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}.$$

Definition 10.1 [Total differential]**.** For $f = f(x, y, z)$, the total differential is

$$df = f_x\,dx + f_y\,dy + f_z\,dz.$$

The differential df is a new kind of object with its own rules, and it is not the same as the change Δf. It encodes how variations of f are related to variations of x, y, z, and it is used in two ways.

1. As a placeholder for the approximation formula: replacing d by Δ gives $\Delta f \approx f_x\,\Delta x + f_y\,\Delta y + f_z\,\Delta z$.
2. Dividing by dt gives the chain rule below.

10.2 The chain rule along a curve

Theorem 10.2 [Chain rule]**.** *If f is smooth and $x = x(t)$, $y = y(t)$, $z = z(t)$ are differentiable, then $f(x(t), y(t), z(t))$ is a function of t and*

$$\frac{df}{dt} = f_x\frac{dx}{dt} + f_y\frac{dy}{dt} + f_z\frac{dz}{dt}.$$

Formally, substitute $dx = x'(t)\,dt$, $dy = y'(t)\,dt$, $dz = z'(t)\,dt$ into df to get $df = (f_x x' + f_y y' + f_z z')\,dt$.

Proof. By the approximation formula (Lecture 8, with its error terms),

$$\Delta f = (f_x + \varepsilon_1)\Delta x + (f_y + \varepsilon_2)\Delta y + (f_z + \varepsilon_3)\Delta z,$$

where $\varepsilon_i \to 0$ as $(\Delta x, \Delta y, \Delta z) \to 0$. Divide by Δt and let $\Delta t \to 0$. Then $\Delta x/\Delta t \to x'(t)$, and similarly for y, z; since x, y, z are continuous, $\Delta x, \Delta y, \Delta z \to 0$, so each $\varepsilon_i \to 0$. □

Example: *Checking against substitution.* Let $w = x^2 y + z$, so $dw = 2xy\,dx + x^2\,dy + dz$. Along $x = t$, $y = e^t$, $z = \sin t$,

$$\frac{dw}{dt} = (2te^t)\cdot 1 + t^2 \cdot e^t + \cos t.$$

Substituting first gives $w = t^2 e^t + \sin t$, whose derivative is the same.

The product and quotient rules are special cases. For $f = uv$ with $u = u(t)$, $v = v(t)$,

$$\frac{d(uv)}{dt} = f_u\,u' + f_v\,v' = vu' + uv'.$$

For $g = u/v$,

$$\frac{d(u/v)}{dt} = g_u\,u' + g_v\,v' = \frac{1}{v}\,u' - \frac{u}{v^2}\,v' = \frac{u'v - uv'}{v^2}.$$

10.3 Several independent variables

Suppose $w = f(x, y)$ with $x = x(u, v)$ and $y = y(u, v)$. Here w is the DEPENDENT VARIABLE, u, v are the INDEPENDENT VARIABLES, and x, y are INTERMEDIATE VARIABLES; computing w is a chain $(u, v) \to (x, y) \to w$.

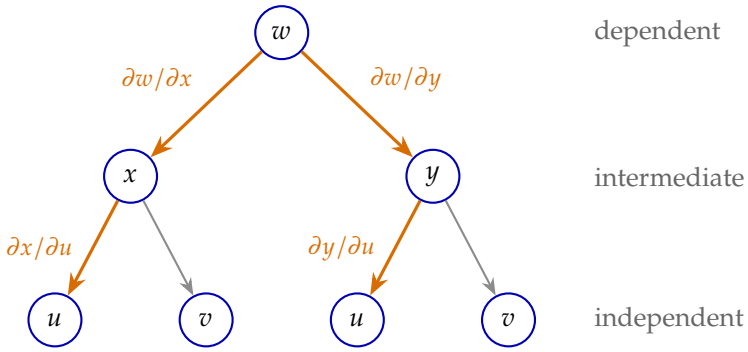


Figure 14: Dependence diagram for $w = f(x, y)$, $x = x(u, v)$, $y = y(u, v)$. Each highlighted path from w down to u contributes one product of partial derivatives to $\partial w/\partial u$.

Substituting $dx = x_u\,du + x_v\,dv$ and $dy = y_u\,du + y_v\,dv$ into $dw = f_x\,dx + f_y\,dy$,

$$dw = (f_x x_u + f_y y_u)\,du + (f_x x_v + f_y y_v)\,dv.$$

Identifying the coefficients of du and dv gives

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial w}{\partial v} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial v}. \tag{10.1}$$

The same follows from the approximation $\Delta w \approx w_x \Delta x + w_y \Delta y$: hold v fixed, divide by Δu, and let $\Delta u \to 0$. One may not "cancel the ∂x" in (10.1): each term on the right is only part of the answer.

Example: *A two-variable chain.* Let $w = x^2 y + y^2 + x$ with $x = u^2 v$, $y = u v^2$. Then

$$w_x = 2xy + 1, \quad w_y = x^2 + 2y, \quad x_u = 2uv, \quad y_u = v^2, \quad x_v = u^2, \quad y_v = 2uv,$$

so

$$\frac{\partial w}{\partial u} = (2xy + 1)\,2uv + (x^2 + 2y)\,v^2, \qquad \frac{\partial w}{\partial v} = (2xy + 1)\,u^2 + (x^2 + 2y)\,2uv.$$

The variables may be left mixed. At $(u, v) = (1, 2)$ we have $x = 2$, $y = 4$, so $\partial w/\partial u = 17{\cdot}4 + 12{\cdot}4 = 116$. Substituting first, $w = u^5 v^4 + u^2 v^4 + u^2 v$ and $\partial w/\partial u = 5u^4 v^4 + 2uv^4 + 2uv$, which is also 116 at $(1, 2)$.

Example: *Polar coordinates.* With $x = r\cos\theta$, $y = r\sin\theta$,

$$f_r = f_x x_r + f_y y_r = \cos\theta\, f_x + \sin\theta\, f_y, \qquad f_\theta = f_x x_\theta + f_y y_\theta = -r\sin\theta\, f_x + r\cos\theta\, f_y.$$

Which variables are independent is often clear only from context. In thermodynamics, for example, among P, T, V, U, S any two may be taken as independent and the others regarded as functions of them, so a partial derivative such as $\partial U/\partial P$ depends on which other variable is held fixed.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 10, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 11: Gradient; directional derivative; tangent plane

Collecting the partial derivatives into one vector, the gradient, turns the chain rule into a dot product. The gradient is perpendicular to level curves and surfaces, gives tangent planes to surfaces, and determines the rate of change in every direction.

11.1 The gradient

In vector notation, with $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$, the chain rule reads

$$\frac{dw}{dt} = w_x\frac{dx}{dt} + w_y\frac{dy}{dt} + w_z\frac{dz}{dt} = \nabla w \cdot \frac{d\mathbf{r}}{dt}.$$

Definition 11.1 [Gradient]. The gradient of $w = f(x, y, z)$ is the vector

$$\nabla w = \operatorname{grad} w = \langle w_x, w_y, w_z\rangle,$$

and for $w = f(x, y)$ it is $\langle w_x, w_y\rangle$.

The gradient turns a scalar function into a vector that depends on the point where it is evaluated, written $\nabla w|_{P_0}$ or $\nabla w(x_0, y_0)$. For $f(x, y)$ the vector ∇f lies in the xy-plane, not in the space of the graph.

Theorem 11.2 [Gradient is normal to level sets]. *At a point P of the level surface $f(x, y, z) = c$, the gradient $\nabla f|_P$ is perpendicular to the surface: it is perpendicular to the velocity vector of every curve on the surface through P. The same holds for level curves $f(x, y) = c$.*

Proof. Let $\mathbf{r}(t)$ be a curve on the level surface with $\mathbf{r}(t_0) = P$. Then $f(\mathbf{r}(t)) = c$ for all t. Differentiating with the chain rule,

$$0 = \frac{d}{dt} f(\mathbf{r}(t))\Big|_{t_0} = \nabla f|_P \cdot \mathbf{r}'(t_0).$$

So $\nabla f|_P$ is perpendicular to $\mathbf{r}'(t_0)$, and every vector tangent to the surface at P arises as such a velocity. □

Example: *Level sets and their normals.*

- For $w = ax + by + cz$, the level surface $w = d$ is a plane, with normal vector $\nabla w = \langle a, b, c\rangle$.
- For $w = x^2 + y^2$, the level curves are circles centered at the origin and $\nabla w = \langle 2x, 2y\rangle$ points radially outward, perpendicular to them. Likewise for $w = (x^2 + y^2)/3$, $\nabla w = \frac{2}{3}\langle x, y\rangle$; at $(1, 2)$

it is $\frac{2}{3}\langle 1,2\rangle$, normal to the level circle $x^2+y^2=5$ through that point.

- The graph of $y=e^x$ is the level curve $w=y-e^x=0$. So $\nabla w=\langle -e^x,1\rangle$, which at $(1,e)$ is $\langle -e,1\rangle$, is perpendicular to the tangent vector $\langle 1,e\rangle$; no negative reciprocal slope is needed.

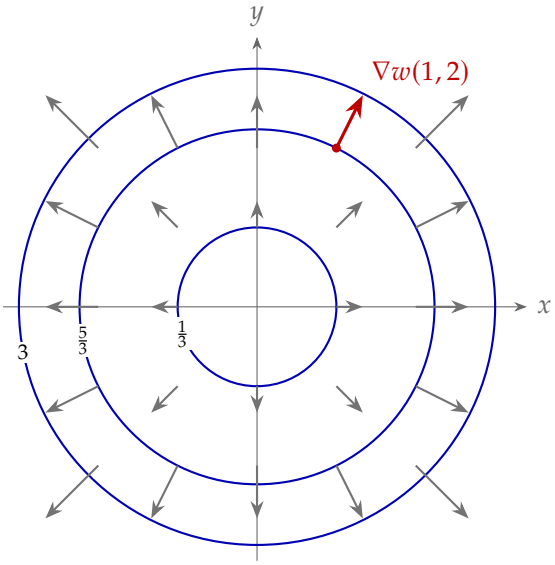


Figure 15: Level curves of $w=(x^2+y^2)/3$ and its gradient field $\nabla w=\frac{2}{3}\langle x,y\rangle$ (arrows drawn at half their length; circles labeled with their values of w). The gradient points radially outward, perpendicular to the level circles; it is highlighted at $(1,2)$ on the circle $x^2+y^2=5$.

11.2 Tangent planes to surfaces

Since the gradient is normal to a level surface, it gives the tangent plane in point-normal form.

Example: *Two tangent planes.* For $x^2+y^2-z^2=4$ at $(2,1,1)$, take $w=x^2+y^2-z^2$; then $\nabla w=\langle 2x,2y,-2z\rangle=\langle 4,2,-2\rangle$, and the tangent plane is

$$4(x-2)+2(y-1)-2(z-1)=0, \qquad \text{i.e.} \qquad 4x+2y-2z=8.$$

Equivalently, $dw=2x\,dx+2y\,dy-2z\,dz=4\,dx+2\,dy-2\,dz$ at the point, so $\Delta w\approx 4\Delta x+2\Delta y-2\Delta z$; the level surface is $\Delta w=0$, and its linear approximation $4\Delta x+2\Delta y-2\Delta z=0$ is the same plane. Here one could also solve $z=\sqrt{x^2+y^2-4}$ and use the tangent plane of a graph, but in general a surface cannot be solved for one variable.

Similarly, for $x^2+2y^2+3z^2=6$ at $(1,1,1)$, the gradient of $w=x^2+2y^2+3z^2$ is $\langle 2x,4y,6z\rangle=\langle 2,4,6\rangle$, and the tangent plane is $2(x-1)+4(y-1)+6(z-1)=0$, i.e. $2x+4y+6z=12$.

11.3 Directional derivatives

The partials w_x, w_y give the rate of change of w in the $\mathbf{i}$ and $\mathbf{j}$ directions. To measure it in an arbitrary direction, fix a unit vector $\hat{\mathbf{u}} = \langle a, b \rangle$ and move along the line

$$x(s) = x_0 + as, \qquad y(s) = y_0 + bs$$

with velocity $\hat{\mathbf{u}}$. Since the speed is 1, s is arclength.

Definition 11.3 [Directional derivative]. The directional derivative of w at P_0 in the direction $\hat{\mathbf{u}}$ is

$$\left.\frac{dw}{ds}\right|_{\hat{\mathbf{u}}} = \lim_{\Delta s \to 0} \frac{\Delta w}{\Delta s},$$

where Δw is the change in w caused by a step of length Δs in the direction $\hat{\mathbf{u}}$.

Geometrically, slice the graph of w by the vertical plane through P_0 in the direction $\hat{\mathbf{u}}$, which is no longer parallel to an axis; the directional derivative is the slope of the slice. For $\hat{\mathbf{u}} = \mathbf{i}$ or $\mathbf{j}$ it is w_x or w_y.

Proposition 11.4 [Directional derivative formula].

$$\left.\frac{dw}{ds}\right|_{\hat{\mathbf{u}}} = \nabla w \cdot \hat{\mathbf{u}}.$$

Proof. By the chain rule along $\mathbf{r}(s)$, $dw/ds = \nabla w \cdot d\mathbf{r}/ds = \nabla w \cdot \hat{\mathbf{u}}$. Equivalently, a step of length Δs has components $\Delta x, \Delta y$ with $\langle \Delta x/\Delta s, \Delta y/\Delta s \rangle = \hat{\mathbf{u}}$; dividing $\Delta w \approx w_x \Delta x + w_y \Delta y$ by Δs gives $\Delta w/\Delta s \approx \nabla w \cdot \hat{\mathbf{u}}$, and the limit $\Delta s \to 0$ gives the formula. □

Example: *A computation.* For $w = x^3 + 3y^2$ at $P_0 = (1, 2)$ in the direction of $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$: $\nabla w = \langle 3x^2, 6y \rangle$, so $\nabla w|_{(1,2)} = \langle 3, 12 \rangle$. The direction must be a unit vector, $\hat{\mathbf{u}} = \mathbf{v}/|\mathbf{v}| = \langle \frac{3}{5}, \frac{4}{5} \rangle$. Then

$$\left.\frac{dw}{ds}\right|_{\hat{\mathbf{u}}} = \langle 3, 12 \rangle \cdot \left\langle \tfrac{3}{5}, \tfrac{4}{5} \right\rangle = \frac{57}{5}.$$

11.4 Direction of fastest increase

Let θ be the angle between ∇w and $\hat{\mathbf{u}}$. Since $|\hat{\mathbf{u}}| = 1$,

$$\left.\frac{dw}{ds}\right|_{\hat{\mathbf{u}}} = \nabla w \cdot \hat{\mathbf{u}} = |\nabla w| \cos\theta.$$

- The maximum, at $\cos\theta = 1$, occurs when $\hat{\mathbf{u}}$ points along ∇w. So the direction of ∇w is the direction of fastest increase of w, and $|\nabla w|$ is the directional derivative in that direction. The

opposite direction gives the fastest decrease, at rate $-|\nabla w|$.

- The directional derivative is 0 when $\hat{\mathbf{u}} \perp \nabla w$, that is, when $\hat{\mathbf{u}}$ is tangent to the level curve, consistent with Theorem 11.2.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 11, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 12: Lagrange multipliers

To minimize or maximize $f(x, y, z)$ when the variables are constrained by an equation $g(x, y, z) = c$, one cannot simply set the partial derivatives of f to zero. At a constrained extremum the level set of f is tangent to the constraint, so the two gradients are parallel.

12.1 The method

Example: *The point of $xy = 3$ closest to the origin.* Minimize $\sqrt{x^2 + y^2}$, or more simply $f(x, y) = x^2 + y^2$, subject to $g(x, y) = xy = 3$. At the minimum, the level curve of f (a circle) is tangent to the hyperbola $g = 3$, so their normal vectors ∇f and ∇g are parallel.

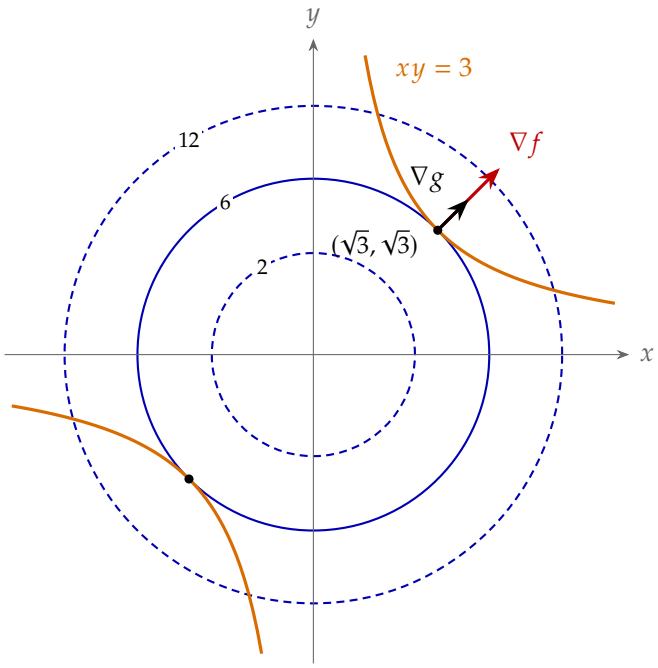


Figure 16: Level circles $x^2 + y^2 = c$ and the constraint curve $xy = 3$. The smallest circle that meets the constraint, $x^2 + y^2 = 6$, is tangent to it at $(\sqrt{3}, \sqrt{3})$ and $(-\sqrt{3}, -\sqrt{3})$, where $\nabla f = 2\nabla g$ (gradients drawn at a quarter of their length). The dashed circles $c = 2$ and $c = 12$ miss and cross the constraint.

Definition 12.1 [Lagrange multiplier equations]. For f subject to $g = c$, the Lagrange multiplier method looks for points where

$$\nabla f = \lambda \nabla g \quad \text{and} \quad g = c$$

for some number λ, the multiplier. In two variables this is the system $f_x = \lambda g_x$, $f_y = \lambda g_y$, $g = c$ in the three unknowns x, y, λ; such points are the critical points of the constrained problem.

Example: *Solving the example.* Here the system is

$$2x = \lambda y, \qquad 2y = \lambda x, \qquad xy = 3.$$

In general such systems are hard and may need a computer. Here the first two equations form a linear system $2x - \lambda y = 0$, $-\lambda x + 2y = 0$ in x, y. Either $x = y = 0$, which is impossible since $xy = 3$, or its determinant $4 - \lambda^2$ is zero, so $\lambda = \pm 2$. For $\lambda = -2$ we get $y = -x$ and $xy = -x^2 = 3$, which has no solution. For $\lambda = 2$ we get $y = x$ and $x^2 = 3$, giving $(\sqrt{3}, \sqrt{3})$ and $(-\sqrt{3}, -\sqrt{3})$, at distance $\sqrt{6}$ from the origin.

12.2 Why the method works

Theorem 12.2 [Lagrange multipliers]. *Let f and g be smooth, and suppose P is a local maximum or minimum of f on the surface $g(x, y, z) = c$, with $\nabla g|_P \neq \mathbf{0}$. Then $\nabla f|_P = \lambda \nabla g|_P$ for some λ.*

Proof. Let $\mathbf{r}(t)$ be any curve on the constraint surface with $\mathbf{r}(0) = P$, and let $h(t) = f(\mathbf{r}(t))$. Since P is a local extremum of f on the surface, h has a local extremum at $t = 0$, so by the chain rule

$$0 = h'(0) = \nabla f|_P \cdot \mathbf{r}'(0).$$

Thus $\nabla f|_P$ is perpendicular to every curve on the surface through P, hence normal to the tangent plane. So is $\nabla g|_P$, because the surface is a level surface of g (Lecture 11). Two normal vectors to the same plane are parallel, and $\nabla g|_P \neq \mathbf{0}$, so $\nabla f|_P = \lambda \nabla g|_P$. □

In terms of directional derivatives: at a constrained extremum, moving along the constraint in any direction $\hat{\mathbf{u}}$ tangent to $\{g = c\}$ gives $df/ds|_{\hat{\mathbf{u}}} = \nabla f \cdot \hat{\mathbf{u}} = 0$.

Geometrically, in two variables, follow the level curves $f = k$ as k decreases from a value whose level curve misses the constraint curve. The first level curve to touch the constraint does so at a local maximum P of f on the constraint, since moving along the constraint from P in either direction lowers f. The two curves are tangent at P, so their normals ∇f and ∇g are parallel. The last level curve to touch gives a local minimum in the same way.

The method does not say whether a critical point is a minimum or a maximum, and the second derivative test does not apply to constrained variables. One decides by a geometric argument or by comparing the values of f at the critical points and on the boundary.

12.3 Examples

Example: *The box with the least material, again.* As in Lecture 8, a box with no top, double-thick sides, triple-thick bottom and single-thick front and back uses $f = 4yz + 2xz + 3xy$ of cardboard, now subject to the volume constraint $V = xyz = 3$ instead of eliminating z. The equations $\nabla f = \lambda \nabla V$ read

$$2z + 3y = \lambda yz, \qquad 4z + 3x = \lambda xz, \qquad 4y + 2x = \lambda xy, \qquad xyz = 3.$$

Solve each of the first three for λ (solving symmetrically):

$$\lambda = \frac{2}{y} + \frac{3}{z} = \frac{4}{x} + \frac{3}{z} = \frac{4}{x} + \frac{2}{y}.$$

The first equality gives $2/y = 4/x$, so $x = 2y$; the second gives $3/z = 2/y$, so $z = \frac{3}{2}y$. Then $xyz = 3y^3 = 3$, so $y = 1$, and the answer is $x = 2$, $y = 1$, $z = \frac{3}{2}$, with $f = 18$, as before.

Example: *A linear function on a sphere.* Minimize $w = y$ on $x^2 + y^2 + z^2 = 1$. Here $\nabla f = \langle 0, 1, 0 \rangle$ and $\nabla g = \langle 2x, 2y, 2z \rangle$, so $\nabla f = \lambda \nabla g$ forces $x = z = 0$, and the constraint gives $y = \pm 1$: the minimum at $(0, -1, 0)$ and the maximum at $(0, 1, 0)$.

Example: *Checking the boundary.* A rectangle in the first quadrant has one corner at the origin O, two sides on the axes, and the opposite corner $P = (x, y)$ on the line $x + 2y = 1$. Maximize its area $f = xy$ subject to $g = x + 2y = 1$. Then $\nabla f = \langle y, x \rangle$, $\nabla g = \langle 1, 2 \rangle$, so $y = \lambda$, $x = 2\lambda$, hence $x = 2y$, and $x + 2y = 1$ gives $P = (\frac{1}{2}, \frac{1}{4})$ with area $\frac{1}{8}$. This is the maximum: the maximum occurs at a critical point or on the boundary, and the boundary points $(1, 0)$ and $(0, \frac{1}{2})$ give rectangles of area 0.

Example: *Boundary at infinity.* Same setup with P on the curve $xy = 1$; minimize the perimeter $f = 2x + 2y$. Now $\nabla g = \langle y, x \rangle$, $\nabla f = \langle 2, 2 \rangle$, so $2 = \lambda y$, $2 = \lambda x$, hence $x = y$, and $xy = 1$ gives $P = (1, 1)$ with perimeter 4. This is the minimum: the boundary points are infinitely far out along the axes, where the perimeter tends to infinity.

Example: *The surface-minimizing pyramid.* Among pyramids with a given triangular base and given volume V, find the one of least total area. Since $V = \frac{1}{3} A_{\text{base}} h$, the height h is fixed, and the top vertex moves in the plane $z = h$. In coordinates (base vertices P_1, P_2, P_3 in the plane $z = 0$, top vertex $P = (x, y, h)$, faces of area $\frac{1}{2}|\overrightarrow{PP_1} \times \overrightarrow{PP_2}|$, and so on) the critical point equations are very hard. Instead use variables adapted to the geometry: let a_1, a_2, a_3 be the side lengths of the base triangle and u_1, u_2, u_3 the distances in the plane from the projection of P to the three sides. By the Pythagorean theorem the face over side i is a triangle with base a_i and height $\sqrt{u_i^2 + h^2}$, so we minimize

$$f(u_1, u_2, u_3) = \tfrac{1}{2} a_1 \sqrt{u_1^2 + h^2} + \tfrac{1}{2} a_2 \sqrt{u_2^2 + h^2} + \tfrac{1}{2} a_3 \sqrt{u_3^2 + h^2}.$$

The u_i are not independent: splitting the base into three triangles with apex at the projection of P and heights u_i gives the constraint

$$g(u_1, u_2, u_3) = \tfrac{1}{2} a_1 u_1 + \tfrac{1}{2} a_2 u_2 + \tfrac{1}{2} a_3 u_3 = A_{\text{base}}.$$

The equations $\nabla f = \lambda \nabla g$ read

$$\frac{a_i}{2} \frac{u_i}{\sqrt{u_i^2 + h^2}} = \lambda \frac{a_i}{2}, \qquad \text{so} \qquad \lambda = \frac{u_1}{\sqrt{u_1^2 + h^2}} = \frac{u_2}{\sqrt{u_2^2 + h^2}} = \frac{u_3}{\sqrt{u_3^2 + h^2}}.$$

The function $u \mapsto u/\sqrt{u^2 + h^2}$ is strictly increasing, so $u_1 = u_2 = u_3$: the projection of the top vertex is equidistant from the three sides, and P lies above the incenter of the base.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 12, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 13: Non-independent variables

Variables are often tied together by an equation, as pressure, volume and temperature are by $pV = nRT$. A partial derivative then depends on which variables are treated as independent, and the notation must say so.

13.1 Implicit partial derivatives

If $g(x, y, z) = c$, we may regard z as a function $z(x, y)$ near a point where $g_z \neq 0$. Taking the differential of the constraint gives

$$g_x\,dx + g_y\,dy + g_z\,dz = 0.$$

Holding y fixed means $dy = 0$, so $dz = -(g_x/g_z)\,dx$ and

$$\frac{\partial z}{\partial x} = -\frac{g_x}{g_z}, \qquad \frac{\partial z}{\partial y} = -\frac{g_y}{g_z}.$$

Example: *An implicit surface.* Let $x^2 + yz + z^3 = 8$ at $(2, 3, 1)$. Then $2x\,dx + z\,dy + (y + 3z^2)\,dz = 0$, which at the point reads $4\,dx + dy + 6\,dz = 0$. Hence $dz = -\frac{4}{6}\,dx - \frac{1}{6}\,dy$, and reading off coefficients, $\partial z/\partial x = -2/3$ and $\partial z/\partial y = -1/6$.

13.2 Specifying what is held fixed

A change of variables can change the meaning of a partial derivative even when the variable itself does not change. Let $f(x, y) = x + y$, so $\partial f/\partial x = 1$. Substitute $x = u$, $y = u + v$; then $f = 2u + v$ and $\partial f/\partial u = 2$. Although $x = u$, the two derivatives differ: $\partial f/\partial x$ varies x with y fixed, while $\partial f/\partial u$ varies u with $v = y - x$ fixed.

Definition 13.1 [Constrained partial derivative]. For variables related by one or more equations, $\left(\frac{\partial w}{\partial x}\right)_y$ denotes the rate of change of w as x varies with y held fixed, the remaining variables doing whatever the constraints require. The subscripts, together with the variable of differentiation, list the independent variables. With more variables one writes, for instance, $\left(\frac{\partial w}{\partial x}\right)_{y,t}$.

In the example above, $\left(\frac{\partial f}{\partial u}\right)_v = \left(\frac{\partial f}{\partial x}\right)_v = 2 \neq 1 = \left(\frac{\partial f}{\partial x}\right)_y$. By default, $\partial/\partial x$ keeps the other original variables fixed.

Example: *Two answers for one derivative.* Let $w = x^2 + y^2 + z^2$ with $z = x^2 + y^2$.

- With x, y independent, substituting for z gives $w = x^2+y^2+(x^2+y^2)^2$, so $\left(\frac{\partial w}{\partial x}\right)_y = 2x+4x^3+4xy^2$. The chain rule agrees: $2x + 2z\, z_x = 2x + 2(x^2 + y^2) \cdot 2x$.
- With x, z independent, $y^2 = z - x^2$ gives $w = z + z^2$, so $\left(\frac{\partial w}{\partial x}\right)_z = 0$.

Geometrically, (x, y, z) lies on the paraboloid $z = x^2 + y^2$ and w is the squared distance from the origin. At $P_0 = (1, 0, 1)$, holding y fixed moves P up the parabola in the xz-plane, where w increases at rate 6. Holding z fixed moves P around the horizontal circle $z = 1$, where w is constant. The two derivatives measure change along different paths (Figure 17).

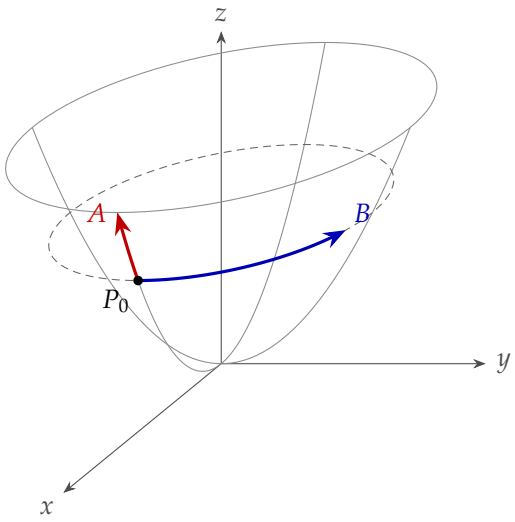


Figure 17: The paraboloid $z = x^2 + y^2$ through $P_0 = (1, 0, 1)$. Holding $y = 0$ fixed, P moves along the parabola $z = x^2$ toward A; holding $z = 1$ fixed, P moves along the circle $x^2 + y^2 = 1$ toward B.

Example: *Area of a right triangle.* A triangle with sides a, b enclosing angle θ has area $A = \frac{1}{2}ab \sin\theta$. If it is a right triangle with hypotenuse b, then $a = b\cos\theta$ (Figure 18). The rate of change of A with respect to θ has three meanings:

1. $\partial A/\partial\theta = A_\theta = \frac{1}{2}ab\cos\theta$, with a, b independent and fixed (the triangle stops being a right triangle);
2. $\left(\frac{\partial A}{\partial\theta}\right)_a$, with a fixed and b adjusting to keep $a = b\cos\theta$;
3. $\left(\frac{\partial A}{\partial\theta}\right)_b$, with b fixed and a adjusting.

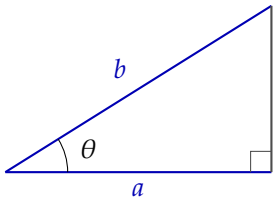


Figure 18: The right triangle with leg a and hypotenuse b enclosing the angle θ, so that $a = b\cos\theta$.

There are three ways to compute $\left(\frac{\partial A}{\partial \theta}\right)_a$, treating A as a function of a and θ with $b = b(a, \theta)$.

- *Substitution.* Here $b = a\sec\theta$, so $A = \frac{1}{2}a^2\tan\theta$ and $\left(\frac{\partial A}{\partial \theta}\right)_a = \frac{1}{2}a^2\sec^2\theta$. This is easiest when one can solve for b, which is not always possible.
- *Total differentials.* We have $dA = \frac{1}{2}ab\cos\theta\,d\theta + \frac{1}{2}b\sin\theta\,da + \frac{1}{2}a\sin\theta\,db$ and, from the constraint, $da = \cos\theta\,db - b\sin\theta\,d\theta$. Setting $da = 0$ gives $db = b\tan\theta\,d\theta$, so

$$dA = \Big(\tfrac{1}{2}ab\cos\theta + \tfrac{1}{2}a\sin\theta\, b\tan\theta\Big)d\theta, \qquad \left(\frac{\partial A}{\partial \theta}\right)_a = \tfrac{1}{2}ab\sec\theta.$$

- *Chain rule.* Since $(\partial\theta/\partial\theta)_a = 1$ and $(\partial a/\partial\theta)_a = 0$,

$$\left(\frac{\partial A}{\partial \theta}\right)_a = A_\theta + A_a\left(\frac{\partial a}{\partial \theta}\right)_a + A_b\left(\frac{\partial b}{\partial \theta}\right)_a = A_\theta + A_b\left(\frac{\partial b}{\partial \theta}\right)_a.$$

  Differentiating $a = b\cos\theta$ with a fixed gives $0 = \left(\frac{\partial b}{\partial \theta}\right)_a\cos\theta - b\sin\theta$, so $\left(\frac{\partial b}{\partial \theta}\right)_a = b\tan\theta$ and again $\left(\frac{\partial A}{\partial \theta}\right)_a = \frac{1}{2}ab\sec\theta$.

All three agree, since $\frac{1}{2}ab\sec\theta = \frac{1}{2}a^2\sec^2\theta$ when $b = a\sec\theta$. The two systematic methods compute the same quantities, written differently.

13.3 The general recipe

Let $f = f(x, y, z)$ with $g(x, y, z) = c$, and suppose we want $\left(\frac{\partial f}{\partial z}\right)_y$. The independent variables are y, z, so x must be eliminated.

Differentials. Set $dy = 0$ in $df = f_x\,dx + f_y\,dy + f_z\,dz$. The constraint $dg = g_x\,dx + g_y\,dy + g_z\,dz = 0$ with $dy = 0$ gives $dx = -(g_z/g_x)\,dz$. Substituting,

$$df = \Big(f_z - \frac{f_x g_z}{g_x}\Big)dz, \qquad \left(\frac{\partial f}{\partial z}\right)_y = f_z - \frac{f_x g_z}{g_x}.$$

Chain rule. Since $(\partial y/\partial z)_y = 0$ and $(\partial z/\partial z)_y = 1$,

$$\left(\frac{\partial f}{\partial z}\right)_y = f_x\left(\frac{\partial x}{\partial z}\right)_y + f_z, \qquad 0 = \left(\frac{\partial g}{\partial z}\right)_y = g_x\left(\frac{\partial x}{\partial z}\right)_y + g_z.$$

The second equation gives $(\partial x/\partial z)_y = -g_z/g_x$, and the first then gives the same answer.

13.4 Differentials versus the chain rule

The DIFFERENTIAL of $w = f(x, y, z)$ is

$$dw = f_x\,dx + f_y\,dy + f_z\,dz, \tag{13.1}$$

where f_x, f_y, f_z are the FORMAL PARTIAL DERIVATIVES, computed as if x, y, z were independent. Differentials obey three rules, treating $dx, dy, dz, \ldots$ like basis vectors:

D1. They can be added, subtracted, and multiplied by scalar functions.

D2. If $x, y, \ldots$ are independent, then $A\,dx + B\,dy + \cdots = A_1\,dx + B_1\,dy + \cdots$ if and only if $A = A_1$, $B = B_1$, and so on.

D3. One differential can be substituted into another (a consequence of the chain rule).

The method needs no bookkeeping of which variables depend on which: eliminate dependent differentials (D1, D3), then read off coefficients (D2). Rule D2 fails for dependent variables; for instance $y\,dx + x\,dy = z\,dt + t\,dz$ when $xy = zt$. There is no meaningful product $du\,dv$ or quotient du/dv of differentials here.

Example: *One constraint, both methods.* Find $\left(\frac{\partial w}{\partial y}\right)_{x,t}$ for $w = x^3y - z^2t$ with $xy = zt$. By the chain rule, $(\partial z/\partial y)_{x,t} = x/t$ from $z = xy/t$, so

$$\left(\frac{\partial w}{\partial y}\right)_{x,t} = x^3 - 2zt\left(\frac{\partial z}{\partial y}\right)_{x,t} = x^3 - 2zt\cdot\frac{x}{t} = x^3 - 2zx.$$

With differentials, $dw = 3x^2y\,dx + x^3\,dy - 2zt\,dz - z^2\,dt$ and $y\,dx + x\,dy = z\,dt + t\,dz$. Multiplying the second equation by $2z$ and adding it to the first eliminates dz:

$$dw = (3x^2y - 2zy)\,dx + (x^3 - 2zx)\,dy + z^2\,dt.$$

Since x, y, t are independent, D2 gives all three derivatives at once: $\left(\frac{\partial w}{\partial x}\right)_{y,t} = 3x^2y - 2zy$, $\left(\frac{\partial w}{\partial y}\right)_{x,t} = x^3 - 2zx$, and $\left(\frac{\partial w}{\partial t}\right)_{x,y} = z^2$: more work than the chain rule, but every derivative at once.

Example: *Two constraints.* Let $w = x^2 - yz + t^2$ with $z^2 = x + y^2$ and $xy = zt$. These determine z and then t as functions of x and y, so $w = w(x, y)$; we find ∇w without computing $w(x, y)$. The differentials are

$$dw = 2x\,dx - z\,dy - y\,dz + 2t\,dt, \quad x\,dy + y\,dx = z\,dt + t\,dz, \quad 2z\,dz = dx + 2y\,dy.$$

Solve the second for dt and substitute into the first, then solve the third for dz and substitute:

$$dw = \left(2x - \frac{y}{2z} + \frac{2ty}{z} - \frac{t^2}{z^2}\right)dx + \left(-z - \frac{y^2}{z} + \frac{2xt}{z} - \frac{2t^2y}{z^2}\right)dy.$$

Since x, y are independent, the two coefficients are w_x and w_y, the components of ∇w.

Example: *Abstract data.* Suppose $P = (1,1,1)$ lies on $g(x,y,z) = 0$, with $(\nabla g)_P = \langle -1,1,2\rangle$ and $(\nabla f)_P = \langle 1,2,1\rangle$. For $w = f(x,y,z(x,y))$, at P we have $dw = dx + 2\,dy + dz$ and $dg = -dx + dy + 2\,dz = 0$. Eliminating $dz = \frac{1}{2}(dx - dy)$ gives $dw = \frac{3}{2}\,dx + \frac{3}{2}\,dy$, so $(\nabla w)_P = \frac{3}{2}\mathbf{i} + \frac{3}{2}\mathbf{j}$.

13.5 Formal and actual partial derivatives

When functions and constraints are given abstractly, distinguish the formal partials $f_x, f_y, \ldots$ from the ACTUAL PARTIAL DERIVATIVES $\partial f/\partial x, \ldots$, which take the relations among the variables into account.

Example: *Formal versus actual.* Let $f(x,y,z) = xy^2z^4$ with $z = 2x + 3y$. The formal partials are $f_x = y^2z^4$, $f_y = 2xyz^4$, $f_z = 4xy^2z^3$. By the chain rule, three actual partials are

$$\begin{aligned}
\left(\frac{\partial f}{\partial x}\right)_y &= f_x + f_z\left(\frac{\partial z}{\partial x}\right)_y = y^2z^4 + 8xy^2z^3,\\
\left(\frac{\partial f}{\partial y}\right)_x &= f_y + f_z\left(\frac{\partial z}{\partial y}\right)_x = 2xyz^4 + 12xy^2z^3,\\
\left(\frac{\partial f}{\partial z}\right)_x &= f_y\left(\frac{\partial y}{\partial z}\right)_x + f_z = \tfrac{2}{3}xyz^4 + 4xy^2z^3.
\end{aligned}$$

Now suppose variables $t, u, v, w, x, y, z, \ldots$ are related, as in thermodynamics, so that any two are independent (and may be chosen as the independent variables) and any three are connected by an equation.

Proposition 13.2 [RULES RELATING PARTIAL DERIVATIVES**].** *Under these assumptions,*

$$\left(\frac{\partial x}{\partial y}\right)_z = \frac{1}{(\partial y/\partial x)_z} \qquad \textit{(reciprocal rule)},$$

$$\left(\frac{\partial x}{\partial y}\right)_z = \frac{(\partial x/\partial t)_z}{(\partial y/\partial t)_z} \qquad \textit{(chain rule)},$$

$$\left(\frac{\partial x}{\partial y}\right)_z\left(\frac{\partial y}{\partial z}\right)_x\left(\frac{\partial z}{\partial x}\right)_y = -1, \quad \textit{that is,} \quad \left(\frac{\partial x}{\partial y}\right)_z = -\frac{(\partial x/\partial z)_y}{(\partial y/\partial z)_x} \qquad \textit{(cyclic rule)}.$$

Proof. With z fixed throughout, each variable becomes a function of one other variable, so the chain rule is the one-variable chain rule $(\partial x/\partial y)_z(\partial y/\partial t)_z = (\partial x/\partial t)_z$, and the reciprocal rule is its case $t = x$.

For the cyclic rule, let $f(x,y,z) = 0$ be the equation relating x, y, z. Taking y, z independent and differentiating with respect to y gives $f_x(\partial x/\partial y)_z + f_y = 0$, so $(\partial x/\partial y)_z = -f_y/f_x$. Permuting the variables cyclically and multiplying,

$$\left(\frac{\partial x}{\partial y}\right)_z\left(\frac{\partial y}{\partial z}\right)_x\left(\frac{\partial z}{\partial x}\right)_y = \left(-\frac{f_y}{f_x}\right)\left(-\frac{f_z}{f_y}\right)\left(-\frac{f_x}{f_z}\right) = -1.$$

□

The second form of the cyclic rule follows from the first by the reciprocal rule. The cyclic rule resembles the chain rule, but a different variable is held fixed in each factor, which changes the answer from 1 to −1.

Example: *Using the reciprocal rule.* Let $w = w(x, r)$ with $r = r(x, \theta)$, and find $\left(\frac{\partial w}{\partial r}\right)_\theta$. The independent variables are r and θ, with x dependent. With θ fixed, the chain rule and the reciprocal rule give

$$\left(\frac{\partial w}{\partial r}\right)_\theta = w_x \left(\frac{\partial x}{\partial r}\right)_\theta + w_r = \frac{w_x}{r_x} + w_r.$$

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 13, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 14: Double integrals

A one-variable integral $\int_a^b f(x)\,dx$ is the area under the graph $y = f(x)$ over $[a, b]$. A double integral is the volume under the graph $z = f(x, y)$ over a plane region R, and it is computed by integrating twice.

14.1 Definition as a limit of sums

Definition 14.1 [Double integral]. Divide the region R into small pieces numbered $1, \dots, n$. Let ΔA_i be the area of the ith piece and pick a point (x_i, y_i) in it. The double integral of f over R is

$$\iint_R f(x, y)\,dA = \lim_{\Delta A_i \to 0} \sum_{i=1}^{n} f(x_i, y_i)\,\Delta A_i,$$

the limit taken as the number of pieces tends to infinity and the size of each piece tends to zero.

The limit must exist and be independent of how the pieces and points are chosen; for the functions and regions considered here it always is. Above the ith piece the solid under the graph is nearly a box of volume $\Delta V_i \approx f(x_i, y_i)\,\Delta A_i$, so for $f \geq 0$ the sum approximates the volume and the integral equals it. Taking $f = 1$ adds up the areas of the pieces: $\text{area}(R) = \iint_R dA$. For example, the volume below $z = 2 - \frac{1}{2}(x + y)$ and above the unit square is $\iint_R \left(2 - \frac{1}{2}(x + y)\right) dA$.

14.2 Iterated integrals

To compute the volume, slice it by planes parallel to the yz-plane. Let $S(x)$ be the area of the slice at a given x. Then

$$\text{volume} = \int_{x_{\min}}^{x_{\max}} S(x)\,dx, \qquad S(x) = \int f(x, y)\,dy.$$

In the inner integral x is a fixed parameter and y is the variable of integration. The result is an iterated integral, written $\int\int f\,dy\,dx$; for small rectangles $\Delta A = \Delta y\,\Delta x$, so $dA = dy\,dx$.

Example: *A paraboloid over the unit square.* For $z = 1 - x^2 - y^2$ over $0 \leq x \leq 1, 0 \leq y \leq 1$,

$$\int_0^1 \int_0^1 (1 - x^2 - y^2)\,dy\,dx.$$

The inner integral, with x constant, is $\left[(1-x^2)y - \frac{1}{3}y^3\right]_0^1 = \frac{2}{3} - x^2$. The outer integral is $\int_0^1 (\frac{2}{3} - x^2)\,dx = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$.

14.3 Limits of integration

For a region that is not a rectangle, the inner limits depend on the outer variable. To set up $\iint_R f\,dy\,dx$:

1. hold x fixed and let y increase, tracing a vertical line;
2. integrate in y from where this line enters R to where it leaves;
3. integrate in x from the smallest to the largest x for which the vertical line meets R.

The inner limits are functions of x; the outer limits are constants. Always begin by sketching the region.

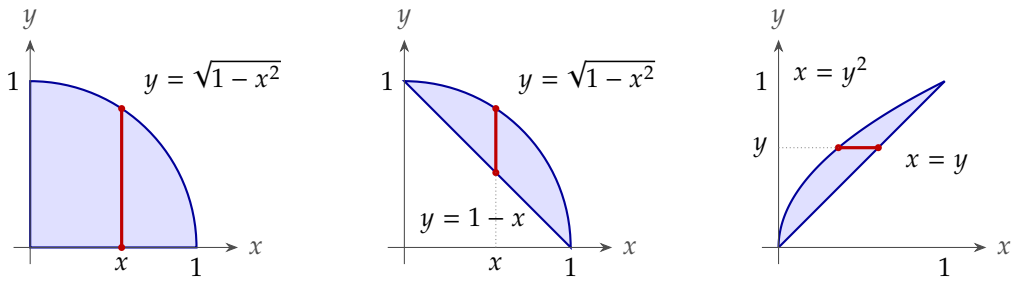


Figure 19: Left: the quarter disk, sliced vertically from $y = 0$ to $y = \sqrt{1-x^2}$. Middle: the region between $x + y = 1$ and the unit circle, with a vertical slice from $y = 1 - x$ to $y = \sqrt{1-x^2}$. Right: the region $x < y < \sqrt{x}$, sliced horizontally from $x = y^2$ to $x = y$.

Example: *The paraboloid over a quarter disk.* Take the same function over the part of $x^2 + y^2 < 1$ in the first quadrant (Figure 19, left). For fixed x the vertical slice runs from $y = 0$ to $y = \sqrt{1-x^2}$; the first slice is $x = 0$ and the last is $x = 1$:

$$\int_0^1 \int_0^{\sqrt{1-x^2}} (1 - x^2 - y^2)\,dy\,dx.$$

The inner integral is $\left[(1-x^2)y - \tfrac{1}{3}y^3\right]_0^{\sqrt{1-x^2}} = \tfrac{2}{3}(1-x^2)^{3/2}$. For the outer integral substitute $x = \sin\theta$, so that $dx = \cos\theta\,d\theta$ and $(1-x^2)^{3/2} = \cos^3\theta$; double-angle formulas give $\int_0^{\pi/2} \cos^4\theta\,d\theta = 3\pi/16$, hence

$$\int_0^1 \tfrac{2}{3}(1-x^2)^{3/2}\,dx = \tfrac{2}{3}\cdot\tfrac{3\pi}{16} = \frac{\pi}{8}.$$

Example: *Between a line and a circle.* Let R lie between $x + y = 1$ and $x^2 + y^2 = 1$ in the first quadrant (Figure 19, middle). A vertical line enters R at $y = 1 - x$ and leaves at $y = \sqrt{1-x^2}$, and the lines meeting R have $0 \le x \le 1$. In the other order, a horizontal line enters at $x = 1 - y$ and leaves at

$x = \sqrt{1-y^2}$:

$$\iint_R f\,dy\,dx = \int_0^1 \int_{1-x}^{\sqrt{1-x^2}} f(x,y)\,dy\,dx, \qquad \iint_R f\,dx\,dy = \int_0^1 \int_{1-y}^{\sqrt{1-y^2}} f(x,y)\,dx\,dy.$$

14.4 Exchanging the order of integration

Over a rectangle the order does not matter, for instance $\int_0^1 \int_0^2 dx\,dy = \int_0^2 \int_0^1 dy\,dx$. In general the limits must be found again from a picture of the region: fix the new inner variable, find where the line enters and leaves, then find the range of the outer variable.

Example: *An inner integral with no formula.* The integral

$$\int_0^1 \int_x^{\sqrt{x}} \frac{e^y}{y}\,dy\,dx$$

cannot be evaluated as written, because e^y/y has no elementary antiderivative. The region is $x < y < \sqrt{x}$ for $0 \le x \le 1$ (Figure 19, right). Fixing y, the left border is $x = y^2$ and the right border is $x = y$, with $0 \le y \le 1$. Hence

$$\int_0^1 \int_{y^2}^{y} \frac{e^y}{y}\,dx\,dy = \int_0^1 \frac{e^y}{y}(y - y^2)\,dy = \int_0^1 (e^y - ye^y)\,dy = \left[2e^y - ye^y\right]_0^1 = e - 2,$$

using $\int ye^y\,dy = ye^y - e^y$ (by parts).

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 14, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 15: Double integrals in polar coordinates; applications

Polar coordinates simplify a double integral when the integrand or the region has a simpler description in terms of r and θ. Double integrals also compute mass, averages, centers of mass, and moments of inertia.

15.1 Polar coordinates and the area element

Polar coordinates (r, θ) are related to rectangular coordinates by

$$x = r\cos\theta, \qquad y = r\sin\theta; \qquad r = \sqrt{x^2 + y^2}, \qquad \text{“}\theta = \tan^{-1}(y/x)\text{”}.$$

The quotation marks indicate that θ is not a single-valued function: adding any multiple of 2π to θ gives the same point, so $(1, 0)$, $(1, 2\pi)$, $(1, 4\pi)$ all describe the point $(1, 0)$, and the origin is $r = 0$ with any θ.

A small polar rectangle, between the circles of radius r and $r + \Delta r$ and the rays at angles θ and $\theta + \Delta\theta$, is nearly a rectangle with sides Δr and $r\,\Delta\theta$ (Figure 20, left). Thus $\Delta A \approx r\,\Delta r\,\Delta\theta$, and in the limit

$$\boxed{dA = r\,dr\,d\theta}.$$

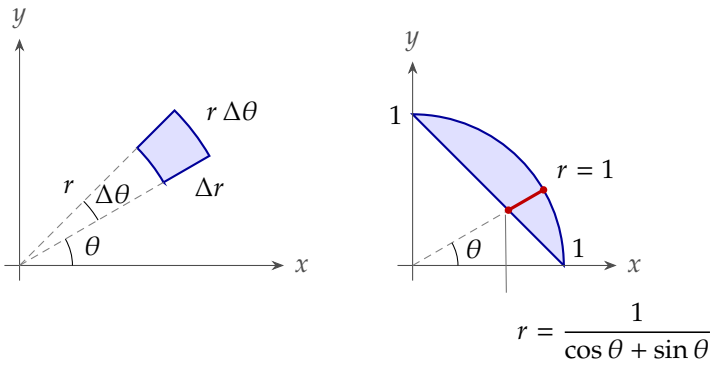


Figure 20: Left: the polar area element, with sides Δr and $r\,\Delta\theta$. Right: the region between $x + y = 1$ and the unit circle, swept by rays; a ray at angle θ enters at $r = 1/(\cos\theta + \sin\theta)$ and leaves at $r = 1$.

To find limits for $\iint_R f\,r\,dr\,d\theta$, sweep the region by rays: for fixed θ, integrate in r from where the ray enters R to where it leaves; then integrate in θ over the angles of the rays that meet R.

Example: *The quarter disk again.* Over the quarter of the unit disk in the first quadrant, $0 \le r \le 1$

and $0 \le \theta \le \pi/2$, so

$$\iint_R (1 - x^2 - y^2)\, dA = \int_0^{\pi/2} \int_0^1 (1 - r^2)\, r\, dr\, d\theta.$$

The inner integral is $\left[\frac{1}{2}r^2 - \frac{1}{4}r^4\right]_0^1 = \frac{1}{4}$ and the outer is $\int_0^{\pi/2} \frac{1}{4}\, d\theta = \pi/8$, matching the rectangular computation of Lecture 14 with far less work. Over the whole unit disk the same integrand gives the volume under $z = 1 - x^2 - y^2$ above the xy-plane: $\int_0^{2\pi} \frac{1}{4}\, d\theta = \pi/2$.

Example: *Between a line and a circle.* The line $x + y = 1$ is $r\cos\theta + r\sin\theta = 1$, or $r = 1/(\cos\theta + \sin\theta)$. For the region between it and the unit circle in the first quadrant (Figure 20, right), a ray enters at the line and leaves at the circle, and the rays meeting R have $0 \le \theta \le \pi/2$:

$$\int_0^{\pi/2} \int_{1/(\cos\theta + \sin\theta)}^1 \cdots\, dr\, d\theta.$$

Example: *Mass of a sector.* Let R be the sector $0 < r < 2$, $0 < \theta < \pi/3$ with density $\delta = xy = r^2 \cos\theta \sin\theta$. Then

$$M = \int_0^{\pi/3} \int_0^2 r^2 \cos\theta \sin\theta\; r\, dr\, d\theta = \int_0^{\pi/3} 4\cos\theta \sin\theta\, d\theta = \left[2\sin^2\theta\right]_0^{\pi/3} = \frac{3}{2}.$$

Example: *Converting an iterated integral.* Let $I = \int_1^2 \int_0^x (x^2 + y^2)^{-3/2}\, dy\, dx$. The region is the trapezoid $0 \le y \le x$, $1 \le x \le 2$ (Figure 21, left). The lines $x = 1$ and $x = 2$ are $r = \sec\theta$ and $r = 2\sec\theta$, and the region lies between the rays $\theta = 0$ and $\theta = \pi/4$. Hence

$$I = \int_0^{\pi/4} \int_{\sec\theta}^{2\sec\theta} \frac{1}{r^3}\, r\, dr\, d\theta = \int_0^{\pi/4} \left[-\frac{1}{r}\right]_{\sec\theta}^{2\sec\theta} d\theta = \int_0^{\pi/4} \tfrac{1}{2}\cos\theta\, d\theta = \frac{\sqrt{2}}{4}.$$

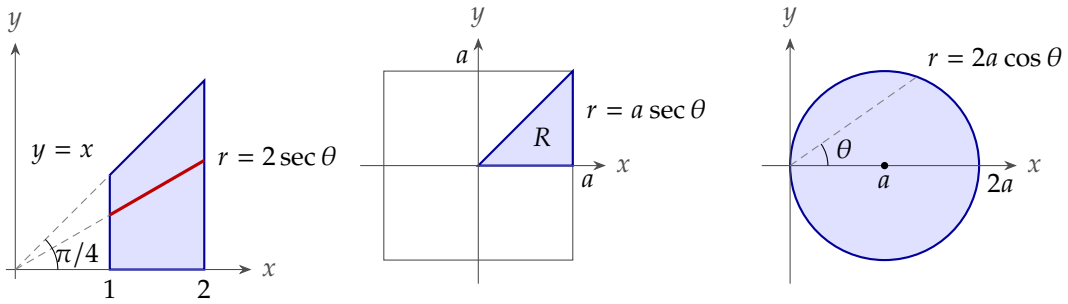


Figure 21: Left: the region $0 \le y \le x$, $1 \le x \le 2$, between the rays $\theta = 0$, $\theta = \pi/4$ and the lines $r = \sec\theta$, $r = 2\sec\theta$; along each ray, r runs from $\sec\theta$ to $2\sec\theta$. Middle: one eighth of the square of side $2a$, the triangle $0 \le \theta \le \pi/4$, $r \le a\sec\theta$. Right: the disk of radius a with the origin on its rim, bounded by $r = 2a\cos\theta$.

15.2 Curves in polar form

A point lies on the graph of $r = f(\theta)$ if any of its polar representations satisfies the equation. The ray $\theta = \pi/3$, the circle $r = 2$ about the origin, the vertical line $x = 2$ ($r = 2\sec\theta$) and the horizontal line $y = 2$ ($r = 2/\sin\theta$) are simple examples. The equation $r = 2a\cos\theta$ describes a circle of radius a centered at $(a, 0)$: multiplying by r gives $x^2 + y^2 = 2ax$, that is, $(x-a)^2 + y^2 = a^2$. It is traced for $-\pi/2 \le \theta \le \pi/2$. Similarly $r = 2a\sin\theta$, for $0 \le \theta \le \pi$, is the circle of radius a centered at $(0, a)$. Other standard curves are the cardioid $r = a(1 + \cos\theta)$, the limaçon $r = a(1 + b\cos\theta)$ with $b > 1$, the lemniscate $r^2 = 2a^2\cos 2\theta$, and the four-leaved rose $r = a\sin 2\theta$.

Negative values of r can be used for plotting but should not be used in integration; use symmetry instead, and integrate only where $r \ge 0$.

15.3 Mass, average value, and center of mass

The area of R is $\iint_R 1\,dA$. A plate with DENSITY $\delta(x, y) = \lim_{\Delta A \to 0} \Delta m/\Delta A$ (mass per unit area, constant for a uniform material) has total mass

$$M = \iint_R \delta\, dA,$$

since a small piece has mass $\delta\, dA$.

Definition 15.1 [AVERAGE VALUE AND CENTER OF MASS]. The AVERAGE VALUE of f over R is $\bar{f} = \frac{1}{\text{area}(R)} \iint_R f\, dA$. The CENTER OF MASS of a plate with density δ is the mass-weighted average position $(\bar{x}, \bar{y})$,

$$\bar{x} = \frac{1}{M}\iint_R x\,\delta\, dA, \qquad \bar{y} = \frac{1}{M}\iint_R y\,\delta\, dA.$$

With $\delta = 1$ it is the CENTROID, the average of x and y with respect to area.

These generalize the discrete weighted average $x_{cm} = \sum m_i x_i / \sum m_i$; two equal masses balance at their midpoint, while masses 2 and 1 at x_1, x_2 balance at $(2x_1 + x_2)/3$.

Example: *A square with density xy.* On the unit square with $\delta = xy$, $M = \int_0^1\int_0^1 xy\, dx\, dy = \frac{1}{4}$ and $\bar{x} = 4\int_0^1\int_0^1 x^2 y\, dy\, dx = 4 \cdot \frac{1}{3} \cdot \frac{1}{2} = \frac{2}{3}$. By symmetry of the region and the density, $\bar{y} = \frac{2}{3}$.

Example: *Average distance from the center of a square.* Center a square of side $2a$ at the origin; its area is $4a^2$. By symmetry the integral of $\sqrt{x^2 + y^2}$ over the square is 8 times its integral over the triangle $0 \le \theta \le \pi/4$, $0 \le r \le a\sec\theta$ (Figure 21, middle). Therefore

$$\text{average} = \frac{8}{4a^2}\int_0^{\pi/4}\int_0^{a\sec\theta} r \cdot r\, dr\, d\theta = \frac{2}{a^2}\int_0^{\pi/4} \frac{a^3\sec^3\theta}{3}\, d\theta = \frac{a}{3}\big(\sqrt{2} + \ln(\sqrt{2}+1)\big),$$

using $\int \sec^3\theta\, d\theta = \frac{1}{2}\big(\sec\theta\tan\theta + \ln(\sec\theta + \tan\theta)\big)$, obtained by integration by parts. For the unit

square, $a = \frac{1}{2}$.

15.4 Moment of inertia

Mass measures resistance to changes in translational motion; the MOMENT OF INERTIA about an axis measures resistance to changes in rotation about that axis. A point mass m at distance r from the axis, rotating with angular speed $\omega = d\theta/dt$, has speed $v = r\omega$ and kinetic energy $\frac{1}{2}mv^2 = \frac{1}{2}(mr^2)\omega^2$, so its moment of inertia is mr^2. Summing over a plate,

$$I = \iint_R (\text{distance to axis})^2 \, \delta \, dA, \qquad I_0 = \iint_R r^2 \, \delta \, dA, \qquad I_x = \iint_R y^2 \, \delta \, dA,$$

where I_0, the POLAR MOMENT OF INERTIA, is taken about the origin (rotation in the plane about the z-axis) and I_x about the x-axis, at distance $|y|$. The rotational kinetic energy is $\frac{1}{2}I_0\omega^2$.

Example: *A disk about its center.* For the disk of radius a with $\delta = 1$,

$$I_0 = \int_0^{2\pi} \int_0^a r^2 \, r \, dr \, d\theta = 2\pi \left[\frac{r^4}{4} \right]_0^a = \frac{\pi a^4}{2}.$$

Example: *The same disk about a point of its circumference.* Put the origin at that point and the diameter along the x-axis. The boundary circle is then $r = 2a\cos\theta$, $-\pi/2 \le \theta \le \pi/2$ (Figure 21, right), and

$$I_0 = \int_{-\pi/2}^{\pi/2} \int_0^{2a\cos\theta} r^2 \, r \, dr \, d\theta = \int_{-\pi/2}^{\pi/2} 4a^4 \cos^4\theta \, d\theta = 4a^4 \cdot \frac{3\pi}{8} = \frac{3}{2}\pi a^4.$$

Example: *The square with density xy.* About the y-axis the distance is x, so $dI = x^2 \, dm = x^3 y \, dx \, dy$ and $I_y = \int_0^1 \int_0^1 x^3 y \, dy \, dx = \frac{1}{8}$. The polar moment of inertia uses $r^2 = x^2 + y^2$: $I_0 = \int_0^1 \int_0^1 (x^3 y + x y^3) \, dy \, dx = \frac{1}{4}$.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 15, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 16: Change of variables

A substitution $u = u(x, y)$, $v = v(x, y)$ can simplify the integrand or the region of a double integral. As with polar coordinates, the area element must be rescaled; the scale factor is a determinant, the Jacobian.

16.1 Scale factors

Example: *Area of an ellipse.* For the ellipse with semiaxes a and b, set $u = x/a$, $v = y/b$. As in one variable, $du = \frac{1}{a}\,dx$ and $dv = \frac{1}{b}\,dy$, so $dx\,dy = ab\,du\,dv$ and

$$\iint_{(x/a)^2+(y/b)^2<1} dx\,dy = \iint_{u^2+v^2<1} ab\,du\,dv = ab \cdot \pi = \pi ab.$$

In general one must find the ratio between $du\,dv$ and $dx\,dy$.

Example: *A linear change of variables.* Let $u = 3x - 2y$, $v = x + y$, and compare the area elements $dA = dx\,dy$ and $dA' = du\,dv$. In matrix form

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

so the unit square in the xy-plane becomes, in the uv-plane, the parallelogram spanned by $\langle 3, 1\rangle$ and $\langle -2, 1\rangle$ (Figure 22). Its area is

$$\left|\det \begin{pmatrix} 3 & 1 \\ -2 & 1 \end{pmatrix}\right| = 5 = \left|\det \begin{pmatrix} 3 & -2 \\ 1 & 1 \end{pmatrix}\right|.$$

The map is linear, so every rectangle is scaled by the same factor: $\Delta A' = 5\,\Delta A$, hence $du\,dv = 5\,dx\,dy$ and $\iint \cdots dx\,dy = \iint \cdots \frac{1}{5}\,du\,dv$.

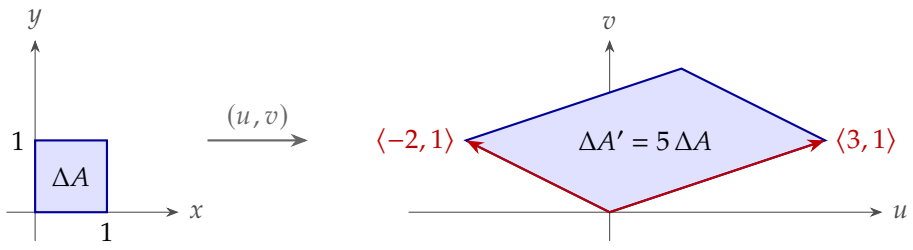


Figure 22: The map $u = 3x - 2y$, $v = x + y$ sends the unit square to the parallelogram spanned by $\langle 3, 1\rangle$ and $\langle -2, 1\rangle$, of area 5.

16.2 The Jacobian

For a general change of variables, the approximation formulas $\Delta u \approx u_x \Delta x + u_y \Delta y$ and $\Delta v \approx v_x \Delta x + v_y \Delta y$ read

$$\begin{pmatrix} \Delta u \\ \Delta v \end{pmatrix} \approx \begin{pmatrix} u_x & u_y \\ v_x & v_y \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix}.$$

A small xy-rectangle therefore becomes approximately a parallelogram in uv-coordinates, scaled by the determinant of this matrix, which now depends on x and y.

Definition 16.1 [Jacobian]. The Jacobian of $u = u(x, y)$, $v = v(x, y)$ is

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}.$$

The area elements are related by $du\, dv = |J|\, dx\, dy$, with the absolute value because area is the absolute value of the determinant.

Usually the change of variables is used in the direction $x = x(u, v)$, $y = y(u, v)$. Then $dA = \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\, dv$ and

$$\iint_R f(x, y)\, dx\, dy = \iint_R g(u, v) \left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\, dv, \qquad g(u, v) = f\big(x(u, v), y(u, v)\big). \tag{16.1}$$

Changing variables thus has three steps: rewrite the integrand, supply the area element, and find the new limits from the region. Picture the new coordinates by the grid of level curves $u = u_0$, $v = v_0$.

Proposition 16.2 [Area element]. $dA = \left|\dfrac{\partial(x, y)}{\partial(u, v)}\right| du\, dv.$

Proof. Nearby grid curves $u = u_0$, $u = u_0 + \Delta u$, $v = v_0$, $v = v_0 + \Delta v$ bound a region that is approximately the parallelogram with vertices $P = (u_0, v_0)$, $Q = (u_0 + \Delta u, v_0)$, $S = (u_0, v_0 + \Delta v)$ (in uv-coordinates), so $\Delta A \approx |\overrightarrow{PQ} \times \overrightarrow{PS}|$. Moving from P to Q changes only u, so by the definition of partial derivatives $\overrightarrow{PQ} \approx x_u \Delta u\, \mathbf{i} + y_u \Delta u\, \mathbf{j}$; similarly $\overrightarrow{PS} \approx x_v \Delta v\, \mathbf{i} + y_v \Delta v\, \mathbf{j}$. The cross product has only a $\mathbf{k}$-component,

$$\begin{vmatrix} x_u \Delta u & y_u \Delta u \\ x_v \Delta v & y_v \Delta v \end{vmatrix} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} \Delta u\, \Delta v,$$

after transposing and factoring Δu and Δv out of the columns. Taking absolute values and letting $\Delta u, \Delta v \to 0$ gives the formula. □

If only $u(x, y)$ and $v(x, y)$ are known, it can be awkward to solve for x and y. The two Jacobians are

reciprocal:

$$\frac{\partial(x,y)}{\partial(u,v)} \cdot \frac{\partial(u,v)}{\partial(x,y)} = 1. \tag{16.2}$$

Indeed, by the chain rule the two matrices of partial derivatives are inverse to each other, and determinants multiply. The reciprocal is expressed in x and y, but it often converts easily to u and v.

Example: *Polar coordinates.* For $x = r\cos\theta$, $y = r\sin\theta$,

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r,$$

so $dx\,dy = r\,dr\,d\theta$; no absolute value is needed since $r \geq 0$.

16.3 Examples

Example: *A substitution through hyperbolas.* Compute $\int_0^1\int_0^1 x^2y\,dx\,dy$ with $u = x$, $v = xy$. This simplifies neither the integrand nor the region; it illustrates the method.

1. *Area element.* $\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} 1 & 0 \\ y & x \end{vmatrix} = x$, so $du\,dv = x\,dx\,dy$ and $dx\,dy = \frac{1}{x}\,du\,dv$.
2. *Integrand.* $x^2y\,dx\,dy = x^2y \cdot \frac{1}{x}\,du\,dv = xy\,du\,dv = v\,du\,dv$.
3. *Limits.* Integrating du first holds $v = xy$ fixed: the slice is a piece of the hyperbola $xy = v$, parametrized by $u = x$. It enters the square at the top edge $y = 1$, where $u = v$, and leaves at the right edge $x = 1$, where $u = 1$. The slices run from $v = 0$ to $v = 1$ (Figure 23). In the uv-plane the region is the triangle $0 \leq v \leq u \leq 1$.

Hence the integral is $\int_0^1\int_v^1 v\,du\,dv = \int_0^1 v(1-v)\,dv = \frac{1}{6}$, the value of the original integral.

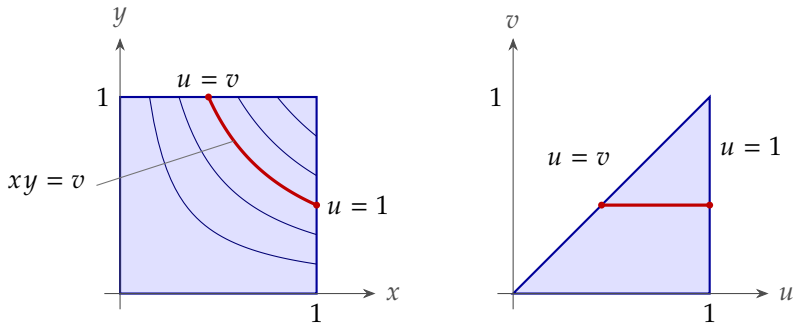


Figure 23: Left: the unit square sliced by the hyperbolas $xy = v$; each slice runs from the top edge ($u = v$) to the right edge ($u = 1$). Right: the same region in the uv-plane, the triangle $0 \leq v \leq u \leq 1$.

Example: *A rotated square.* Evaluate $\iint_R \left(\frac{x-y}{x+y+2}\right)^2 dx\,dy$ over the square R with vertices $(\pm1, 0)$, $(0, \pm1)$,

bounded by $x+y=\pm1$ and $x-y=\pm1$. The boundaries and the integrand suggest

$$u=x+y,\quad v=x-y;\qquad x=\frac{u+v}{2},\quad y=\frac{u-v}{2}.$$

From the second pair, $\frac{\partial(x,y)}{\partial(u,v)}=\begin{vmatrix}1/2 & 1/2\\ 1/2 & -1/2\end{vmatrix}=-\frac12$, so $dA=\frac12\,du\,dv$, this time with the absolute value needed. The region becomes $-1\le u,v\le 1$, and

$$\iint_R\Big(\frac{x-y}{x+y+2}\Big)^2\,dx\,dy=\int_{-1}^{1}\int_{-1}^{1}\Big(\frac{v}{u+2}\Big)^2\frac12\,du\,dv.$$

The inner integral is $\left[-\frac{v^2}{2(u+2)}\right]_{u=-1}^{1}=\frac{v^2}{3}$ and the outer is $\left[\frac{v^3}{9}\right]_{-1}^{1}=\frac29$.

Example: *Using the reciprocal Jacobian.* Evaluate $\iint_R\frac{y}{x}\,dx\,dy$, where R is bounded by $x^2-y^2=1$, $x^2-y^2=4$, $y=0$, and $y=x/2$ (Figure 24, left). The boundaries are level curves of x^2-y^2 and y/x, so take $u=x^2-y^2$, $v=y/x$. Without solving for x,y,

$$\frac{\partial(u,v)}{\partial(x,y)}=\begin{vmatrix}2x & -2y\\ -y/x^2 & 1/x\end{vmatrix}=2-\frac{2y^2}{x^2}=2-2v^2,\qquad\text{so}\qquad\frac{\partial(x,y)}{\partial(u,v)}=\frac{1}{2(1-v^2)}$$

by (16.2). The region is $1\le u\le 4$, $0\le v\le\frac12$, and

$$\iint_R\frac{y}{x}\,dx\,dy=\int_0^{1/2}\int_1^4\frac{v}{2(1-v^2)}\,du\,dv=\Big[-\tfrac34\ln(1-v^2)\Big]_0^{1/2}=-\frac34\ln\frac34,$$

which is positive, as it must be for a positive integrand.

16.4 Putting in the limits

When R is bounded by level curves of u and v, the new limits are constants. Otherwise one must find the uv-equations of the boundary curves, in one of two ways:

- *Method 1:* eliminate x and y from the equations $u=u(x,y)$, $v=v(x,y)$ and the xy-equation of the curve;
- *Method 2:* solve for $x=x(u,v)$, $y=y(u,v)$ and substitute into the xy-equation of the curve.

Method 1 applies more widely but needs a separate calculation for each boundary curve.

Example: *A triangle.* With $u=x+y$, $v=x-y$, rewrite $\int_0^1\int_0^x dy\,dx$ as an iterated integral $du\,dv$. By the previous examples, $\frac{\partial(x,y)}{\partial(u,v)}=-\frac12$, so the Jacobian factor is $\frac12$. The region is the triangle with vertices $(0,0)$, $(1,0)$, $(1,1)$. The diagonal $y=x$ is the level curve $v=0$. For the other sides, Method 1 gives: $y=0$ implies $u=x=v$; and $x=1$ implies $u=1+y$, $v=1-y$, so $u+v=2$. Method 2 gives the same from $x=\frac12(u+v)$, $y=\frac12(u-v)$. Holding v fixed and increasing u, a line enters R at

$u = v$ and leaves at $u = 2 - v$, for $0 \le v \le 1$ (Figure 24, middle):

$$\int_0^1 \int_v^{2-v} \frac{1}{2}\, du\, dv = \int_0^1 (1 - v)\, dv = \frac{1}{2},$$

the area of the triangle.

Example: *An unbounded region.* With $u = x^2 - y^2$, $v = y/x$ as before, set up $\iint_R \frac{dx\,dy}{x^2}$, where R is the infinite region in the first quadrant below $y = 1/x$ and to the right of $x^2 - y^2 = 1$ (Figure 24, right). Eliminating $y = vx$ from $u = x^2 - y^2$ gives $u = x^2(1 - v^2)$, so $x^2 = u/(1 - v^2)$. In R, $0 \le v < 1$, so the Jacobian factor $\frac{1}{2(1-v^2)}$ is positive and

$$\iint_R \frac{dx\,dy}{x^2} = \iint_R \frac{1 - v^2}{u} \cdot \frac{1}{2(1 - v^2)}\, du\, dv = \iint_R \frac{du\, dv}{2u}.$$

The x-axis is $v = 0$ and the left boundary is $u = 1$. For the curve $xy = 1$, Method 1 with $y = vx$ gives $vx^2 = 1$, so $v = 1/x^2$ and $u = x^2 - 1/x^2 = \frac{1}{v} - v$. Holding v fixed, the ray $y = vx$ enters R at $u = 1$ and leaves at $u = \frac{1}{v} - v$. The rays meeting R run from $v = 0$ to the slope $v = a$ of the line through the corner P where $xy = 1$ meets $x^2 - y^2 = 1$:

$$\iint_R \frac{dx\,dy}{x^2} = \int_0^a \int_1^{1/v - v} \frac{du\, dv}{2u}.$$

To find a, substitute $y = ax$ into both curves: $ax^2 = 1$ and $x^2(1 - a^2) = 1$, so $a = 1 - a^2$ and $a = \frac{\sqrt{5}-1}{2}$.

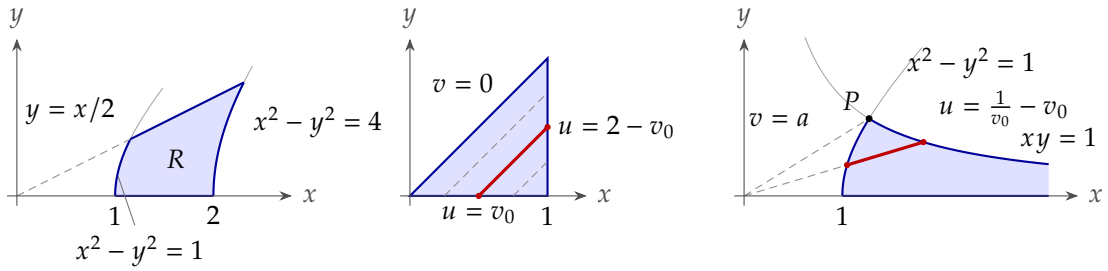


Figure 24: Left: the region between $x^2 - y^2 = 1$, $x^2 - y^2 = 4$, $y = 0$ and $y = x/2$, which becomes $1 \le u \le 4$, $0 \le v \le \frac{1}{2}$. Middle: the triangle under $y = x$ over $[0, 1]$, with a line $v = v_0$ from $u = v_0$ to $u = 2 - v_0$. Right: the unbounded region below $xy = 1$ and right of $x^2 - y^2 = 1$; the ray $v = v_0$ runs from $u = 1$ to $u = 1/v_0 - v_0$, and the corner P lies on the ray $v = a$.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 16, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 17: Vector fields and line integrals in the plane

A vector field assigns a vector to each point of the plane. Integrating a force field along a trajectory gives the work it does, a line integral.

17.1 Vector fields

Definition 17.1 [Vector field]. A vector field in the plane is a function

$$\mathbf{F}(x, y) = M(x, y)\,\mathbf{i} + N(x, y)\,\mathbf{j} = \langle M, N \rangle,$$

assigning to each point (x, y) of its domain a vector, drawn with its tail at (x, y). It is continuous, differentiable, or continuously differentiable in a region if M and N are, the last meaning that M_x, M_y, N_x, N_y exist and are continuous.

Common fields are continuously differentiable except perhaps at isolated points or along curves, and such points matter. Examples include velocity fields (wind flow) and force fields (gravity). Four simple fields are drawn in Figure 25: the constant field $2\mathbf{i} + \mathbf{j}$; the field $x\,\mathbf{i}$, horizontal and growing with $|x|$; the radial field $x\,\mathbf{i} + y\,\mathbf{j}$, pointing away from the origin with length r; and $-y\,\mathbf{i} + x\,\mathbf{j}$, which is $\langle x, y \rangle$ rotated 90° counterclockwise, so it circles the origin counterclockwise with length r.

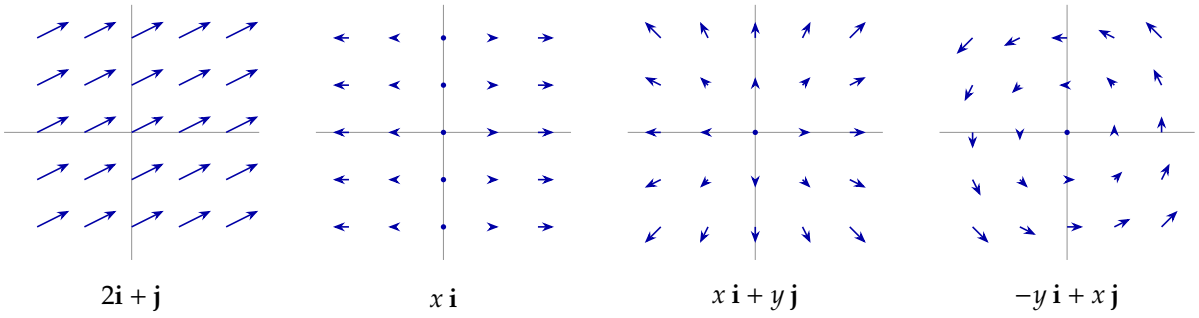


Figure 25: The fields $2\mathbf{i} + \mathbf{j}$, $x\,\mathbf{i}$, $x\,\mathbf{i} + y\,\mathbf{j}$, and $-y\,\mathbf{i} + x\,\mathbf{j}$, each drawn on the same grid with arrows scaled by a common factor.

17.2 Gradient, force, and flow fields

If $w = f(x, y)$ is differentiable, its gradient $\nabla w = w_x\,\mathbf{i} + w_y\,\mathbf{j}$ is a gradient field. Its direction is the direction $\hat{\mathbf{u}}$ in which the directional derivative $dw/ds|_{\hat{\mathbf{u}}} = \nabla w \cdot \hat{\mathbf{u}}$ is greatest, its length is that greatest value, and it is perpendicular to the contour curves $f(x, y) = c$.

Example: *The gradient of r.* For $w = \sqrt{x^2 + y^2} = r$,

$$\nabla w = \frac{x}{r}\,\mathbf{i} + \frac{y}{r}\,\mathbf{j} = \frac{x\,\mathbf{i} + y\,\mathbf{j}}{r},$$

defined and continuously differentiable on the plane minus the origin. Every vector is a unit vector pointing radially outward, consistent with w increasing fastest, at rate 1, in the outward direction, and the field is perpendicular to the contour circles of w.

A FORCE FIELD gives at each point the force on a unit test charge or unit mass placed there, for instance the electrostatic field of static charges or the gravitational field of a mass distribution.

Example: *A point charge.* The electrostatic field of a unit positive charge at the origin points radially away from it with magnitude c/r^2. The vector $x\,\mathbf{i} + y\,\mathbf{j}$ has the right direction and length r, so multiplying by c/r^3 gives

$$\mathbf{F} = \frac{c\,x}{r^3}\,\mathbf{i} + \frac{c\,y}{r^3}\,\mathbf{j} = c\,\frac{x\,\mathbf{i} + y\,\mathbf{j}}{(x^2 + y^2)^{3/2}}.$$

Now consider a steady two-dimensional flow in a shallow tank: the motion is horizontal and does not change with time. The fluid may be compressible or not, its density may vary, and fluid may be added or removed at points. The VELOCITY FIELD $\mathbf{v}(x, y)$ gives the velocity of the flow at each point, and with density $\delta(x, y)$ (mass per unit area) the FLOW FIELD is $\mathbf{F} = \delta\,\mathbf{v}$. Its direction is the direction of flow, and

$|\mathbf{F}|$ = rate of mass transport per unit length, across a line perpendicular to the flow.

Indeed, in time Δt the fluid crossing a segment of length Δl perpendicular to the flow fills a rectangle of area $|\mathbf{v}|\,\Delta t\,\Delta l$, of mass $\delta|\mathbf{v}|\,\Delta t\,\Delta l = |\mathbf{F}|\,\Delta l\,\Delta t$. For constant density the flow and velocity fields differ only by a constant factor.

Example: *A source and a line charge.* Let $\mathbf{F} = (x\,\mathbf{i} + y\,\mathbf{j})/(x^2 + y^2)$, defined away from the origin, radial, with $|\mathbf{F}| = r/r^2 = 1/r$. As a flow field, across the circle of radius a the transport rate is $1/a$ per unit length, so $\frac{1}{a} \cdot 2\pi a = 2\pi$ units of mass cross every circle per unit time: $\mathbf{F}$ is the flow of a SOURCE of magnitude 2π at the origin. Since $\delta|\mathbf{v}| = 1/r$, an incompressible fluid (constant density) slows down like $1/r$, while a compressible gas flowing at constant speed has density decreasing like $1/r$.

As a force field, $\mathbf{F}$ is, up to a constant factor, the electrostatic field of a uniformly charged long straight wire along the z-axis. By symmetry that field has no $\mathbf{k}$-component and its magnitude depends only on the distance r to the wire; it is TWO-DIMENSIONAL, since it looks the same in every horizontal plane. Its magnitude decreases like $1/r$, not like $1/r^2$ as for a point charge. Likewise $-\mathbf{F}$ is, up to a constant, the gravitational field of a uniform mass distribution along the z-axis.

Example: *A rotating fluid.* A fluid of density 1 rotating counterclockwise about the origin with angular velocity ω moves perpendicular to $x\,\mathbf{i} + y\,\mathbf{j}$, counterclockwise, in the direction of $-y\,\mathbf{i} + x\,\mathbf{j}$, which has length r. The speed is $|\mathbf{v}| = \omega r$, so $\mathbf{v} = -\omega y\,\mathbf{i} + \omega x\,\mathbf{j}$.

17.3 Work and line integrals

A constant force $\mathbf{F}$ acting over a displacement $\Delta\mathbf{r}$ does work $|\mathbf{F}|\cos\theta\,|\Delta\mathbf{r}| = \mathbf{F}\cdot\Delta\mathbf{r}$, since only the component of the force along the displacement does work. Summing over small pieces of a trajectory C gives the work along C.

Definition 17.2 [Line integral]. The line integral of $\mathbf{F}$ along a curve C is

$$W = \int_C \mathbf{F}\cdot d\mathbf{r} = \lim_{\Delta\mathbf{r}_i\to 0}\sum_i \mathbf{F}\cdot\Delta\mathbf{r}_i.$$

If C is parametrized by $\mathbf{r}(t)$, $t_1 \le t \le t_2$, it is evaluated as

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_{t_1}^{t_2} \mathbf{F}\big(\mathbf{r}(t)\big)\cdot\frac{d\mathbf{r}}{dt}\,dt.$$

With $\mathbf{F} = \langle M, N\rangle$ and $d\mathbf{r} = \langle dx, dy\rangle$ (a differential: dividing by dt gives the velocity $d\mathbf{r}/dt$), the line integral is written

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C M\,dx + N\,dy.$$

This is not a sum of an integral in x and an integral in y; it is a single integral along C, evaluated by expressing everything in terms of one parameter, which reduces it to a one-variable integral.

Example: *A rotation field along a parabola.* Let $\mathbf{F} = -y\,\mathbf{i} + x\,\mathbf{j}$ and let C be the parabola $y = x^2$ from $(0,0)$ to $(1,1)$, parametrized by $x = t$, $y = t^2$, $0 \le t \le 1$ (Figure 26, left). Then $\mathbf{F} = \langle -t^2, t\rangle$ and $d\mathbf{r}/dt = \langle 1, 2t\rangle$, so

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_0^1 \langle -t^2, t\rangle\cdot\langle 1, 2t\rangle\,dt = \int_0^1 t^2\,dt = \frac{1}{3}.$$

In differential notation, $dx = dt$ and $dy = 2t\,dt$, so $\int_C -y\,dx + x\,dy = \int_0^1(-t^2 + 2t^2)\,dt = \frac{1}{3}$, the same computation.

The definition does not involve a parametrization, so the value does not depend on the choice of one (the parametrization $x = \sin\theta$, $y = \sin^2\theta$ gives the same integral under $t = \sin\theta$). In practice, choose the simplest parametrization.

Example: *Dependence on the path.* Let $I = \int_C x^2y\,dx + (x - 2y)\,dy$ from $(0,0)$ to $(1,1)$.

- Along the parabola $x = t$, $y = t^2$: $I = \int_0^1 \big(t^4 + (t - 2t^2)2t\big)dt = \int_0^1 (t^4 + 2t^2 - 4t^3)\,dt = -\frac{2}{15}$.
- Along the segment $x = t$, $y = t$: $I = \int_0^1 \big(t^3 + (t - 2t)\big)dt = \int_0^1 (t^3 - t)\,dt = -\frac{1}{4}$.
- Along the parabola parametrized by $x = \sin t$, $y = \sin^2 t$, $0 \le t \le \pi/2$: $dx = \cos t\,dt$,

$dy = 2\sin t\cos t\,dt$, and

$$I = \int_0^{\pi/2} (\sin^4 t + 2\sin^2 t - 4\sin^3 t)\cos t\,dt = \left[\frac{\sin^5 t}{5} + \frac{2}{3}\sin^3 t - \sin^4 t\right]_0^{\pi/2} = -\frac{2}{15}.$$

In general a line integral depends on the path but not on its parametrization.

Reversing the direction of C changes the sign: $\int_{-C} \mathbf{F}\cdot d\mathbf{r} = -\int_C \mathbf{F}\cdot d\mathbf{r}$, where $-C$ is C traversed backward. For a closed curve one writes $\oint_C \mathbf{F}\cdot d\mathbf{r}$. A curve made of several pieces is integrated piece by piece, and a coordinate can serve as the parameter.

Example: *A two-piece path.* Let $I = \int_C y\,dx + (x+2y)\,dy$, where C runs from $(0,1)$ to $(1,1)$ along C_1 and then down to $(1,0)$ along C_2 (Figure 26, right). On C_1, $y = 1$ and $dy = 0$, with x from 0 to 1: $\int_{C_1} = \int_0^1 dx = 1$. On C_2, $x = 1$ and $dx = 0$, with y from 1 to 0: $\int_{C_2} = \int_1^0 (1+2y)\,dy = -2$. Hence $I = 1 - 2 = -1$.

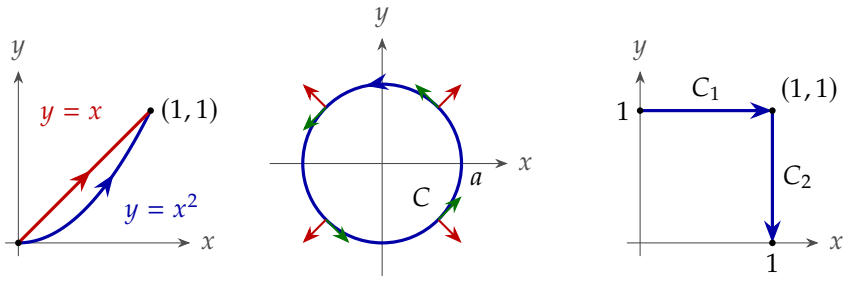


Figure 26: Left: the parabola $y = x^2$ and the segment $y = x$ from $(0,0)$ to $(1,1)$. Middle: a circle of radius a with the fields $x\,\mathbf{i} + y\,\mathbf{j}$ (red, normal to C) and $-y\,\mathbf{i} + x\,\mathbf{j}$ (green, tangent to C). Right: the path $C = C_1 + C_2$ from $(0,1)$ to $(1,1)$ to $(1,0)$.

17.4 The geometric approach

The velocity is $\frac{d\mathbf{r}}{dt} = \frac{ds}{dt}\,\hat{\mathbf{T}}$, where s is arclength and $\hat{\mathbf{T}}$ the unit tangent vector. Hence $d\mathbf{r} = \hat{\mathbf{T}}\,ds$ and $\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C \mathbf{F}\cdot\hat{\mathbf{T}}\,ds$, which is sometimes easier to evaluate.

Example: *Fields along a circle.* Let C be the circle of radius a about the origin, counterclockwise (Figure 26, middle). For $\mathbf{F} = x\,\mathbf{i} + y\,\mathbf{j}$, $\mathbf{F}$ is normal to C, so $\mathbf{F}\cdot\hat{\mathbf{T}} = 0$ and $\oint_C \mathbf{F}\cdot d\mathbf{r} = 0$. For $\mathbf{F} = -y\,\mathbf{i} + x\,\mathbf{j}$, $\mathbf{F}$ is tangent to C and points along $\hat{\mathbf{T}}$, so $\mathbf{F}\cdot\hat{\mathbf{T}} = |\mathbf{F}| = a$ and

$$\oint_C \mathbf{F}\cdot d\mathbf{r} = \oint_C a\,ds = a(2\pi a) = 2\pi a^2.$$

The parametrization $x = a\cos\theta$, $y = a\sin\theta$ gives $\int_0^{2\pi} (a^2\sin^2\theta + a^2\cos^2\theta)\,d\theta = 2\pi a^2$, the same answer.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 17, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 18: Path independence and conservative fields

For a gradient field, a line integral is computed by evaluating a potential function at the endpoints, exactly as in the fundamental theorem of calculus. Such fields do no work around closed loops.

18.1 A line integral around a sector

Recall that for $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ along a curve C,

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C M\,dx + N\,dy = \int_C \mathbf{F}\cdot\hat{\mathbf{T}}\,ds.$$

Example: *Around a sector.* Let $\mathbf{F} = y\,\mathbf{i} + x\,\mathbf{j}$ and let $C = C_1 + C_2 + C_3$ enclose the sector of the unit disk between the angles 0 and $\pi/4$ (Figure 27, left). We compute $\int_{C_i} y\,dx + x\,dy$ on each piece.

1. On the x-axis, $x = t$, $y = 0$, $dx = dt$, $dy = 0$, $0 \le t \le 1$, so $\int_{C_1} = \int_0^1 0\,dt = 0$. Geometrically, $\mathbf{F} = x\,\mathbf{j}$ is perpendicular to $\hat{\mathbf{T}} = \mathbf{i}$ there.
2. On the arc, $x = \cos\theta$, $y = \sin\theta$, $0 \le \theta \le \pi/4$:

$$\int_{C_2} y\,dx + x\,dy = \int_0^{\pi/4} \left(-\sin^2\theta + \cos^2\theta\right) d\theta = \int_0^{\pi/4} \cos 2\theta\,d\theta = \left[\tfrac{1}{2}\sin 2\theta\right]_0^{\pi/4} = \frac{1}{2}.$$

3. The segment C_3 runs from $(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ to $(0,0)$. Its reverse $-C_3$ is $x = y = t$, $0 \le t \le \frac{1}{\sqrt{2}}$, and work along $-C_3$ is the opposite of work along C_3:

$$\int_{C_3} y\,dx + x\,dy = -\int_0^{1/\sqrt{2}} 2t\,dt = -\left[t^2\right]_0^{1/\sqrt{2}} = -\frac{1}{2}.$$

In total, $\oint_C \mathbf{F}\cdot d\mathbf{r} = 0 + \frac{1}{2} - \frac{1}{2} = 0$.

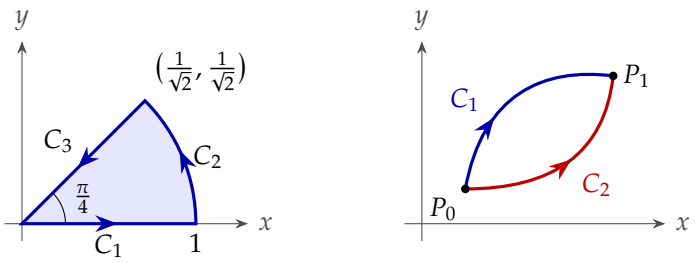


Figure 27: Left: the boundary $C = C_1 + C_2 + C_3$ of the sector of the unit disk between angles 0 and $\pi/4$. Right: two paths C_1 and C_2 from P_0 to P_1; $C_1 - C_2$ is a closed curve.

18.2 The fundamental theorem for line integrals

Definition 18.1 [Gradient field and potential]**.** A vector field of the form $\mathbf{F} = \nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$ is a gradient field, and f is a potential function for $\mathbf{F}$. For example, $\nabla(x^3y^4) = \langle 3x^2y^4, 4x^3y^3\rangle$.

Theorem 18.2 [Fundamental theorem for line integrals]**.** *If C is a curve from P_0 to P_1, then*

$$\int_C \nabla f \cdot d\mathbf{r} = f(P_1) - f(P_0).$$

Equivalently, in differentials, $\int_C f_x\,dx + f_y\,dy = \int_C df = f(P_1) - f(P_0)$.

Proof. Parametrize C by $\big(x(t), y(t)\big)$, $t_0 \le t \le t_1$. By the chain rule,

$$\int_C \nabla f \cdot d\mathbf{r} = \int_{t_0}^{t_1} \Big(f_x \frac{dx}{dt} + f_y \frac{dy}{dt}\Big)dt = \int_{t_0}^{t_1} \frac{d}{dt} f\big(x(t), y(t)\big)\,dt = \Big[f\big(x(t), y(t)\big)\Big]_{t_0}^{t_1} = f(P_1) - f(P_0).$$

The last step is the one-variable fundamental theorem of calculus. □

In the sector example, $f(x, y) = xy$ has $\nabla f = \langle y, x\rangle = \mathbf{F}$, so each $\int_{C_i}$ is the change in xy between its endpoints: $0 - 0$, $\frac{1}{2} - 0$, and $0 - \frac{1}{2}$.

Example: *A potential along a segment.* Let $f(x, y) = xy^3 + x^2$, so $\mathbf{F} = \nabla f = \langle y^3 + 2x,\ 3xy^2\rangle$, and let C be the segment from $(0, 0)$ to $(1, 2)$. Directly, with $x = x$, $y = 2x$, $0 \le x \le 1$,

$$\int_C (y^3 + 2x)\,dx + 3xy^2\,dy = \int_0^1 \big((8x^3 + 2x) + 12x^3 \cdot 2\big)\,dx = \int_0^1 (32x^3 + 2x)\,dx = 9.$$

By the theorem, the answer is $f(1, 2) - f(0, 0) = 9$ immediately.

18.3 Consequences: path independence and conservativeness

For a gradient field the theorem has two consequences.

- Path independence: if C_1 and C_2 have the same endpoints, then $\int_{C_1} \nabla f \cdot d\mathbf{r} = \int_{C_2} \nabla f \cdot d\mathbf{r}$, since both equal $f(P_1) - f(P_0)$. The line integral depends only on the endpoints, not on the trajectory.
- Conservativeness: if C is a closed curve, then $\oint_C \nabla f \cdot d\mathbf{r} = f(P) - f(P) = 0$, as in the sector example.

These hold only for gradient fields. The field $\mathbf{F} = -y\,\mathbf{i} + x\,\mathbf{j}$ is tangent to the counterclockwise circle of radius a, where $\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi a^2 \neq 0$ (Lecture 17). So $\mathbf{F}$ is not conservative, and therefore not a gradient field.

Physical interpretation. If a force field is the gradient of a potential f, its work is the change in

potential. Examples are the gravitational field with the gravitational potential and the electric field with the electric potential (voltage); physicists use the opposite sign convention, $\mathbf{F} = -\nabla f$. Conservativeness means that no energy can be extracted from motion along a closed trajectory: the change in kinetic energy equals the work of the force, which equals the change in potential energy. This is conservation of energy, and it is why such fields are called CONSERVATIVE.

18.4 Four equivalent properties

A differential $M\,dx + N\,dy$ is EXACT if $M\,dx + N\,dy = df = f_x\,dx + f_y\,dy$ for some function f; then $\int_C M\,dx + N\,dy = \int_C df = f(P_1) - f(P_0)$.

Theorem 18.3 [CONSERVATIVE FIELDS]. *For a continuous vector field $\mathbf{F} = \langle M, N\rangle$ defined on the whole plane (or on a connected open region), the following are equivalent:*

1. $\mathbf{F}$ *is conservative:* $\oint_C \mathbf{F}\cdot d\mathbf{r} = 0$ *for every closed curve C;*
2. $\int_C \mathbf{F}\cdot d\mathbf{r}$ *is path independent: it is the same for any two curves with the same endpoints;*
3. $\mathbf{F}$ *is a gradient field:* $\mathbf{F} = \nabla f$ *for some f;*
4. $M\,dx + N\,dy$ *is an exact differential.*

Proof. (1)⇒(2). Let C_1 and C_2 both run from P_0 to P_1 (Figure 27, right). Then $C = C_1 - C_2$, which follows C_1 and returns along C_2 backward, is closed, so

$$0 = \oint_{C_1 - C_2} \mathbf{F}\cdot d\mathbf{r} = \int_{C_1} \mathbf{F}\cdot d\mathbf{r} - \int_{C_2} \mathbf{F}\cdot d\mathbf{r}.$$

(2)⇒(1). A closed curve C through P and the constant path at P have the same endpoints, so by path independence $\oint_C \mathbf{F}\cdot d\mathbf{r}$ equals the integral along the constant path, which is 0.
(3)⇒(2) is Theorem 18.2.
(2)⇒(3). Fix a base point Q (for instance the origin) and define $f(x,y) = \int_Q^{(x,y)} \mathbf{F}\cdot d\mathbf{r}$ along any curve; by path independence this is well defined. To compute f_x, reach $(x+h, y)$ by first going to (x, y) and then along the horizontal segment, on which $dy = 0$:

$$f(x+h, y) - f(x, y) = \int_x^{x+h} M(t, y)\,dt,$$

so $f_x = M$ by the one-variable fundamental theorem. Likewise, using a vertical segment, $f_y = N$. Hence $\nabla f = \mathbf{F}$.
(3)⇔(4). The two statements say the same thing, since $\nabla f \cdot d\mathbf{r} = f_x\,dx + f_y\,dy = df$. □

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 18, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 19: Gradient fields and potential functions

Let $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ be a vector field in the plane with M, N continuous. Recall that the following are equivalent, and a field with these properties is called conservative:

$$\mathbf{F} = \nabla f \iff \int_P^Q \mathbf{F}\cdot d\mathbf{r} \text{ is path-independent} \iff \oint_C \mathbf{F}\cdot d\mathbf{r} = 0 \text{ for all closed } C.$$

None of these conditions is easy to check directly. This lecture gives a test in terms of partial derivatives, two methods for finding a potential, and the curl, which measures the failure of the test.

19.1 A test for gradient fields

Proposition 19.1 [Criterion for gradient fields]. *Let $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ be continuously differentiable in a region D. If $\mathbf{F} = \nabla f$ in D, then $M_y = N_x$ in D.*

Proof. If $\mathbf{F} = \nabla f$ then $M = f_x$ and $N = f_y$, so $M_y = f_{xy}$ and $N_x = f_{yx}$. These mixed partials are continuous because M_y and N_x are, so they are equal. Hence $M_y = N_x$. □

Theorem 19.2 [Converse in the whole plane]. *If $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ is defined and continuously differentiable at every point of the plane and $N_x = M_y$ everywhere, then $\mathbf{F}$ is a gradient field.*

The proof uses Green's theorem and is given in Lecture 20. The hypothesis that $\mathbf{F}$ is defined everywhere cannot be dropped, as the second example below shows.

Example: *Applying the test.*

- $\mathbf{F} = -y\,\mathbf{i} + x\,\mathbf{j}$ has $N_x = 1$ and $M_y = -1$, so it is not a gradient field.
- $\mathbf{F} = (4x^2 + axy)\,\mathbf{i} + (3y^2 + 4x^2)\,\mathbf{j}$ has $N_x = 8x$ and $M_y = ax$. It is a gradient field exactly when $a = 8$.
- $\mathbf{F} = axy\,\mathbf{i} + (x^2 + by)\,\mathbf{j}$ has $M_y = ax$ and $N_x = 2x$. It is a gradient field exactly when $a = 2$; b is arbitrary.

Example: *Two fields undefined at the origin.* Let

$$\mathbf{F} = \frac{x\,\mathbf{i} + y\,\mathbf{j}}{x^2 + y^2}, \qquad \mathbf{G} = \frac{-y\,\mathbf{i} + x\,\mathbf{j}}{x^2 + y^2}.$$

Direct differentiation gives

$$\frac{\partial}{\partial y}\Big(\frac{x}{x^2+y^2}\Big) = \frac{-2xy}{(x^2+y^2)^2} = \frac{\partial}{\partial x}\Big(\frac{y}{x^2+y^2}\Big), \qquad \frac{\partial}{\partial y}\Big(\frac{-y}{x^2+y^2}\Big) = \frac{y^2-x^2}{(x^2+y^2)^2} = \frac{\partial}{\partial x}\Big(\frac{x}{x^2+y^2}\Big),$$

so both fields pass the test wherever they are defined. Neither is defined at $(0,0)$, so Theorem 19.2 does not apply, and the two fields behave differently. One checks that $\mathbf{F} = \nabla \ln\sqrt{x^2+y^2} = \nabla \ln r$, so $\mathbf{F}$ is a gradient field. But on the unit circle $x = \cos t$, $y = \sin t$,

$$\oint_C \mathbf{G}\cdot d\mathbf{r} = \oint_C \frac{-y\,dx + x\,dy}{x^2+y^2} = \int_0^{2\pi} \frac{\sin^2 t + \cos^2 t}{\sin^2 t + \cos^2 t}\,dt = 2\pi \neq 0,$$

so $\mathbf{G}$ is not conservative.

19.2 Finding the potential

Suppose the test shows that $\mathbf{F} = \nabla f$. The potential is determined up to an additive constant: if f is a potential, so is $f + c$. Physicists use $-f$ as the potential, so that potential differences measure work done against the field rather than by it.

Method 1: line integrals. Reading the fundamental theorem for line integrals backwards,

$$f(x_1, y_1) = \int_{(x_0,y_0)}^{(x_1,y_1)} \mathbf{F}\cdot d\mathbf{r} + f(x_0, y_0),$$

and since the integral is path-independent we may choose the simplest path. Usually $(x_0, y_0) = (0,0)$, if $\mathbf{F}$ is defined there, and the path goes along the x-axis and then vertically (Figure 28); on each piece either $dy = 0$ or $dx = 0$.

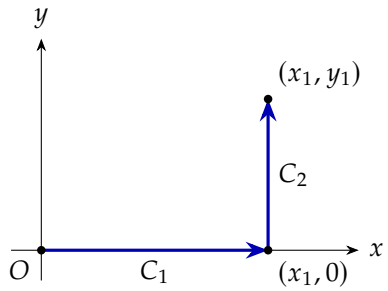


Figure 28: The path $C = C_1 + C_2$ from $(0,0)$ to (x_1, y_1): along the x-axis to $(x_1, 0)$, then vertically.

Example: *Method 1 for* $\mathbf{F} = (4x^2 + 8xy)\,\mathbf{i} + (3y^2 + 4x^2)\,\mathbf{j}$. On C_1, $0 \le x \le x_1$, $y = 0$, $dy = 0$; on C_2, $0 \le y \le y_1$, $x = x_1$, $dx = 0$. So

$$\int_{C_1} \mathbf{F}\cdot d\mathbf{r} = \int_0^{x_1} 4x^2\,dx = \tfrac{4}{3}x_1^3, \qquad \int_{C_2} \mathbf{F}\cdot d\mathbf{r} = \int_0^{y_1} (3y^2 + 4x_1^2)\,dy = y_1^3 + 4x_1^2 y_1,$$

and $f(x, y) = \frac{4}{3}x^3 + 4x^2y + y^3 + c$. Checking $\nabla f = \mathbf{F}$ confirms the answer.

Method 2: antiderivatives. We need $f_x = M$ and $f_y = N$. Integrate the first equation in x, holding y fixed; the constant of integration may depend on y. Then differentiate in y and compare with the second equation.

Example: *Method 2 for the same field.* From $f_x = 4x^2+8xy$, $f = \frac{4}{3}x^3+4x^2y+g(y)$. Then $f_y = 4x^2+g'(y)$, and comparing with $f_y = 3y^2 + 4x^2$ gives $g'(y) = 3y^2$, so $g(y) = y^3 + c$ and $f = \frac{4}{3}x^3 + 4x^2y + y^3 + c$, as before.

Method 1 keeps in view what f means and extends to three dimensions without extra bookkeeping. Method 2 needs no line integrals but its steps must be carried out in the right order.

19.3 Exact differentials

The integrand $M\,dx + N\,dy$ of a line integral is called a DIFFERENTIAL. The TOTAL DIFFERENTIAL of a differentiable function f is $df = f_x\,dx + f_y\,dy$; for example $d(x^2y^3) = 2xy^3\,dx + 3x^2y^2\,dy$. A differential $M\,dx + N\,dy$ is EXACT in a region D if $M\,dx + N\,dy = df$ for some f in D, that is, if $M\,\mathbf{i} + N\,\mathbf{j} = \nabla f$. The results above translate directly.

Proposition 19.3 [EXACTNESS CRITERION]. *Let M and N be continuously differentiable in a region D. If $M\,dx + N\,dy$ is exact in D, then $M_y = N_x$. If D is the whole plane and $M_y = N_x$, then $M\,dx + N\,dy$ is exact.*

A potential for an exact differential is found by either method above.

19.4 Curl

For fields defined everywhere we now have

$$N_x = M_y \iff \mathbf{F} \text{ is a gradient field} \iff \oint_C \mathbf{F}\cdot d\mathbf{r} = 0 \text{ for every closed curve } C,$$

where the first implication from left to right needs $\mathbf{F}$ to be defined everywhere (or, as Lecture 22 shows, in a simply connected region). The quantity measuring the failure of this test has a name.

Definition 19.4 [CURL]. The CURL (two-dimensional curl) of $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ is the scalar function

$$\operatorname{curl}\mathbf{F} = N_x - M_y.$$

Thus, by Proposition 19.1, $\mathbf{F} = \nabla f$ implies $\operatorname{curl}\mathbf{F} = 0$. For example, $\mathbf{F} = x^3y^2\,\mathbf{i}+x\,\mathbf{j}$ has $\operatorname{curl}\mathbf{F} = 1-2x^3y$. The two-dimensional curl is a scalar; it will reappear as the $\mathbf{k}$-component of the three-dimensional curl.

Interpretation for a velocity field. The curl is twice the angular velocity of the rotational component of the motion. The uniform translation $\langle a, b\rangle$ and the expanding motion $\langle x, y\rangle$ have curl 0, while the rotation $\langle -y, x\rangle$, at unit angular velocity, has curl 2.

Interpretation for a force field. The curl is the torque exerted on a test mass, and measures how $\mathbf{F}$ imparts rotational motion. In the analogy

$$\frac{\text{force}}{\text{mass}} = \text{acceleration} = \frac{d}{dt}(\text{velocity}),$$
$$\frac{\text{torque}}{\text{moment of inertia}} = \text{angular acceleration} = \frac{d}{dt}(\text{angular velocity}),$$

force drives translation and torque drives rotation.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 19, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 20: Green's theorem

Green's theorem converts a line integral around a closed curve into a double integral over the region it encloses.

20.1 Statement and examples

Let C be a simple closed curve (it never crosses itself) that is piecewise smooth (smooth except for finitely many corners), and let R be its interior. C is positively oriented if it is traversed with R on the left; for a circle this means counterclockwise.

Theorem 20.1 [Green's theorem]. *Let C be a positively oriented, piecewise smooth, simple closed curve enclosing a region R, and let $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ be continuously differentiable on R and C. Then*

$$\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_R \operatorname{curl}\mathbf{F}\,dA, \qquad \textit{that is,} \qquad \oint_C M\,dx + N\,dy = \iint_R (N_x - M_y)\,dA.$$

The theorem can be used in either direction: to replace a hard line integral by an easy double integral, or the reverse.

Example: *A line integral replaced by a double integral.* Let C be the unit circle centered at $(2,0)$, counterclockwise, and R the disk it bounds. With $M = ye^{-x}$ and $N = \frac{1}{2}x^2 - e^{-x}$,

$$\oint_C ye^{-x}\,dx + \left(\tfrac{1}{2}x^2 - e^{-x}\right)dy = \iint_R \left((x + e^{-x}) - e^{-x}\right)dA = \iint_R x\,dA = \text{area}(R)\cdot\bar{x} = \pi\cdot 2 = 2\pi.$$

Parametrizing by $x = 2 + \cos\theta$, $y = \sin\theta$ would lead to a much harder integral.

Example: *Another use of the centroid.* Let C be the circle of radius a centered at $(a,0)$, counterclockwise. Then

$$\oint_C 3x^2y^2\,dx + 2x^2(1+xy)\,dy = \iint_R \left(4x + 6x^2y - 6x^2y\right)dA = 4\iint_R x\,dA.$$

Since $\bar{x} = \frac{1}{\text{area}}\iint_R x\,dA = a$, we get $\iint_R x\,dA = \pi a^3$, so the line integral is $4\pi a^3$.

20.2 Proof of the criterion for gradient fields

Theorem 20.2 [Gradient fields in the whole plane]. *Let $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ be defined and continuously differentiable in the whole plane. Then $\mathbf{F}$ is conservative (equivalently, a gradient field) if and only if* $\operatorname{curl}\mathbf{F} = N_x - M_y = 0$.

Proof. If $N_x = M_y$, then for every simple closed curve C enclosing a region R, Green's theorem gives $\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_R 0\,dA = 0$, so $\mathbf{F}$ is conservative. (A closed curve that crosses itself finitely often splits into simple closed loops; Lecture 22 gives the details.) Conversely, if $\mathbf{F}$ is conservative, then $\iint_R \operatorname{curl}\mathbf{F}\,dA = 0$ for every such region R. If $\operatorname{curl}\mathbf{F}$ were nonzero at some point, by continuity it would have one sign on a small disk around that point, and its integral over that disk would be nonzero. Hence $\operatorname{curl}\mathbf{F} = 0$. □

The argument needs $\mathbf{F}$ and its curl to be defined everywhere inside C. For $\mathbf{G} = (-y\,\mathbf{i} + x\,\mathbf{j})/(x^2+y^2)$ from Lecture 19, the curl is zero wherever it is defined, yet the line integral around the unit circle is 2π; Green's theorem cannot be applied to a region containing the origin.

20.3 Proof of Green's theorem

Two preliminary remarks reduce the work.

1. The theorem splits into two identities,
$$\oint_C M\,dx = -\iint_R M_y\,dA \qquad \text{and} \qquad \oint_C N\,dy = \iint_R N_x\,dA.$$
2. *Additivity.* If the theorem holds for R_1 and R_2, with boundaries C_1 and C_2, then it holds for $R = R_1 \cup R_2$ glued along a common edge. Indeed $\iint_R = \iint_{R_1} + \iint_{R_2}$, and $\oint_C = \oint_{C_1} + \oint_{C_2}$ because the common edge is traversed once in each direction, so its contributions cancel (Figure 29, right).

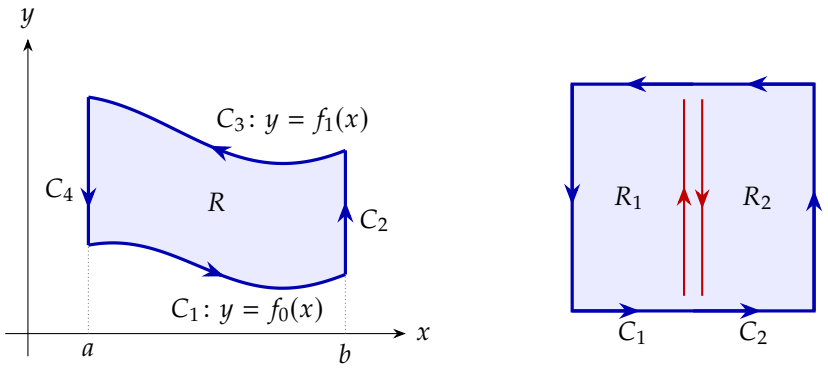


Figure 29: Left: a vertically simple region $a < x < b$, $f_0(x) < y < f_1(x)$, with its boundary split into four pieces. Right: when two regions share an edge, the edge is traversed in opposite directions and the two line integrals along it cancel.

Main step. We prove $\oint_C M\,dx = -\iint_R M_y\,dA$ for a VERTICALLY SIMPLE region $a < x < b$, $f_0(x) < y < f_1(x)$. Split C into the lower curve C_1, the right segment C_2, the upper curve C_3, and the left segment C_4. On C_2 and C_4, x is constant, so $dx = 0$ and these contribute nothing. On C_1 use x as parameter,

$a \le x \le b$, $y = f_0(x)$; on C_3, x runs from b to a with $y = f_1(x)$. Hence

$$\oint_C M\,dx = \int_{C_1} + \int_{C_3} = \int_a^b M(x, f_0(x))\,dx - \int_a^b M(x, f_1(x))\,dx.$$

On the other side, integrating first in y,

$$-\iint_R M_y\,dA = -\int_a^b \int_{f_0(x)}^{f_1(x)} M_y\,dy\,dx = -\int_a^b \big(M(x, f_1(x)) - M(x, f_0(x))\big)\,dx,$$

which is the same.

Conclusion. Any region can be subdivided into vertically simple pieces R_i with boundaries C_i. Each satisfies $\oint_{C_i} M\,dx = -\iint_{R_i} M_y\,dA$, so by additivity $\oint_C M\,dx = -\iint_R M_y\,dA$. Subdividing into horizontally simple pieces proves $\oint_C N\,dy = \iint_R N_x\,dA$ in the same way, which completes the proof. One can equally approximate R by a union of small rectangles, which are both vertically and horizontally simple: the interior edges cancel and only the outer boundary remains.

20.4 Area as a line integral

Taking $\mathbf{F} = x\,\mathbf{j}$ or $\mathbf{F} = -y\,\mathbf{i}$, whose curl is 1, gives

$$\text{area}(R) = \iint_R 1\,dA = \oint_C x\,dy = \oint_C -y\,dx.$$

This is the principle of the PLANIMETER, a mechanical device that computes such a line integral as its arm traces a closed curve on paper, and so measures the enclosed area.

Example: *Area under one arch of a cycloid.* Let C_2 be the arch $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$, $0 \le \theta \le 2\pi$, and C_1 the segment of the x-axis from 0 to $2\pi a$. The region under the arch is bounded by $C = C_1 - C_2$, positively oriented (Figure 30). Since $y = 0$ on C_1,

$$\text{area} = \oint_{C_1 - C_2} -y\,dx = \int_{C_1} 0\,dx - \int_{C_2} -y\,dx = \int_0^{2\pi} a^2(1 - \cos\theta)^2\,d\theta = 3\pi a^2,$$

using $\int_0^{2\pi}(1 - 2\cos\theta + \cos^2\theta)\,d\theta = 2\pi + \pi$.

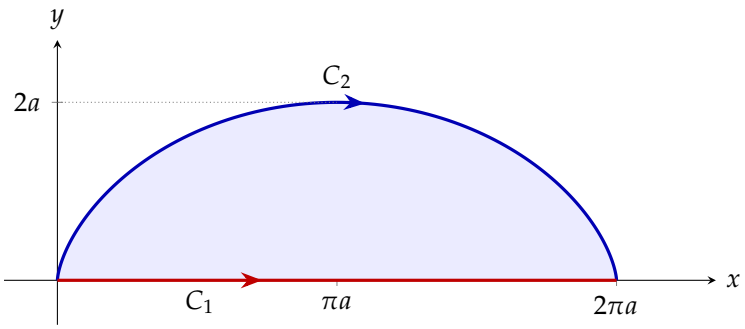


Figure 30: One arch of the cycloid (C_2, traversed with increasing θ) and the segment C_1 of the x-axis. The boundary $C_1 - C_2$ encloses the shaded region counterclockwise.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 20, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 21: Flux; normal form of Green's theorem

21.1 Flux across a plane curve

Let C be a curve in the plane, with unit tangent $\hat{\mathbf{T}}$ in its direction of travel, and let $\hat{\mathbf{n}}$ be the unit normal obtained by rotating $\hat{\mathbf{T}}$ by 90° *clockwise*, so that $\hat{\mathbf{n}}$ points to the right of the direction of travel.

Definition 21.1 [Flux]. The flux of a vector field $\mathbf{F}$ across C is

$$\int_C \mathbf{F} \cdot \hat{\mathbf{n}}\, ds.$$

There are now two kinds of line integral. The work $\int_C \mathbf{F} \cdot \hat{\mathbf{T}}\, ds$ sums the component of $\mathbf{F}$ along C; the flux sums the component of $\mathbf{F}$ perpendicular to C. If C is cut into small pieces of length Δs_i, the flux is approximately $\sum_i (\mathbf{F} \cdot \hat{\mathbf{n}})\, \Delta s_i$.

Physical interpretation. If $\mathbf{F}$ is the velocity field of a fluid flow (in a shallow tank of unit depth), the flux is the amount of fluid crossing C per unit time. On a small piece of C of length Δs, $\mathbf{F}$ is nearly constant, and what crosses the piece in unit time fills a parallelogram with sides Δs and $\mathbf{F}$ (Figure 31, left). Its area is $\Delta s \cdot \text{height} = \Delta s\, (\mathbf{F} \cdot \hat{\mathbf{n}})$: the component of $\mathbf{F}$ along C carries nothing across. Summing over the pieces, $\int_C \mathbf{F} \cdot \hat{\mathbf{n}}\, ds$ is the total flow across C per unit time, counted positively for flow to the right of C and negatively for flow to the left, as seen by a point travelling along C.

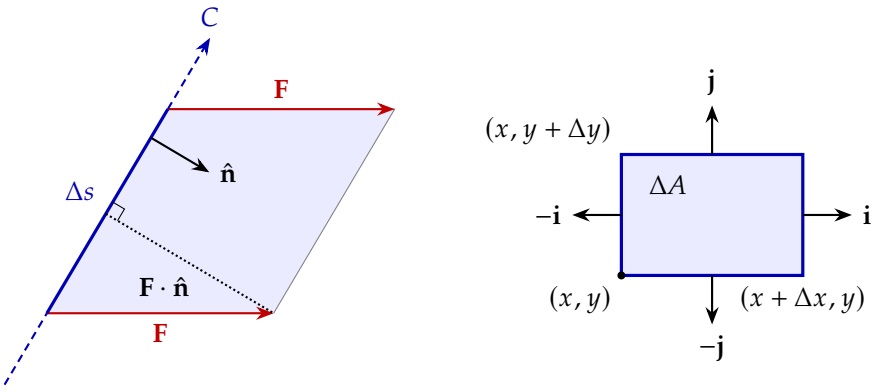


Figure 31: Left: in unit time, the fluid crossing a piece of C of length Δs fills a parallelogram with sides Δs and $\mathbf{F}$; its height is $\mathbf{F} \cdot \hat{\mathbf{n}}$. Right: a small rectangle with its outward normals, used to interpret the divergence.

Example: *Flux across a circle, geometrically.* Let C be the circle of radius a centered at the origin, counterclockwise, so $\hat{\mathbf{n}}$ points outward. For $\mathbf{F} = x\,\mathbf{i} + y\,\mathbf{j}$, along C the field is parallel to $\hat{\mathbf{n}}$ with

$|\mathbf{F}| = a$, so $\mathbf{F} \cdot \hat{\mathbf{n}} = a$ and

$$\int_C \mathbf{F} \cdot \hat{\mathbf{n}}\, ds = \int_C a\, ds = a \,\text{length}(C) = 2\pi a^2.$$

The flux of $-y\,\mathbf{i} + x\,\mathbf{j}$ across C is zero, since that field is tangent to C.

Flux in coordinates. Since $d\mathbf{r} = \hat{\mathbf{T}}\, ds = \langle dx, dy\rangle$ and $\hat{\mathbf{n}}$ is $\hat{\mathbf{T}}$ rotated clockwise, $\hat{\mathbf{n}}\, ds = \langle dy, -dx\rangle$. So for $\mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j}$,

$$\int_C \mathbf{F} \cdot \hat{\mathbf{n}}\, ds = \int_C \langle P, Q\rangle \cdot \langle dy, -dx\rangle = \int_C -Q\, dx + P\, dy,$$

or $\int_C -N\, dx + M\, dy$ if $\mathbf{F} = \langle M, N\rangle$. This is evaluated like any line integral, by expressing x, y, dx, dy in terms of a parameter.

Example: *A point source.* Let $\mathbf{F} = (x\,\mathbf{i} + y\,\mathbf{j})/(x^2 + y^2)$ and C the circle of radius a centered at the origin. Geometrically, $\mathbf{F}$ is radial and outward, so $\mathbf{F} \cdot \hat{\mathbf{n}} = |\mathbf{F}| = 1/\sqrt{x^2 + y^2} = 1/a$ and the flux is $\frac{1}{a} \cdot 2\pi a = 2\pi$. In coordinates, with $x = a\cos t$, $y = a \sin t$,

$$\oint_C \frac{x\, dy - y\, dx}{x^2 + y^2} = \int_0^{2\pi} \frac{a^2\cos^2 t + a^2 \sin^2 t}{a^2}\, dt = 2\pi.$$

The field describes a flow from a single source of strength 2π at the origin, which is why the flux across every circle centered at the origin is the same, whatever its radius.

21.2 Green's theorem for flux

Theorem 21.2 [Green's theorem, normal form]. *Let C be a positively oriented simple closed curve enclosing a region R, and let $\mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j}$ be continuously differentiable on R. Then*

$$\oint_C \mathbf{F} \cdot \hat{\mathbf{n}}\, ds = \iint_R \operatorname{div} \mathbf{F}\, dA, \qquad \textit{where} \qquad \operatorname{div} \mathbf{F} = P_x + Q_y$$

is the divergence *of* $\mathbf{F}$.

Because C is counterclockwise, $\hat{\mathbf{n}}$ points out of R, so the left side is the flux of $\mathbf{F}$ out of R; flow into R counts negatively.

Proof. We have $\oint_C \mathbf{F} \cdot \hat{\mathbf{n}}\, ds = \oint_C -Q\, dx + P\, dy$. Set $M = -Q$ and $N = P$ and apply the usual (tangential) Green's theorem $\oint_C M\, dx + N\, dy = \iint_R (N_x - M_y)\, dA$:

$$\oint_C -Q\, dx + P\, dy = \iint_R \big(P_x - (-Q_y)\big)\, dA = \iint_R \operatorname{div} \mathbf{F}\, dA. \qquad \square$$

This proof by renaming the components is why the components were called P, Q. With $\mathbf{F} = \langle M, N\rangle$ the statement reads $\oint_C -N\, dx + M\, dy = \iint_R (M_x + N_y)\, dA$. Mathematically it is the same theorem as

the tangential form; what changes is the interpretation.

Example: *The circle again.* For $\mathbf{F} = x\,\mathbf{i} + y\,\mathbf{j}$, $\operatorname{div}\mathbf{F} = 2$, so the flux out of the circle of radius a is $\iint_R 2\,dA = 2\,\text{area}(R) = 2\pi a^2$, as found geometrically. If the circle is translated so that it is no longer centered at the origin, a direct computation of the flux becomes harder, but the theorem shows that the total flux is still $2\pi a^2$.

21.3 Interpretation of the divergence

For a small rectangle of area $\Delta A = \Delta x\,\Delta y$ with corner (x, y), $\operatorname{div}\mathbf{F}$ is nearly constant, and the theorem gives

$$\text{flux out of the rectangle} \approx \operatorname{div}\mathbf{F}\,\Delta A.$$

This also follows directly, without Green's theorem (Figure 31, right). Write $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$. The outward normal is $\mathbf{j}$ on the top and $-\mathbf{j}$ on the bottom, so

$$\text{flux across top and bottom} \approx \big(N(x, y + \Delta y) - N(x, y)\big)\Delta x \approx \frac{\partial N}{\partial y}\,\Delta y\,\Delta x,$$

and similarly the flux across the left and right sides is about $\frac{\partial M}{\partial x}\,\Delta x\,\Delta y$. Adding, the flux out of the rectangle is about $(M_x + N_y)\,\Delta x\,\Delta y$.

For an incompressible fluid, a net outward flow from the rectangle is possible only if fluid is being added inside it by a source (a sink is a negative source). By conservation of mass the flux out of the rectangle equals its SOURCE RATE. Dividing by ΔA and letting the rectangle shrink,

$$\operatorname{div}\mathbf{F}(x, y) = \text{source rate per unit area at } (x, y), \qquad \text{source rate for } R = \iint_R \operatorname{div}\mathbf{F}\,dA.$$

So the normal form of Green's theorem says that the total flux out of C equals the total source rate inside. This does not prove conservation of mass; that law was used to interpret the divergence.

The two forms side by side:

tangential form	normal form
$\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_R \operatorname{curl}\mathbf{F}\,dA$	$\oint_C \mathbf{F}\cdot\hat{\mathbf{n}}\,ds = \iint_R \operatorname{div}\mathbf{F}\,dA$
$\oint_C M\,dx + N\,dy = \iint_R (N_x - M_y)\,dA$	$\oint_C -N\,dx + M\,dy = \iint_R (M_x + N_y)\,dA$
work of $\mathbf{F}$ around C	flux of $\mathbf{F}$ across C = source rate in R

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 21, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 22: Simply connected regions

Green's theorem, in both forms,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R \operatorname{curl} \mathbf{F}\, dA, \qquad \oint_C \mathbf{F} \cdot \hat{\mathbf{n}}\, ds = \iint_R \operatorname{div} \mathbf{F}\, dA,$$

requires $\mathbf{F}$ to be defined and continuously differentiable on the *entire* region R enclosed by C. This lecture identifies the regions where that is automatic and extends the theorem to regions with holes.

22.1 The failure at a point

Let

$$\mathbf{F} = \frac{-y\,\mathbf{i} + x\,\mathbf{j}}{x^2 + y^2}$$

and C the unit circle, counterclockwise. A direct computation gives

$$\operatorname{curl} \mathbf{F} = \frac{\partial}{\partial x}\Big(\frac{x}{x^2+y^2}\Big) - \frac{\partial}{\partial y}\Big(\frac{-y}{x^2+y^2}\Big) = \frac{y^2-x^2}{(x^2+y^2)^2} - \frac{y^2-x^2}{(x^2+y^2)^2} = 0$$

wherever $\mathbf{F}$ is defined. Yet $\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi$ (Lecture 19), while $\iint_R \operatorname{curl} \mathbf{F}\, dA$ would be $\iint_R 0\, dA = 0$. There is no contradiction: R contains the origin, where $\mathbf{F}$ is not defined, so Green's theorem does not apply.

Definition 22.1 [Simply connected region]. A connected region D in the plane is simply connected if for every simple closed curve C lying in D, the interior of C also lies entirely in D. Otherwise D is multiply connected.

Informally, a simply connected region has no holes. The whole plane, the half-plane $x \geq 0$, and a disk are simply connected; the plane with the origin removed is not, since a circle around the origin lies in it but its interior does not. If the domain where $\mathbf{F}$ is defined and continuously differentiable is simply connected, Green's theorem applies safely: whenever $\mathbf{F}$ is defined on C, it is also defined on the region bounded by C.

22.2 Green's theorem for regions with holes

Suppose the boundary of R consists of several simple closed curves $C_1, \ldots, C_m$, all inside the domain of $\mathbf{F}$, each oriented so that R is on its left (equivalently, $\hat{\mathbf{n}}$ points away from R): the outer boundary counterclockwise, the boundaries of holes clockwise.

Theorem 22.2 [Extended Green's theorem]. *With these orientations,*

$$\oint_{C_1} \mathbf{F} \cdot d\mathbf{r} + \cdots + \oint_{C_m} \mathbf{F} \cdot d\mathbf{r} = \iint_R \operatorname{curl} \mathbf{F}\, dA.$$

Proof. Cut R along finitely many segments into pieces $R_1, \ldots, R_k$, each bounded by a single simple closed curve B_j, and apply Green's theorem to each piece: $\oint_{B_j} \mathbf{F} \cdot d\mathbf{r} = \iint_{R_j} \operatorname{curl} \mathbf{F}\, dA$. Adding, the right sides give $\iint_R \operatorname{curl} \mathbf{F}\, dA$. On the left, each cut is traversed twice in opposite directions, so those contributions cancel and only the boundary curves $C_1, \ldots, C_m$ remain. □

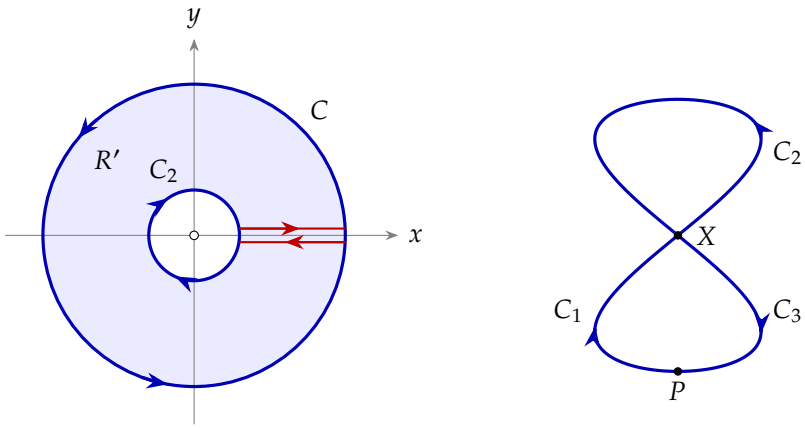


Figure 32: Left: the annulus R' between the unit circle C (counterclockwise) and a small circle C_2 around the origin (clockwise), joined by a cut along the x-axis traversed once in each direction. Right: a closed curve through P that crosses itself at X splits into the simple closed loops C_2 and $C_1 + C_3$.

Example: *Line integrals around the origin.* For the field $\mathbf{F}$ above, apply Theorem 22.2 to the annulus R' between the unit circle C and a small circle C_2 around the origin (Figure 32, left). Equivalently, use the single closed curve C' that goes counterclockwise around C, in along the x-axis, clockwise around C_2, and back out along the x-axis. Since $\operatorname{curl} \mathbf{F} = 0$ on R',

$$0 = \iint_{R'} 0\, dA = \oint_{C'} \mathbf{F} \cdot d\mathbf{r} = \oint_{C} \mathbf{F} \cdot d\mathbf{r} - \oint_{C_2} \mathbf{F} \cdot d\mathbf{r},$$

with both circles now counterclockwise. So the line integral is the same, 2π, around C, around C_2, and around any simple closed curve encircling the origin once counterclockwise.

In normal form the same phenomenon is clearer: a flow with $\operatorname{div} \mathbf{F} = 0$ everywhere except at one point can still have nonzero flux across a curve around that point, because the point may be a source adding fluid to the flow.

22.3 The curl test in simply connected regions

Green's theorem still holds for regions with holes, but the implication $\operatorname{curl}\mathbf{F} = 0 \Rightarrow \mathbf{F} = \nabla f$ does not: for a curve C around a hole, the interior of C is not contained in the region, so Green's theorem cannot be applied to C. Assuming the region is simply connected removes the difficulty.

Theorem 22.3 [Gradient fields in simply connected regions]. *Let $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ be continuously differentiable in a simply connected region D. If $\operatorname{curl}\mathbf{F} = 0$ (that is, $M_y = N_x$) in D, then $\mathbf{F} = \nabla f$ for some f in D.*

Proof. It suffices to show $\oint_C \mathbf{F}\cdot d\mathbf{r} = 0$ for every closed curve C in D. If C is simple, its interior R lies in D because D is simply connected, so $\mathbf{F}$ is continuously differentiable on R and Green's theorem gives $\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_R 0\,dA = 0$. If C crosses itself finitely many times, it splits into simple closed loops. For example, if C runs from P along C_1 to a crossing point X, around a loop C_2 back to X, and along C_3 back to P (Figure 32, right), then C_2 and $C_1 + C_3$ are simple closed curves, so

$$\oint_C \mathbf{F}\cdot d\mathbf{r} = \left(\int_{C_1} + \int_{C_3}\right)\mathbf{F}\cdot d\mathbf{r} + \oint_{C_2} \mathbf{F}\cdot d\mathbf{r} = 0 + 0.$$

Repeating this at each crossing handles any finite number of crossings. (Curves crossing themselves infinitely often require approximation and are omitted.) □

Theorem 22.4 [Curl theorem]. *Let $\mathbf{F} = M\,\mathbf{i} + N\,\mathbf{j}$ be continuously differentiable in a simply connected region D. The following are equivalent:*

1. *$\int_P^Q \mathbf{F}\cdot d\mathbf{r} = \int_P^Q M\,dx + N\,dy$ is path-independent for all P, Q in D;*
2. *$\oint_C \mathbf{F}\cdot d\mathbf{r} = 0$ for every simple closed curve C in D;*
3. *$\mathbf{F} = \nabla f$, that is, $M\,dx + N\,dy = df$, for some f in D;*
4. *$\operatorname{curl}\mathbf{F} = 0$, that is, $M_y = N_x$, in D.*

Without the hypotheses, part of this survives. Statements 1, 2, and 3 are equivalent even if $\mathbf{F}$ is only continuous and D is not simply connected. Each of them implies 4 if $\mathbf{F}$ is continuously differentiable, for any D. Only the implication from 4 to the others needs D to be simply connected.

22.4 Examples

Example: *A field that fails the test.* $\mathbf{F} = xy\,\mathbf{i} + x^2\,\mathbf{j}$ has $\operatorname{curl}\mathbf{F} = 2x - x = x \neq 0$, so it is not a gradient field.

Example: *An exact differential on two half-planes.* Consider $(y\,dx - x\,dy)/y^2$, so $M = 1/y$ and $N = -x/y^2$. These are continuously differentiable wherever $y \neq 0$, that is, on the upper and lower half-planes, each simply connected. In each, $M_y = -1/y^2 = N_x$, so the differential is exact there,

and either method of Lecture 19 gives $f = x/y + c$. The constant need not be the same on the two half-planes, since they do not touch, so the general potential is

$$f(x,y) = \begin{cases} x/y + c, & y > 0, \\ x/y + c', & y < 0, \end{cases} \qquad c, c' \text{ arbitrary.}$$

Example: *Radial fields.* Let $\mathbf{F} = r^n(x\,\mathbf{i} + y\,\mathbf{j})$, $r = \sqrt{x^2+y^2}$, n an integer. Using $r_x = x/r$ and $r_y = y/r$,

$$N_x = nr^{n-1}\frac{x}{r}\,y = nr^{n-2}xy = M_y,$$

so curl $\mathbf{F} = 0$ wherever $\mathbf{F}$ is defined.

Case $n \geq 0$. $\mathbf{F}$ is continuously differentiable on the whole plane, so it is a gradient field. Integrate along the ray $x = x_1 t$, $y = y_1 t$, $0 \leq t \leq 1$, from the origin to (x_1, y_1). With $r_1 = \sqrt{x_1^2 + y_1^2}$, we have $r^n = r_1^n t^n$ and $x\,dx + y\,dy = r_1^2 t\,dt$, so

$$f(x_1,y_1) = \int_C r^n(x\,dx + y\,dy) = \int_0^1 r_1^{n+2} t^{n+1}\,dt = \frac{r_1^{n+2}}{n+2}, \qquad \text{that is,} \qquad f = \frac{r^{n+2}}{n+2}.$$

Case $n < 0$. $\mathbf{F}$ is undefined at the origin and its domain is not simply connected, so Theorem 22.3 does not apply directly. Nevertheless, for $n \neq -2$ one checks by differentiation that $f = r^{n+2}/(n+2)$ still satisfies $\nabla f = \mathbf{F}$, and for $n = -2$, $f = \ln r$ works.

So $\mathbf{F}$ is a gradient field for every n. In particular, up to constant factors, the field of a positive point charge at the origin and that of a uniform positive charge along the z-axis are both gradient fields:

$$\frac{x\,\mathbf{i} + y\,\mathbf{j}}{r^3} = \nabla\left(-\frac{1}{r}\right) \quad (n = -3), \qquad \frac{x\,\mathbf{i} + y\,\mathbf{j}}{r^2} = \nabla(\ln r) \quad (n = -2).$$

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 22, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 23: Triple integrals in rectangular and cylindrical coordinates

A triple integral $\iiint_R f\,dV$ integrates a function over a solid region R in space, where dV is the volume element. As for double integrals, it is computed as an iterated integral, and the main work is setting up the bounds.

23.1 Setting up bounds

To set up $\iiint_D f\,dz\,dy\,dx$ over a region D:

1. Fix (x, y) and let z increase along the vertical line through (x, y). Integrate from the z-value where this line enters D to the value where it leaves D.
2. Find the shadow of D, the region R in the xy-plane consisting of the points above or below which D lies, and supply the remaining bounds for a double integral over R, in rectangular or polar coordinates.

Example: *The region between two paraboloids.* Let D lie between $z = x^2+y^2$ (below) and $z = 4-x^2-y^2$ (above). For fixed (x, y), z runs from x^2+y^2 to $4-x^2-y^2$. The region is widest where the paraboloids meet; in general the shadow is where the top surface lies above the bottom one, $4-x^2-y^2 > x^2+y^2$, that is, $x^2 + y^2 < 2$. So the shadow is the disk of radius $\sqrt{2}$ (Figure 33), the intersection lies in the plane $z = 2$, and

$$\text{vol}(D) = \int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} \int_{x^2+y^2}^{4-x^2-y^2} dz\,dy\,dx.$$

The disk suggests polar coordinates in the xy-plane, with $x^2 + y^2 = r^2$:

$$\text{vol}(D) = \int_0^{2\pi} \int_0^{\sqrt{2}} \int_{r^2}^{4-r^2} dz\,r\,dr\,d\theta = 2\pi \int_0^{\sqrt{2}} (4 - 2r^2)\,r\,dr = 2\pi\left[2r^2 - \tfrac{r^4}{2}\right]_0^{\sqrt{2}} = 4\pi.$$

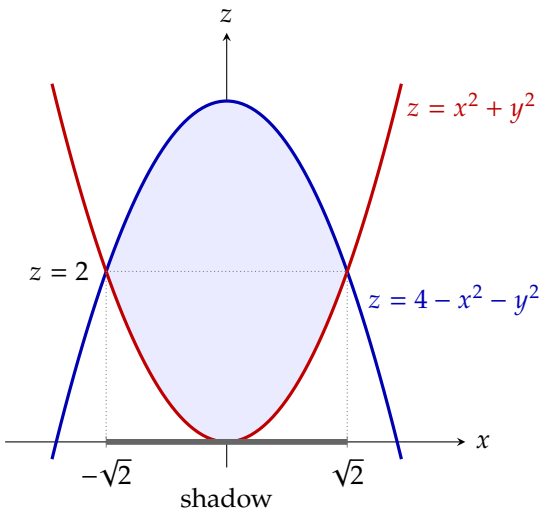


Figure 33: The slice $y = 0$ of the region between $z = x^2 + y^2$ and $z = 4 - x^2 - y^2$. The paraboloids meet in the plane $z = 2$, and the shadow of the region is the disk $x^2 + y^2 < 2$.

23.2 Cylindrical coordinates

Cylindrical coordinates (r, θ, z) use polar coordinates in the xy-plane and keep z:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z.$$

Here r is the distance from the z-axis and θ the angle from the xz-plane. The cylinder of radius a about the z-axis is $r = a$, and $\theta = 0$ is a vertical half-plane.

The volume elements are

$$dV = dx\,dy\,dz \quad \text{(rectangular)}, \qquad dV = r\,dr\,d\theta\,dz \quad \text{(cylindrical)}.$$

Both follow from a small box with base area ΔA and height Δz, whose volume is $\Delta V = \Delta A\,\Delta z$, with $\Delta A \approx r\,\Delta r\,\Delta\theta$ in polar coordinates.

23.3 Applications

For a solid R with density δ:

- mass $M = \iiint_R \delta\,dV$;
- average value of f: $\bar{f} = \frac{1}{\mathrm{vol}(R)} \iiint_R f\,dV$, and weighted average $\bar{f} = \frac{1}{M} \iiint_R f\,\delta\,dV$;
- center of mass $(\bar{x}, \bar{y}, \bar{z})$, where $\bar{x} = \frac{1}{M} \iiint_R x\,\delta\,dV$, and similarly for $\bar{y}, \bar{z}$;

- moment of inertia about an axis: $I = \iiint_R (\text{distance from the axis})^2 \, \delta \, dV$. In particular

$$I_z = \iiint_R (x^2 + y^2)\, \delta \, dV = \iiint_R r^2 \, \delta \, dV, \quad I_x = \iiint_R (y^2 + z^2)\, \delta \, dV, \quad I_y = \iiint_R (x^2 + z^2)\, \delta \, dV.$$

I_z is consistent with the polar moment of inertia I_0 of a plane region, and setting $z = 0$ recovers the earlier I_x and I_y of plane regions. Symmetry often avoids computation: for the region between the paraboloids, $\bar{x} = \bar{y} = 0$.

Example: *Moment of inertia of a cone.* Let R be the solid cone between $z = ar$ and $z = b$, with $\delta = 1$. Integrate in the order $dr\, d\theta\, dz$: for fixed z, the slice is the disk $r \le z/a$, and the slices run from $z = 0$ to $z = b$ (Figure 34, left). Then

$$I_z = \iiint_R r^2 \, dV = \int_0^b \int_0^{2\pi} \int_0^{z/a} r^2 \, r \, dr \, d\theta \, dz = 2\pi \int_0^b \frac{z^4}{4a^4} \, dz = \frac{\pi b^5}{10a^4}.$$

Example: *A sphere cut by a plane.* Find the volume of the region where $z > 1 - y$ and $x^2 + y^2 + z^2 < 1$. The bottom surface is the plane $z = 1 - y$ and the top is the sphere $z = \sqrt{1 - x^2 - y^2}$, so the inner integral is $\int_{1-y}^{\sqrt{1-x^2-y^2}} dz$. The shadow is where $1 - y < \sqrt{1 - x^2 - y^2}$; squaring, $(1 - y)^2 < 1 - x^2 - y^2$, that is, $x^2 < 2y - 2y^2$, or $-\sqrt{2y - 2y^2} < x < \sqrt{2y - 2y^2}$. This has solutions exactly when $2y - 2y^2 > 0$, that is, $0 < y < 1$, which matches the slice by the yz-plane (Figure 34, right): the region runs from $(y, z) = (0, 1)$ to $(1, 0)$. So

$$\text{vol} = \int_0^1 \int_{-\sqrt{2y-2y^2}}^{\sqrt{2y-2y^2}} \int_{1-y}^{\sqrt{1-x^2-y^2}} dz \, dx \, dy.$$

Lecture 24 evaluates this volume in spherical coordinates.

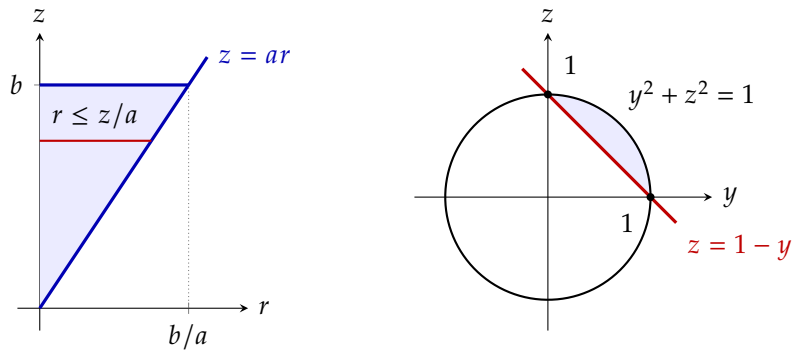


Figure 34: Left: the cone $ar \le z \le b$ in the (r, z) half-plane, drawn for $a = 3/2$, $b = 3$; at height z the slice is the disk $r \le z/a$. Right: the slice $x = 0$ of the region above the plane $z = 1 - y$ inside the unit sphere.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 23, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 24: Spherical coordinates; surface area

24.1 Spherical coordinates

The spherical coordinates (ρ, ϕ, θ) of a point are

- $\rho \geq 0$, the distance to the origin;
- $\phi \in [0, \pi]$, the angle down from the positive z-axis;
- $\theta \in [0, 2\pi]$, the same angle as in cylindrical coordinates.

In the vertical half-plane through the point (Figure 35), $z = \rho\cos\phi$ and $r = \rho\sin\phi$, where r is the distance to the z-axis. With $x = r\cos\theta$ and $y = r\sin\theta$ this gives

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi,$$

and conversely

$$\rho = \sqrt{x^2+y^2+z^2} = \sqrt{r^2+z^2}, \qquad \phi = \cos^{-1}(z/\rho), \qquad \theta = \tan^{-1}(y/x) \text{ (in the correct quadrant).}$$

For example, $(1,0,0)$ has $\rho = 1$, $\phi = \pi/2$, $\theta = 0$; $(0,1,0)$ has $\rho = 1$, $\phi = \pi/2$, $\theta = \pi/2$; and $(0,0,1)$ has $\rho = 1$, $\phi = 0$, with θ arbitrary.

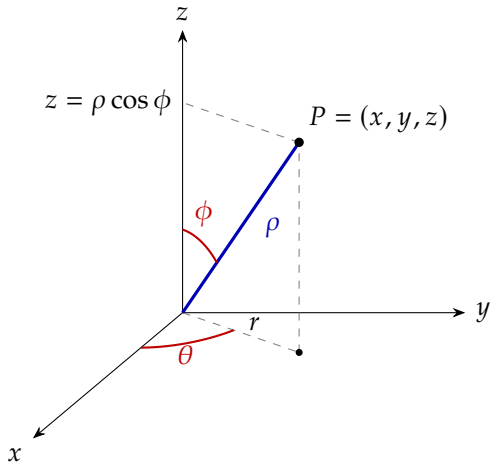


Figure 35: Spherical coordinates of a point P: $\rho = |OP|$, ϕ is the angle from the z-axis, and θ is the polar angle of the projection of P to the xy-plane, at distance $r = \rho\sin\phi$ from the origin.

The surface $\rho = a$ is the sphere of radius a centered at the origin. On it, ϕ is like latitude, except that it is 0 at the north pole, $\pi/2$ on the equator, and π at the south pole; θ is like longitude. The

surface $\phi = \pi/4$ is the cone $z = r = \sqrt{x^2 + y^2}$, and $\phi = \pi/2$ is the xy-plane.

24.2 Surface and volume elements

Proposition 24.1 [Area element on a sphere]. *On the sphere of radius a, the area element is*

$$dS = a^2 \sin\phi \, d\phi \, d\theta.$$

Proof. A small "rectangle" on the sphere corresponding to increments $\Delta\phi$ and $\Delta\theta$ has two kinds of sides. One is an arc of a great circle of radius a, of length $a\,\Delta\phi$. The other is an arc of a horizontal circle of radius $r = a\sin\phi$, of length $r\,\Delta\theta = a\sin\phi\,\Delta\theta$. Hence $\Delta S \approx a^2 \sin\phi\,\Delta\phi\,\Delta\theta$. □

Proposition 24.2 [Volume element in spherical coordinates].

$$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta.$$

Proof. A small box between the spheres of radii ρ and $\rho + \Delta\rho$ has base $\Delta S \approx \rho^2 \sin\phi\,\Delta\phi\,\Delta\theta$ and height $\Delta\rho$, so $\Delta V \approx \Delta\rho \cdot \rho\,\Delta\phi \cdot \rho\sin\phi\,\Delta\theta = \rho^2\sin\phi\,\Delta\rho\,\Delta\phi\,\Delta\theta$. □

The same element follows from the general change of variables in a triple integral. For new coordinates (u, v, w), the small region bounded by nearby level surfaces is approximately a parallelepiped, and

$$dV = \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| du\, dv\, dw, \qquad \frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix}.$$

For $(u, v, w) = (\rho, \phi, \theta)$ this determinant equals $\rho^2 \sin\phi$, but the geometric derivation above is quicker.

24.3 Setting up bounds

To set up $\iiint_D f\, \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$:

1. Fix ϕ and θ; letting ρ increase gives a ray from the origin. Integrate from the value of ρ where the ray enters D to the value where it leaves.
2. Fix θ and let ϕ vary; the rays form a fan. Integrate over the values of ϕ whose rays meet D.
3. Integrate over the values of θ whose fans meet D.

Example: *A cap of the unit ball.* The region of Lecture 23 inside the unit sphere above the plane $z = 1 - y$ becomes, after rotating the coordinate system, the part of the unit ball above the plane $z = 1/\sqrt{2}$ (the plane is at distance $1/\sqrt{2}$ from the origin). Along a ray, ρ runs from the plane, $\rho\cos\phi = 1/\sqrt{2}$, that is, $\rho = \frac{1}{\sqrt{2}}\sec\phi$, to the sphere $\rho = 1$. In a vertical slice $\theta = \text{const}$ (Figure 36, left), the edge of the region is where $z = r = 1/\sqrt{2}$, so $0 \le \phi \le \pi/4$. Thus

$$\text{vol} = \int_0^{2\pi}\int_0^{\pi/4}\int_{\sec\phi/\sqrt{2}}^{1} \rho^2\sin\phi\, d\rho\, d\phi\, d\theta.$$

The inner integral is $\frac{1}{3}\sin\phi - \frac{1}{6\sqrt{2}}\sec^3\phi\sin\phi$. Since $\int_0^{\pi/4}\sin\phi\, d\phi = 1 - \frac{1}{\sqrt{2}}$ and $\int_0^{\pi/4}\frac{\sin\phi}{\cos^3\phi}\, d\phi = \left[\frac{1}{2\cos^2\phi}\right]_0^{\pi/4} = \frac{1}{2}$,

$$\text{vol} = 2\pi\left(\frac{1}{3} - \frac{1}{3\sqrt{2}} - \frac{1}{12\sqrt{2}}\right) = \frac{2\pi}{3} - \frac{5\pi}{6\sqrt{2}}.$$

Example: *A solid of revolution.* Rotate the circle of radius 1 centered at $(y, z) = (1, 0)$ in the yz-plane about the z-axis, and let D be the part of the resulting solid in the first octant. In the yz-plane, for a point P on the circle, the triangle with vertices O, P, and $A = (2, 0)$ has a right angle at P (it is inscribed in a semicircle), and its angle at A equals ϕ, since both are complementary to the angle at O. So $\rho = |OP| = |OA|\sin\phi = 2\sin\phi$, and by rotation this is the equation of the whole surface. A ray enters D at $\rho = 0$ and leaves at $\rho = 2\sin\phi$; the first octant gives $0 \le \phi \le \pi/2$ and $0 \le \theta \le \pi/2$:

$$\text{vol}(D) = \int_0^{\pi/2}\int_0^{\pi/2}\int_0^{2\sin\phi} \rho^2\sin\phi\, d\rho\, d\phi\, d\theta.$$

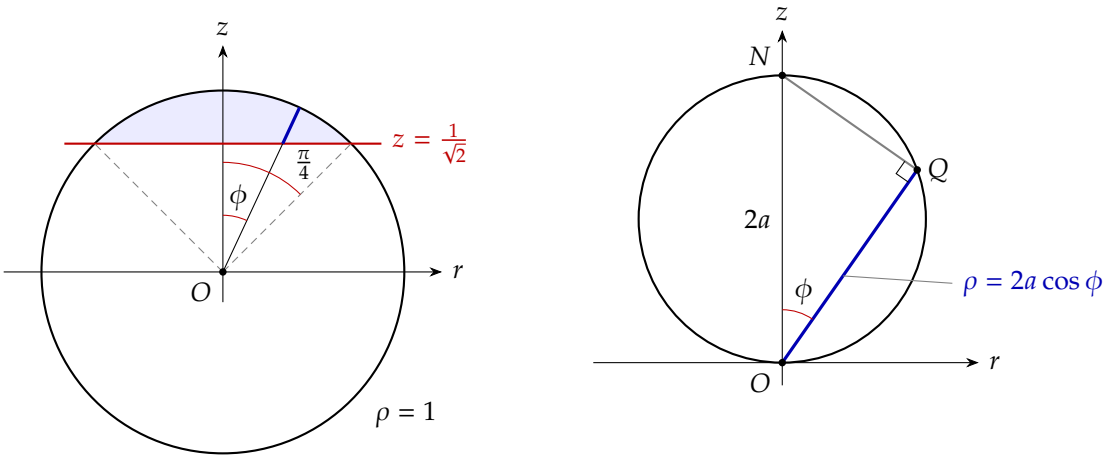


Figure 36: Vertical slices $\theta = \text{const}$. Left: the cap of the unit ball above $z = 1/\sqrt{2}$; a ray at angle ϕ enters at $\rho = \sec\phi/\sqrt{2}$ and leaves at $\rho = 1$, and the rays meeting the cap have $\phi \le \pi/4$. Right: the sphere of radius a tangent to the xy-plane at the origin; the right triangle ONQ inscribed in the circle gives $\rho = 2a\cos\phi$.

24.4 Gravitational attraction

A mass ΔM at (x, y, z) exerts on a mass m at the origin a force of magnitude $G\,\Delta M\,m/\rho^2$ in the direction $\langle x, y, z\rangle/\rho$, where G is the gravitational constant:

$$\mathbf{F} = \frac{G\,\Delta M\,m}{\rho^3}\langle x, y, z\rangle.$$

For a solid R with density δ, summing the contributions of the pieces $\Delta M = \delta\,\Delta V$ gives

$$\mathbf{F} = \iiint_R \frac{Gm\langle x, y, z\rangle}{\rho^3}\,\delta\,dV, \qquad \text{so} \qquad F_z = Gm\iiint_R \frac{z}{\rho^3}\,\delta\,dV,$$

and similarly for F_x and F_y.

General setup. Put the test mass at the origin and place the solid so that the z-axis is an axis of symmetry. By symmetry $\mathbf{F} = \langle 0, 0, F_z\rangle$, and in spherical coordinates, with $z = \rho\cos\phi$,

$$F_z = Gm\iiint_R \frac{\rho\cos\phi}{\rho^3}\,\delta\,\rho^2\sin\phi\,d\rho\,d\phi\,d\theta = Gm\iiint_R \delta\cos\phi\sin\phi\,d\rho\,d\phi\,d\theta.$$

Theorem 24.3 [Newton]. *The gravitational attraction of a spherical planet of uniform density δ on an exterior point mass is the same as that of a point mass M, the mass of the planet, at its center.*

Proof. We compute the attraction on a test mass on the surface. Let the planet have radius a, centered at $(0, 0, a)$ so that it is tangent to the xy-plane at the origin, where the mass m sits. In a vertical slice (Figure 36, right), a ray at angle ϕ from the origin meets the sphere again at a point Q; the triangle with vertices the two poles O, $N = (0, 0, 2a)$ and Q has its right angle at Q and hypotenuse $|ON| = 2a$, so the sphere is $\rho = 2a\cos\phi$. The planet lies above the xy-plane, so $0 \le \phi \le \pi/2$. Hence

$$\begin{aligned} F_z &= Gm\delta\int_0^{2\pi}\int_0^{\pi/2}\int_0^{2a\cos\phi} \cos\phi\sin\phi\,d\rho\,d\phi\,d\theta \\ &= 2\pi Gm\delta\int_0^{\pi/2} 2a\cos^2\phi\sin\phi\,d\phi = \frac{4}{3}\pi Gm\delta a. \end{aligned}$$

With $M = \frac{4}{3}\pi a^3\delta$ this is $F_z = GMm/a^2$, the attraction of a point mass M at the center, at distance a. □

The analogous statement fails for other solids: a uniform solid of mass M does not in general attract like a point mass M at its center of mass. For two equal point masses, a test mass on the axis between them is pulled toward the closer one, not toward the midpoint.

Example: *A half ball with variable density.* Let R be the upper half of the ball of radius a centered at the origin, with density $\delta = \sqrt{x^2 + y^2} = r = \rho\sin\phi$, attracting a unit mass at the origin. The solid

and density are symmetric about the z-axis, so

$$F_z = G\int_0^{2\pi}\int_0^{\pi/2}\int_0^{a} \rho\sin^2\phi\cos\phi\,d\rho\,d\phi\,d\theta = G\cdot 2\pi\cdot\frac{1}{3}\cdot\frac{a^2}{2} = \frac{\pi G a^2}{3}.$$

Example: *A spherical cap.* Slice a solid ball of radius $a\sqrt{2}$, density 1, by a plane at distance a from its center, and let V be the smaller piece. Find its attraction on a unit mass at the center. Center the ball at the origin with the cap along the positive z-axis. A ray enters V through the flat side $z = a$, that is, $\rho = a/\cos\phi$, and leaves through the sphere $\rho = a\sqrt{2}$; the rays meeting V have $0 \le \phi \le \pi/4$, where $a/\cos\phi = a\sqrt{2}$. Then

$$\begin{aligned} F_z &= G\int_0^{2\pi}\int_0^{\pi/4}\int_{a/\cos\phi}^{a\sqrt{2}} \sin\phi\cos\phi\,d\rho\,d\phi\,d\theta \\ &= 2\pi G a\int_0^{\pi/4}\left(\sqrt{2}-\frac{1}{\cos\phi}\right)\sin\phi\cos\phi\,d\phi = 2\pi G a\left(\frac{3\sqrt{2}}{4}-1\right), \end{aligned}$$

using $\int_0^{\pi/4}\sin\phi\cos\phi\,d\phi = \frac{1}{4}$ and $\int_0^{\pi/4}\sin\phi\,d\phi = 1-\frac{1}{\sqrt{2}}$.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 24, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 25: Vector fields in 3D; surface integrals and flux

In space, flux becomes a double integral over a surface. Setting one up requires two parameters for the surface and an expression for $\hat{\mathbf{n}}\, dS$ in terms of them.

25.1 Vector fields in space

A vector field in space assigns to each point a vector

$$\mathbf{F}(x, y, z) = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k},$$

where P, Q, R are functions of x, y, z. The field is continuous in a domain D if P, Q, R are, and continuously differentiable if all nine first partial derivatives $P_x, P_y, P_z, Q_x, \ldots, R_z$ exist and are continuous in D.

The two standard interpretations are force fields (gravitational, electric, magnetic) and velocity fields of a steady fluid flow, where the flow at each point does not change with time. For a flow with velocity $\mathbf{v}$ and density δ, the flow field $\mathbf{F} = \delta\mathbf{v}$ points in the direction of flow, and $|\mathbf{F}|$ is the mass transported per unit time per unit area across a small piece of plane perpendicular to the flow. Gradients of temperature or pressure give a third kind of example.

Example: *Standard fields.* Write $\rho = \sqrt{x^2 + y^2 + z^2}$.

- A unit positive charge at the origin produces a field directed radially outward with magnitude $1/\rho^2$. Since $\langle x, y, z\rangle$ is radial of length ρ,

  $$\mathbf{F} = \frac{\langle x, y, z\rangle}{\rho^3}.$$

  Gravitational attraction toward a mass at the origin is $\mathbf{F} = -c\,\langle x, y, z\rangle/\rho^3$.

- A fluid rotating about the z-axis with angular velocity ω (counterclockwise seen from above, by the right-hand rule) has velocity $\mathbf{v} = \omega(-y\,\mathbf{i} + x\,\mathbf{j})$, independent of z. For rotation about the y-axis the flow has no $\mathbf{j}$-component and is clockwise in the xz-plane when x and z are drawn as a right-handed pair, so $\mathbf{v} = \omega(z\,\mathbf{i} - x\,\mathbf{k})$.

- A gas streaming radially outward at constant speed from a source of constant strength at the origin must have density proportional to $1/\rho^2$, since the area of a sphere grows like ρ^2. Hence $\mathbf{F} = \delta\mathbf{v} = c\,\langle x, y, z\rangle/\rho^3$, the same shape as the point-charge field.

25.2 Surface integrals and flux

For a function f on a surface S, subdivide S into small pieces of area ΔS_i, choose a point (x_i, y_i, z_i) in each, and form $\sum_i f(x_i, y_i, z_i)\,\Delta S_i$. For a smooth surface of finite extent and reasonable subdivisions, these sums converge as the pieces shrink; the limit is the SURFACE INTEGRAL $\iint_S f\,dS$.

A surface is ORIENTED when it has two sides and one of them is designated positive; the unit normal $\hat{\mathbf{n}}$ is the one based on the positive side. There are always two choices of $\hat{\mathbf{n}}$, and the choice decides which direction of crossing counts positively. A CLOSED SURFACE, such as a sphere or a cube, encloses a region of space and is oriented by convention with $\hat{\mathbf{n}}$ pointing outward.

Definition 25.1 [FLUX]. The FLUX of a vector field $\mathbf{F}$ through an oriented surface S is

$$\iint_S \mathbf{F}\cdot\hat{\mathbf{n}}\,dS = \iint_S \mathbf{F}\cdot d\mathbf{S}, \qquad d\mathbf{S} = \hat{\mathbf{n}}\,dS.$$

In the plane, flux was a line integral across a curve; in space it is a double integral through a surface. The vector $d\mathbf{S}$ is often easier to compute directly than $\hat{\mathbf{n}}$ and dS separately.

If $\mathbf{F}$ is the velocity of a fluid flow, the flux is the amount of fluid crossing S per unit time. Over a small piece ΔS, the fluid crossing in unit time fills a parallelepiped with base ΔS and slanted side $\mathbf{F}$ (Figure 37). Its volume is base times height, $(\mathbf{F}\cdot\hat{\mathbf{n}})\,\Delta S$, and summing over pieces gives the flux. Flow along $\hat{\mathbf{n}}$ counts positively and flow in the opposite direction negatively. For a flow field $\delta\mathbf{v}$ the flux is the net mass transport rate. For a force field nothing flows, but the same integral is still called the flux.

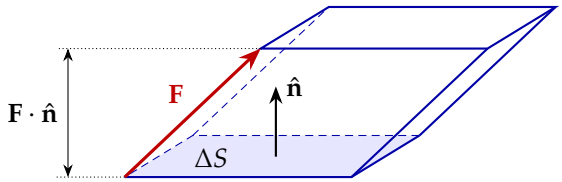


Figure 37: The fluid crossing a small piece ΔS in unit time fills a parallelepiped with base ΔS and side $\mathbf{F}$. Its height is the normal component $\mathbf{F}\cdot\hat{\mathbf{n}}$.

Example: *Two fields through a sphere.* Let S be the sphere of radius a centered at the origin, oriented outward, so $\hat{\mathbf{n}} = \frac{1}{a}\langle x, y, z\rangle$.

- For $\mathbf{F} = x\,\mathbf{i} + y\,\mathbf{j} + z\,\mathbf{k}$, $\mathbf{F}\cdot\hat{\mathbf{n}} = \frac{1}{a}(x^2+y^2+z^2) = a$ on S, so the flux is $\iint_S a\,dS = a\cdot 4\pi a^2 = 4\pi a^3$, with no integration needed.
- For $\mathbf{H} = z\,\mathbf{k}$, $\mathbf{H}\cdot\hat{\mathbf{n}} = z^2/a$. With $z = a\cos\phi$ and $dS = a^2\sin\phi\,d\phi\,d\theta$ (derived below),

$$\iint_S \frac{z^2}{a}\,dS = \int_0^{2\pi}\int_0^{\pi} a\cos^2\phi\; a^2\sin\phi\,d\phi\,d\theta = 2\pi a^3\int_0^{\pi}\cos^2\phi\sin\phi\,d\phi = \frac{4}{3}\pi a^3.$$

25.3 Setting up flux integrals

Sometimes a geometric argument gives the answer, as in the first example. In general, describe S with two parameters and express $\mathbf{F} \cdot \hat{\mathbf{n}}\, dS$ in terms of them; the integrand must then use only those parameters. For the basic surfaces:

- *Horizontal plane* $z = a$: use x, y; $\hat{\mathbf{n}} = \pm\mathbf{k}$ and $dS = dx\, dy$. Planes parallel to the xz- or yz-plane are similar.
- *Sphere of radius a centered at the origin*: use ϕ, θ and substitute $\rho = a$; $\hat{\mathbf{n}} = \frac{1}{a}\langle x, y, z\rangle$ and $dS = a^2 \sin\phi\, d\phi\, d\theta$.
- *Cylinder of radius a about the z-axis*: use z, θ and substitute $r = a$; $\hat{\mathbf{n}} = \frac{1}{a}\langle x, y, 0\rangle$ points horizontally away from the axis, and $dS = a\, dz\, d\theta$.
- *Graph* $z = f(x, y)$: use x, y; then $\hat{\mathbf{n}}\, dS = \langle -f_x, -f_y, 1\rangle\, dx\, dy$.

The cylinder is cut by horizontal circles and vertical lines into curved rectangles of height Δz and width $a\,\Delta\theta$, so $\Delta S \approx (\Delta z)(a\,\Delta\theta)$. For the sphere, the volume element factors as area times thickness: $dV = \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta = dS \cdot d\rho$. Dividing by $d\rho$ and setting $\rho = a$ gives $dS = a^2 \sin\phi\, d\phi\, d\theta$, the product of the side $a\, d\phi$ along a meridian and the side $a \sin\phi\, d\theta$ along a circle of latitude of radius $a \sin\phi$ (Figure 38).

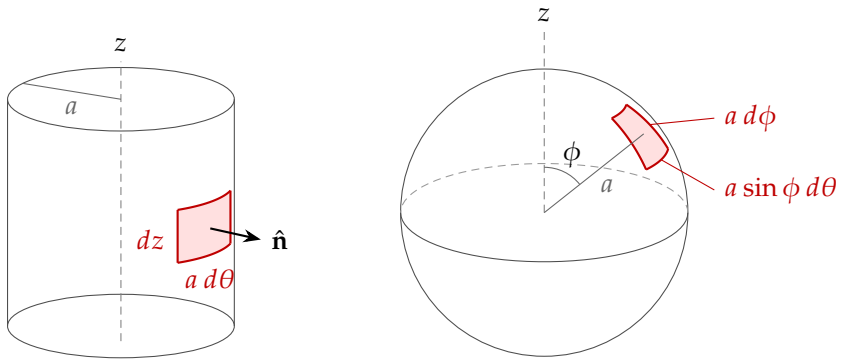


Figure 38: Area elements. Left: on the cylinder of radius a, $dS = (dz)(a\, d\theta)$ and $\hat{\mathbf{n}}$ points horizontally away from the axis. Right: on the sphere of radius a, $dS = (a\, d\phi)(a \sin\phi\, d\theta)$.

Example: *Flux through a quarter cylinder.* Find the outward flux of $\mathbf{F} = z\,\mathbf{i} + x\,\mathbf{j} + y\,\mathbf{k}$ through the part of the cylinder $x^2 + y^2 = a^2$ with $0 \le \theta \le \pi/2$ and $0 \le z \le h$. Here $\hat{\mathbf{n}} = \frac{1}{a}\langle x, y, 0\rangle$ and $dS = a\, dz\, d\theta$, so with $x = a\cos\theta$, $y = a\sin\theta$,

$$\mathbf{F} \cdot \hat{\mathbf{n}}\, dS = \frac{zx + xy}{a}\, a\, dz\, d\theta = (az\cos\theta + a^2 \sin\theta\cos\theta)\, dz\, d\theta.$$

The inner integral over $0 \le z \le h$ is $\frac{ah^2}{2}\cos\theta + a^2 h \sin\theta\cos\theta$, and then

$$\iint_S \mathbf{F}\cdot\hat{\mathbf{n}}\,dS = \left[\frac{ah^2}{2}\sin\theta + a^2 h\frac{\sin^2\theta}{2}\right]_0^{\pi/2} = \frac{ah}{2}(a+h).$$

Example: *Flux through an octant of a sphere.* Find the outward flux of $\mathbf{F} = xz\,\mathbf{i} + yz\,\mathbf{j} + z^2\,\mathbf{k}$ through the part of the sphere $x^2 + y^2 + z^2 = a^2$ in the first octant. With $\hat{\mathbf{n}} = \frac{1}{a}\langle x, y, z\rangle$,

$$\mathbf{F}\cdot\hat{\mathbf{n}} = \frac{z(x^2+y^2+z^2)}{a} = az = a^2\cos\phi,$$

and $dS = a^2\sin\phi\,d\phi\,d\theta$, so

$$\iint_S \mathbf{F}\cdot\hat{\mathbf{n}}\,dS = a^4\int_0^{\pi/2}\int_0^{\pi/2}\cos\phi\sin\phi\,d\phi\,d\theta = a^4\cdot\frac{\pi}{2}\cdot\frac{1}{2} = \frac{\pi a^4}{4}.$$

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 25, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 26: Divergence theorem

For a general surface it is simplest to compute the vector $d\mathbf{S} = \hat{\mathbf{n}}\, dS$ directly, as a cross product of two infinitesimal sides. For closed surfaces, the divergence theorem converts flux into a triple integral.

26.1 Flux through a graph

Let S be the graph of $z = f(x, y)$ over a region R of the xy-plane, and use x, y as parameters. The part of S above a small rectangle $\Delta x \times \Delta y$ is, in linear approximation, a parallelogram with vertices $(x, y, f(x, y))$, $(x + \Delta x, y, f(x + \Delta x, y))$, $(x, y + \Delta y, f(x, y + \Delta y))$, and a fourth. Since $f(x+\Delta x, y) \approx f + f_x \Delta x$ and $f(x, y+\Delta y) \approx f + f_y \Delta y$, its sides are $\mathbf{A} = \Delta x\, \langle 1, 0, f_x \rangle$ and $\mathbf{B} = \Delta y\, \langle 0, 1, f_y \rangle$ (Figure 39). The cross product has the normal direction and the parallelogram's area as its length:

$$\Delta\mathbf{S} = \mathbf{A} \times \mathbf{B} = \Delta x\, \Delta y \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & f_x \\ 0 & 1 & f_y \end{vmatrix} = \langle -f_x, -f_y, 1 \rangle\, \Delta x\, \Delta y.$$

Proposition 26.1 [Surface element of a graph]. *For $z = f(x, y)$ with the upward normal,*

$$\boxed{\hat{\mathbf{n}}\, dS = \langle -f_x, -f_y, 1 \rangle\, dx\, dy}, \qquad \hat{\mathbf{n}} = \frac{\langle -f_x, -f_y, 1 \rangle}{\sqrt{f_x^2 + f_y^2 + 1}}, \qquad dS = \sqrt{f_x^2 + f_y^2 + 1}\, dx\, dy.$$

For the downward normal, change the sign of $\hat{\mathbf{n}}\, dS$.

The factor $\sqrt{f_x^2 + f_y^2 + 1}$ converts the area dA of the projection to the xy-plane into the area dS on the surface. Neither $\langle -f_x, -f_y, 1 \rangle$ nor $dx\, dy$ alone is $\hat{\mathbf{n}}$ or dS; only the product is meaningful. After substituting, the integrand must be written in x, y only (replace z by $f(x, y)$), and the limits come from the projection R.

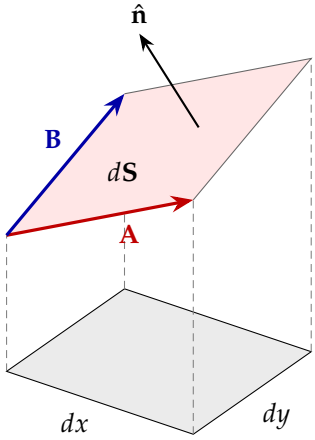


Figure 39: Above a rectangle $dx \times dy$ the graph is approximately the parallelogram spanned by $\mathbf{A} = \langle dx, 0, f_x\, dx\rangle$ and $\mathbf{B} = \langle 0, dy, f_y\, dy\rangle$; $d\mathbf{S} = \mathbf{A} \times \mathbf{B}$. Slopes are schematic.

Example: *A paraboloid.* Let S be the part of $z = x^2 + y^2$ above the unit disk, with the normal pointing up (into the paraboloid), and $\mathbf{F} = z\,\mathbf{k}$. Geometrically the flux should be positive. Here $\hat{\mathbf{n}}\, dS = \langle -2x, -2y, 1\rangle\, dx\, dy$, so

$$\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint z\, dx\, dy = \iint_{x^2+y^2\le 1} (x^2+y^2)\, dx\, dy = \int_0^{2\pi}\int_0^1 r^2\, r\, dr\, d\theta = \frac{\pi}{2}.$$

Example: *A slanted triangle.* The plane $2x - 2y + z = 1$ meets the coordinate axes at $x = \frac{1}{2}$, $y = -\frac{1}{2}$, $z = 1$; let S be the triangle cut off by the coordinate planes, oriented upward, and $\mathbf{F} = \langle x, y, z\rangle$. With $z = 1 - 2x + 2y$, $d\mathbf{S} = \langle 2, -2, 1\rangle\, dx\, dy$, and

$$\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint_R \big(2x - 2y + (1 - 2x + 2y)\big)\, dx\, dy = \iint_R 1\, dx\, dy = \frac{1}{8},$$

the area of the projection R, the triangle with vertices $(0,0)$, $(\frac{1}{2}, 0)$, $(0, -\frac{1}{2})$.

26.2 Parametric and implicit surfaces

If S is given parametrically by $\mathbf{r} = \mathbf{r}(u, v)$, a small piece corresponding to $\Delta u, \Delta v$ is approximately the parallelogram with sides $\mathbf{r}(u + \Delta u, v) - \mathbf{r}(u, v) \approx \mathbf{r}_u\, \Delta u$ and $\mathbf{r}_v\, \Delta v$. Hence

$$d\mathbf{S} = \pm \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v}\, du\, dv,$$

which contains all the previous formulas as special cases.

If S is the level surface $g(x, y, z) = 0$, then $\mathbf{N} = \nabla g$ is a (non-unit) normal vector, and $\hat{\mathbf{n}} = \pm\mathbf{N}/|\mathbf{N}|$. For a slanted plane $ax + by + cz = d$, $\mathbf{N} = \langle a, b, c\rangle$. To find dS, project to the xy-plane. If α is the angle between the surface element and the horizontal (equal to the angle between $\mathbf{N}$ and $\mathbf{k}$),

projection shrinks one direction by $|\cos\alpha|$ and preserves the other, so

$$\Delta A = |\cos\alpha|\,\Delta S = \frac{|\mathbf{N}\cdot\mathbf{k}|}{|\mathbf{N}|}\,\Delta S.$$

Proposition 26.2 [Surface element of a level surface]. *For $g(x,y,z)=0$ with $\mathbf{N}=\nabla g$ and $\mathbf{N}\cdot\mathbf{k}\neq 0$,*

$$dS = \frac{|\mathbf{N}|}{|\mathbf{N}\cdot\mathbf{k}|}\,dx\,dy, \qquad \boxed{\hat{\mathbf{n}}\,dS = \pm\frac{\mathbf{N}}{\mathbf{N}\cdot\mathbf{k}}\,dx\,dy},$$

with the sign chosen to match the orientation.

If S is vertical, $\mathbf{N}\cdot\mathbf{k}=0$ and S cannot be projected to the xy-plane; project to the xz- or yz-plane instead. For a graph, $g = z - f(x,y)$ gives $\mathbf{N}=\langle -f_x,-f_y,1\rangle$ and $\mathbf{N}\cdot\mathbf{k}=1$, recovering the earlier formula. Conversely, if $z=z(x,y)$ solves $g(x,y,z)=0$, differentiating with respect to x gives $g_x + g_z z_x = 0$, so $z_x = -g_x/g_z$, and similarly $z_y = -g_y/g_z$; substituting into $\langle -z_x,-z_y,1\rangle$ gives $\nabla g/g_z$ again.

Example: *A quarter ellipsoid.* For the part of $4x^2+y^2+4z^2=4$ in the first octant, oriented upward, and $\mathbf{F}=\langle x,y,z\rangle$: $\mathbf{N}=\langle 8x,2y,8z\rangle$, so $d\mathbf{S}=\frac{\langle 8x,2y,8z\rangle}{8z}\,dx\,dy$ and

$$\iint_S \mathbf{F}\cdot d\mathbf{S} = \iint \frac{8x^2+2y^2+8z^2}{8z}\,dx\,dy = \iint_R \frac{dx\,dy}{\sqrt{1-x^2-(y/2)^2}},$$

since $8x^2+2y^2+8z^2=8$ on S. Here R is the part of the ellipse $4x^2+y^2\le 4$ in the first quadrant. The substitution $u=y/2$ turns R into a quarter disk, and polar coordinates give $2\cdot\frac{\pi}{2}\int_0^1 \frac{r\,dr}{\sqrt{1-r^2}} = \pi$.

26.3 General surface integrals

The integral $\iint_S f\,dS$ has uses besides flux. Taking $f=1$ gives the surface area of S. For a thin shell with density δ per unit area, the mass is $\iint_S \delta\,dS$ and the center of mass has $\bar{x}=\frac{1}{\text{mass}}\iint_S x\,\delta\,dS$, and similarly for $\bar{y},\bar{z}$; for a charge density the same integral gives the total charge. The average value of f on S is $\frac{1}{\text{area}(S)}\iint_S f\,dS$. The area element is $|d\mathbf{S}|$:

$$dS = \sqrt{z_x^2+z_y^2+1}\,dx\,dy \quad \text{for } z=z(x,y), \qquad dS = \frac{|\nabla g|}{|g_z|}\,dx\,dy \quad \text{for } g(x,y,z)=c.$$

Example: *Average distance from the North Pole.* On a sphere of radius a with the pole at $(0,0,a)$, the distance along the surface from the pole to the point with angle ϕ is $a\phi$, measured along a meridian. Over the northern hemisphere,

$$\iint_S a\phi\,dS = \int_0^{2\pi}\int_0^{\pi/2} a\phi\,a^2\sin\phi\,d\phi\,d\theta = 2\pi a^3\int_0^{\pi/2}\phi\sin\phi\,d\phi = 2\pi a^3,$$

by parts. The hemisphere has area $2\pi a^2$, so the average distance is exactly a.

Example: *Area of a saddle.* The piece of $z = xy$ above the unit disk has area

$$\iint_R \sqrt{y^2 + x^2 + 1}\, dx\, dy = \int_0^{2\pi} \int_0^1 \sqrt{r^2+1}\, r\, dr\, d\theta = 2\pi \cdot \frac{1}{3}(r^2+1)^{3/2}\Big|_0^1 = \frac{2\pi}{3}\left(2\sqrt{2} - 1\right).$$

26.4 The divergence theorem

A CLOSED SURFACE here is one connected piece that does not intersect itself and encloses a single finite region D, its INTERIOR; S is the BOUNDARY of D. A sphere, a cube, and a torus are closed surfaces; a plane, a sphere with a point removed, an infinite cylinder, and a surface crossing itself are not. A closed surface is oriented with $\hat{\mathbf{n}}$ pointing out of D.

Definition 26.3 [DIVERGENCE]. The DIVERGENCE of $\mathbf{F} = P\,\mathbf{i} + Q\,\mathbf{j} + R\,\mathbf{k}$ is the scalar function $\operatorname{div}\mathbf{F} = P_x + Q_y + R_z$.

Theorem 26.4 [DIVERGENCE THEOREM]. *Let S be a closed surface bounding a region D, with $\hat{\mathbf{n}}$ pointing outward, and let $\mathbf{F}$ be continuously differentiable on a domain containing D. Then*

$$\boxed{\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_D \operatorname{div}\mathbf{F}\, dV}.$$

This is the three-dimensional analogue of the normal form of Green's theorem, and is also called the Gauss–Green or Gauss theorem. The volume element is written dV because the triple integral is often computed in cylindrical or spherical coordinates.

Example: *Checks on a sphere of radius a.*

- For $\mathbf{F} = z\,\mathbf{k}$, $\operatorname{div}\mathbf{F} = 1$, so the outward flux is $\operatorname{vol}(D) = \frac{4}{3}\pi a^3$, as computed directly in Lecture 25.
- For $\mathbf{F} = \langle x, y, z\rangle$, $\operatorname{div}\mathbf{F} = 3$, and $\iiint_D 3\, dV = 4\pi a^3$, which matches the direct calculation.
- For $\mathbf{F} = x^3\,\mathbf{i} + y^3\,\mathbf{j} + z^3\,\mathbf{k}$, $\operatorname{div}\mathbf{F} = 3\rho^2$. Slicing D into thin spherical shells of volume $4\pi\rho^2\, d\rho$,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = 3\int_0^a \rho^2 \cdot 4\pi\rho^2\, d\rho = \frac{12\pi a^5}{5}.$$

Example: *Replacing a surface.* Let S_1 be the part of $z = 1 - x^2 - y^2$ above the xy-plane, oriented upward, and $\mathbf{F} = yz\,\mathbf{i} + xz\,\mathbf{j} + xy\,\mathbf{k}$. Then $\operatorname{div}\mathbf{F} = 0$. Let S_2 be the unit disk in the xy-plane with $\hat{\mathbf{n}} = \mathbf{k}$, and S_2' the same disk oriented downward. Then $S_1 \cup S_2'$ is closed with outward normal (Figure 40), and the divergence theorem gives

$$0 = \iint_{S_1} \mathbf{F}\cdot d\mathbf{S} - \iint_{S_2} \mathbf{F}\cdot d\mathbf{S}, \qquad \text{so} \qquad \iint_{S_1} \mathbf{F}\cdot d\mathbf{S} = \iint_{S_2} \mathbf{F}\cdot\mathbf{k}\, dS = \iint_{S_2} xy\, dx\, dy = 0 \quad \text{(by symmetry)}.$$

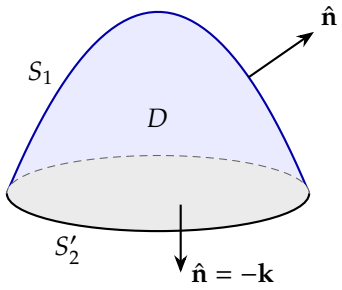


Figure 40: The cap S_1 of $z = 1 - x^2 - y^2$ and the downward-oriented disk S_2' together bound the region D, with outward normals.

26.5 Divergence as source rate

Think of $\mathbf{F}$ as a flow field. The left side of the divergence theorem is the net rate at which fluid leaves D through S. The right side is the total SOURCE RATE in D, because div $\mathbf{F}$ at a point is the rate per unit volume at which fluid is added there (negative for a sink):

$$\text{flux out of } S = \text{total sources in } D.$$

To see why, take a small box with corner $P_0 = (x_0, y_0, z_0)$ and edges $\Delta x, \Delta y, \Delta z$ parallel to the axes. On the top face $\hat{\mathbf{n}} = \mathbf{k}$ and on the bottom face $\hat{\mathbf{n}} = -\mathbf{k}$, so the net flux through these two faces is approximately

$$\big[R(x_0, y_0, z_0 + \Delta z) - R(x_0, y_0, z_0)\big]\Delta x\, \Delta y \approx \left(\frac{\partial R}{\partial z}\right)_0 \Delta x\, \Delta y\, \Delta z.$$

The two side faces contribute $(\partial Q/\partial y)_0\, \Delta V$ and the front and back faces $(\partial P/\partial x)_0\, \Delta V$. The source rate for the box equals its net outward flux, approximately $(\operatorname{div}\mathbf{F})_0\, \Delta V$, and dividing by the volume and letting the box shrink gives

$$\text{source rate at } P_0 = \lim_{\text{box}\to 0} \frac{\text{flux out of box}}{\text{volume of box}} = (\operatorname{div}\mathbf{F})_0.$$

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 26, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 27: Divergence theorem (cont.): applications and proof

27.1 The del operator

The divergence theorem reads $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_D \operatorname{div} \mathbf{F}\, dV$ for a closed surface S bounding D, oriented outward. Its integrand is conveniently written with the symbolic vector

$$\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle,$$

called del, where the "product" of $\partial/\partial x$ with a function P means $\partial P/\partial x$. Then

$$\nabla f = \langle f_x, f_y, f_z \rangle \text{ (the gradient)}, \qquad \nabla \cdot \mathbf{F} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle \cdot \langle P, Q, R \rangle = P_x + Q_y + R_z = \operatorname{div} \mathbf{F}.$$

27.2 Physical interpretation

Let $\mathbf{F}$ be the velocity field of an incompressible fluid, so that a given mass of fluid always occupies the same volume. Then $\operatorname{div} \mathbf{F}$ is the source rate, the flux generated per unit volume, and

$$\iiint_D \operatorname{div} \mathbf{F}\, dV = \iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS$$

says that the net amount of fluid leaving D per unit time equals the total amount of sources minus sinks in D.

27.3 Proof of the divergence theorem

Both sides are additive in $\mathbf{F} = \langle P, Q, R \rangle$, so it is enough to prove separately

$$\iint_S R\, \mathbf{k} \cdot d\mathbf{S} = \iiint_D R_z\, dV,$$

and the analogous identities for $P\,\mathbf{i}$ with P_x and $Q\,\mathbf{j}$ with Q_y.

Proof. First suppose D is vertically simple: it is described by $z_1(x, y) \le z \le z_2(x, y)$ for (x, y) in a region U of the xy-plane, so its top and bottom surfaces are graphs (Figure 41). Vertical

sides are allowed, as for a cylinder. Integrating first in z,

$$\iiint_D R_z \, dV = \iint_U \Big(\int_{z_1(x,y)}^{z_2(x,y)} R_z \, dz \Big) dx \, dy = \iint_U \big(R(x,y,z_2(x,y)) - R(x,y,z_1(x,y)) \big) \, dx \, dy.$$

Now compute the flux of $R\,\mathbf{k}$ through the three parts of S:

- On the top, oriented upward, $d\mathbf{S} = \langle -\partial z_2/\partial x, -\partial z_2/\partial y, 1\rangle \, dx \, dy$, so $\iint_{\text{top}} R\,\mathbf{k} \cdot d\mathbf{S} = \iint_U R(x,y,z_2(x,y)) \, dx \, dy$.
- On the bottom, oriented downward, $d\mathbf{S} = -\langle -\partial z_1/\partial x, -\partial z_1/\partial y, 1\rangle \, dx \, dy$, so $\iint_{\text{bottom}} R\,\mathbf{k} \cdot d\mathbf{S} = -\iint_U R(x,y,z_1(x,y)) \, dx \, dy$.
- On the vertical sides, $\hat{\mathbf{n}}$ is horizontal and $R\,\mathbf{k}$ is vertical, so the flux is 0.

The sum is the double integral above. For a general region, decompose D into vertically simple pieces (a solid torus, for example, can be cut into such pieces). The triple integrals over the pieces add up to the triple integral over D. The flux integrals add up to the flux through S, because each internal boundary is counted twice with opposite normals and cancels. The same argument with regions simple in the x- or y-direction proves the identities for $P\,\mathbf{i}$ and $Q\,\mathbf{j}$, and adding the three gives the theorem. □

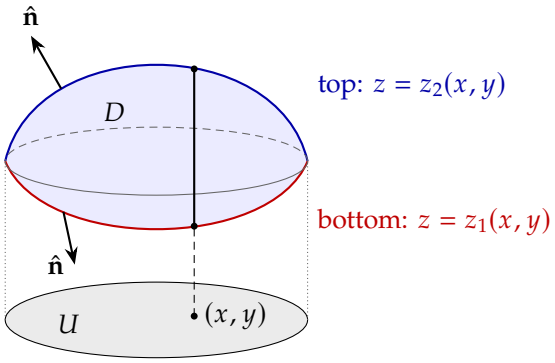


Figure 41: A vertically simple region D (schematic): every vertical line over U enters D through the bottom surface $z = z_1(x,y)$, where $\hat{\mathbf{n}}$ points down, and leaves through the top surface $z = z_2(x,y)$, where $\hat{\mathbf{n}}$ points up.

27.4 The diffusion equation

The DIFFUSION EQUATION governs the spread of smoke in still air or of dye in a solution:

$$\boxed{\frac{\partial u}{\partial t} = k\nabla^2 u = k\Big(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\Big)},$$

where $u(x,y,z,t)$ is the concentration and $k > 0$ a constant. With u a temperature, the same equation is the HEAT EQUATION. Let $\mathbf{F}$ be the flow of smoke. The equation combines two inputs.

1. Physics: $\mathbf{F} = -k\nabla u$. Smoke flows from high to low concentration, faster where the concentration changes more abruptly.
2. Conservation: if D is bounded by a closed surface S, the flux out of D is minus the rate of change of the amount inside,

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS = -\frac{d}{dt} \iiint_D u\, dV.$$

Differentiating under the integral sign, the right side is $-\iiint_D \frac{\partial u}{\partial t}\, dV$; this is justified by viewing the integral as a limit of sums $\sum u(x_i, y_i, z_i, t)\, \Delta V_i$, whose derivative is the sum of the derivatives. By the divergence theorem the left side is $\iiint_D \operatorname{div} \mathbf{F}\, dV$. Dividing by $\operatorname{vol}(D)$,

$$-\frac{1}{\operatorname{vol}(D)} \iiint_D \frac{\partial u}{\partial t}\, dV = \frac{1}{\operatorname{vol}(D)} \iiint_D \operatorname{div} \mathbf{F}\, dV$$

for every region D: the two functions have the same average over any region. Shrinking D to a point gives $\partial u/\partial t = -\operatorname{div} \mathbf{F}$, and combining with the first input,

$$\frac{\partial u}{\partial t} = -\operatorname{div} \mathbf{F} = k \operatorname{div}(\nabla u) = k\nabla^2 u.$$

27.5 Partial differential equations of physics

A PARTIAL DIFFERENTIAL EQUATION (PDE) is an equation involving an unknown function of several variables and some of its partial derivatives; a solution is a differentiable function satisfying it. For example, every $w = x^n y^n$ solves $x\, w_x - y\, w_y = 0$. A PDE typically has far too many solutions to list, and the problem is to find the one that satisfies extra BOUNDARY CONDITIONS or INITIAL CONDITIONS. Three equations are basic.

- The LAPLACE EQUATION $\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} = 0$ is satisfied by every steady-state temperature distribution $w = T(x, y, z)$ (one that does not change with time). It is also satisfied by the gravitational potential of a distribution of masses in any region free of mass, and by the electrostatic potential in any region free of charge. The potential ϕ is the work done against the field to carry a unit test mass or charge from a fixed reference point to (x, y, z), and the field is recovered as $\mathbf{F} = -\nabla\phi$. The two-dimensional version, which drops the z term, describes steady temperatures in a flat plate with insulated faces. With heat sources and sinks that do not change over time, the temperature satisfies the POISSON EQUATION $\nabla^2 w = f(x, y, z)$ for a given f.
- The WAVE EQUATION, in one and two space dimensions,

$$\frac{\partial^2 w}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 w}{\partial t^2}, \qquad \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} = \frac{1}{c^2}\frac{\partial^2 w}{\partial t^2},$$

  where c is the speed of the wave in the medium. For each time t_0, the solution $w(x, t_0)$ or

$w(x, y, t_0)$ gives the shape of the wave.

- The HEAT OR DIFFUSION EQUATION,

$$\frac{\partial^2 w}{\partial x^2} = \frac{1}{a^2}\frac{\partial w}{\partial t}, \qquad \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} = \frac{1}{a^2}\frac{\partial w}{\partial t},$$

  the case $k = a^2$ of the equation derived above. The right side has only a first time derivative, which gives it an entirely different character from the wave equation. Here w is the temperature in a rod or plate with insulated sides or faces, or the concentration of a substance diffusing along a tube. In a steady state, $\partial w/\partial t = 0$ and the heat equation becomes the Laplace equation.

All of these equations are second order: they involve no derivatives beyond the second.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 27, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 28: Line integrals in space, curl, exactness and potentials

28.1 Line integrals in space

A curve C in space is given by a vector function $\mathbf{r}(t) = x(t)\,\mathbf{i} + y(t)\,\mathbf{j} + z(t)\,\mathbf{k}$. The curves used here are finite, connected, piecewise smooth, and oriented (a positive direction is chosen); C is closed if it returns to its starting point. The velocity is $\mathbf{v} = d\mathbf{r}/dt$, with $|\mathbf{v}| = |ds/dt|$ where s is arclength, and the unit tangent in the positive direction is $\mathbf{T} = d\mathbf{r}/ds$.

For a force field $\mathbf{F} = \langle P, Q, R\rangle$ and $d\mathbf{r} = \langle dx, dy, dz\rangle$, the work done by $\mathbf{F}$ along C is the line integral

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C P\,dx + Q\,dy + R\,dz = \int_{t_0}^{t_1} \mathbf{F}\cdot\frac{d\mathbf{r}}{dt}\,dt = \int_C \mathbf{F}\cdot\mathbf{T}\,ds,$$

the integral of the tangential component of $\mathbf{F}$. It is the limit of sums $\sum_k \mathbf{F}\cdot\Delta\mathbf{r}_k$ over inscribed polygons. To evaluate it, express x, y, z in terms of a single parameter: a curve has one degree of freedom.

Example: *Two paths, one answer.* Let $\mathbf{F} = \langle yz, xz, xy\rangle$ and let C be $x = t^3$, $y = t^2$, $z = t$, $0 \le t \le 1$. Then $dx = 3t^2\,dt$, $dy = 2t\,dt$, $dz = dt$, and

$$\int_C yz\,dx + xz\,dy + xy\,dz = \int_0^1 (3t^5 + 2t^5 + t^5)\,dt = \int_0^1 6t^5\,dt = 1.$$

Along the broken path C' from $(0,0,0)$ to $(1,0,0)$ to $(1,1,0)$ to $(1,1,1)$: in the xy-plane $z = 0$, so $\mathbf{F} = xy\,\mathbf{k}$ is perpendicular to $d\mathbf{r}$ and does no work. On the last segment $x = y = 1$, $dx = dy = 0$, $\mathbf{F} = \langle z, z, 1\rangle$ and $d\mathbf{r} = \langle 0, 0, dz\rangle$, giving $\int_0^1 dz = 1$. The answers agree because $\mathbf{F} = \nabla(xyz)$.

Example: *Two paths, two answers.* For $\mathbf{F} = y\,\mathbf{i} + z\,\mathbf{j} + x\,\mathbf{k}$ from $(1,1,1)$ to $(2,4,8)$: along the segment $x = 1 + t$, $y = 1 + 3t$, $z = 1 + 7t$, $0 \le t \le 1$,

$$\int_C y\,dx + z\,dy + x\,dz = \int_0^1 \big[(1+3t) + 3(1+7t) + 7(1+t)\big]\,dt = \int_0^1 (11 + 31t)\,dt = 26.5.$$

Along the twisted cubic $x = t$, $y = t^2$, $z = t^3$, $1 \le t \le 2$,

$$\int_1^2 \big(t^2 + t^3\cdot 2t + t\cdot 3t^2\big)\,dt = \Big[\frac{t^3}{3} + \frac{3t^4}{4} + \frac{2t^5}{5}\Big]_1^2 = \frac{1559}{60} \approx 25.98.$$

So for this field the line integral depends on the path.

28.2 Gradient fields and path independence

The line integral of $\mathbf{F}$ is PATH-INDEPENDENT in a region D if $\int_C \mathbf{F}\cdot d\mathbf{r}$ depends only on the endpoints of C for curves in D; equivalently, $\oint_C \mathbf{F}\cdot d\mathbf{r} = 0$ for every closed curve C in D. A field of the form ∇f is a GRADIENT FIELD.

Theorem 28.1 [FUNDAMENTAL THEOREMS FOR LINE INTEGRALS]. *1. If f is continuously differentiable in D, then for any curve in D from P_0 to P_1, $\int_{P_0}^{P_1} \nabla f\cdot d\mathbf{r} = f(P_1) - f(P_0)$.*

2. If $\mathbf{F}$ is continuous and its line integral is path-independent in D, then $f(x,y,z) = \int_{(x_0,y_0,z_0)}^{(x,y,z)} \mathbf{F}\cdot d\mathbf{r}$ satisfies $\nabla f = \mathbf{F}$ in D.

The first follows from the chain rule, $\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f\cdot \mathbf{r}'(t)$, and the one-variable fundamental theorem; the second is proved as in the plane. Together they show that in D

$$\mathbf{F} = \nabla f \iff \int_P^Q \mathbf{F}\cdot d\mathbf{r} \text{ is path-independent} \iff \oint_C \mathbf{F}\cdot d\mathbf{r} = 0 \text{ for every closed } C.$$

Such force fields are called CONSERVATIVE, since no work is done around closed paths. Then f is the (mathematical) potential; the physical potential is $-f$.

Example: *No integration needed.* For $f = (x+y^2)z$, $\mathbf{F} = \nabla f = z\,\mathbf{i} + 2yz\,\mathbf{j} + (x+y^2)\,\mathbf{k}$. The helix $x = \cos t$, $y = \sin t$, $z = t$, $0 \le t \le \pi$, runs from $(1,0,0)$ to $(-1,0,\pi)$, so $\int_C \mathbf{F}\cdot d\mathbf{r} = (x+y^2)z\big|_{(1,0,0)}^{(-1,0,\pi)} = -\pi$.

28.3 The criterion for gradient fields

If $\mathbf{F} = \langle P, Q, R\rangle = \langle f_x, f_y, f_z\rangle$ with $\mathbf{F}$ continuously differentiable, then $f_{xy} = f_{yx}$, $f_{xz} = f_{zx}$, $f_{yz} = f_{zy}$, so

$$P_y = Q_x, \qquad P_z = R_x, \qquad Q_z = R_y.$$

Theorem 28.2 [CRITERION FOR GRADIENT FIELDS]. *Let $\mathbf{F} = \langle P, Q, R\rangle$ be continuously differentiable on all of space (or on a simply connected region). Then $\mathbf{F}$ is a gradient field if and only if $P_y = Q_x$, $P_z = R_x$, and $Q_z = R_y$.*

The forward direction is the computation above. The converse needs a condition on the domain and is proved with Stokes' theorem.

Example: *Finding a potential by successive integration.* For which a, b is $\mathbf{F} = axy\,\mathbf{i} + (x^2+z^3)\,\mathbf{j} + (byz^2 - 4z^3)\,\mathbf{k}$ a gradient field? $P_y = ax$ and $Q_x = 2x$ force $a = 2$; $P_z = 0 = R_x$; $Q_z = 3z^2$ and $R_y = bz^2$ force

$b = 3$. Now solve $f_x = 2xy$, $f_y = x^2 + z^3$, $f_z = 3yz^2 - 4z^3$ one equation at a time:

$$\begin{aligned} f_x = 2xy &\;\Rightarrow\; f = x^2y + g(y,z), \\ f_y = x^2 + g_y = x^2 + z^3 &\;\Rightarrow\; g_y = z^3 \;\Rightarrow\; g = yz^3 + h(z), \\ f_z = 3yz^2 + h'(z) = 3yz^2 - 4z^3 &\;\Rightarrow\; h'(z) = -4z^3 \;\Rightarrow\; h = -z^4 + c. \end{aligned}$$

Hence $f = x^2y + yz^3 - z^4 + c$. Each integration constant depends on the variables not yet used.

The other method uses the second fundamental theorem: $f(x_1, y_1, z_1) = f(0,0,0) + \int_{(0,0,0)}^{(x_1,y_1,z_1)} \mathbf{F} \cdot d\mathbf{r}$ along a convenient path, such as three segments parallel to the axes (Figure 42).

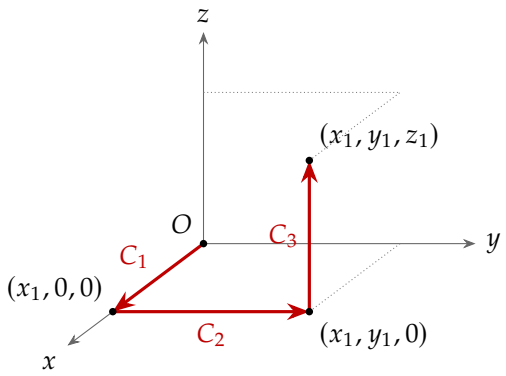


Figure 42: A path of three segments parallel to the axes, from the origin to (x_1, y_1, z_1): on C_1 only x varies, on C_2 only y, on C_3 only z.

Example: *Finding a potential by a line integral.* The field $\mathbf{F} = y\,\mathbf{i} + (x + cyz)\,\mathbf{j} + (y^2 + z^2)\,\mathbf{k}$ has $P_y = 1 = Q_x$, $P_z = 0 = R_x$, and $Q_z = cy$, $R_y = 2y$, so it is a gradient field exactly when $c = 2$. Along C_1 ($y = z = 0$, $dy = dz = 0$) the integrand is 0; along C_2 ($x = x_1$, $z = 0$) it is $(x_1 + 2y \cdot 0)\,dy$; along C_3 ($x = x_1$, $y = y_1$) it is $(y_1^2 + z^2)\,dz$. So

$$f(x_1, y_1, z_1) = 0 + \int_0^{y_1} x_1\,dy + \int_0^{z_1} (y_1^2 + z^2)\,dz = x_1y_1 + y_1^2z_1 + \tfrac{1}{3}z_1^3,$$

and, dropping subscripts, $f = xy + y^2z + \frac{1}{3}z^3 + c$. One checks $\nabla f = \mathbf{F}$; successive integration gives the same result.

In the language of differentials, $P\,dx + Q\,dy + R\,dz$ is EXACT in D if it equals $df = f_x\,dx + f_y\,dy + f_z\,dz$ for some f in D. The criterion reads: exactness implies $R_y = Q_z$, $P_z = R_x$, $Q_x = P_y$, and the converse holds when D is all of space.

28.4 Curl

The CURL measures how much $\mathbf{F}$ fails to be conservative:

$$\boxed{\operatorname{curl}\mathbf{F} = (R_y - Q_z)\,\mathbf{i} + (P_z - R_x)\,\mathbf{j} + (Q_x - P_y)\,\mathbf{k}} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix} = \nabla \times \mathbf{F}.$$

The determinant is symbolic, expanded along the first row with "$\frac{\partial}{\partial x}$ times Q" meaning Q_x. The criterion becomes: $\mathbf{F} = \nabla f$ implies $\operatorname{curl}\mathbf{F} = \mathbf{0}$, with the converse on all of space. For a plane field $\mathbf{F} = P(x, y)\,\mathbf{i} + Q(x, y)\,\mathbf{j}$, viewed in space, $\operatorname{curl}\mathbf{F} = (Q_x - P_y)\,\mathbf{k}$; its $\mathbf{k}$-component is the scalar curl used in the plane.

In del notation, $\operatorname{grad} f = \nabla f$, $\operatorname{div}\mathbf{F} = \nabla \cdot \mathbf{F}$ (a scalar), $\operatorname{curl}\mathbf{F} = \nabla \times \mathbf{F}$ (a vector), and the LAPLACIAN is $\nabla^2 f = \nabla \cdot \nabla f = f_{xx} + f_{yy} + f_{zz}$, so Laplace's equation is $\nabla^2 f = 0$. Two identities hold for twice continuously differentiable f and $\mathbf{F}$:

$$\boxed{\nabla \times \nabla f = \mathbf{0}}, \qquad \boxed{\nabla \cdot (\nabla \times \mathbf{F}) = 0}.$$

The first is the criterion above. For the second, $(R_y - Q_z)_x + (P_z - R_x)_y + (Q_x - P_y)_z = 0$ because the mixed partials cancel in pairs. The notation suggests both, since $\mathbf{A} \times \mathbf{A} = \mathbf{0}$ and $\mathbf{A} \cdot (\mathbf{A} \times \mathbf{F}) = 0$ for ordinary vectors.

28.5 Curl and rotation

For a velocity field, curl measures angular velocity. Rotation about the z-axis with constant angular velocity ω, with trajectories circles centered on the axis, is $\mathbf{v} = \langle -\omega y, \omega x, 0\rangle$, and

$$\nabla \times \mathbf{v} = 0\,\mathbf{i} + 0\,\mathbf{j} + (\omega + \omega)\,\mathbf{k} = 2\omega\,\mathbf{k}.$$

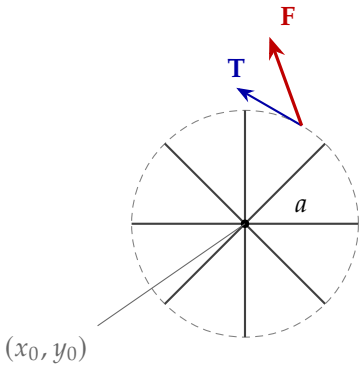
The length of the curl is twice the angular velocity and its direction is the axis of rotation. A general motion can be complicated, but the curl picks out its rotation component.

For a plane flow this can be seen with a small paddle wheel of radius a centered at (x_0, y_0) with vertical axis (Figure 43). A single blade would move at the tangential velocity $\mathbf{F} \cdot \mathbf{T}$, which varies around the circle C; with many blades the wheel turns at the average tangential velocity. By Green's theorem, with R the disk inside C,

$$\text{speed} = \frac{1}{2\pi a} \oint_C \mathbf{F} \cdot \mathbf{T}\, ds = \frac{1}{2\pi a} \iint_R \operatorname{curl}\mathbf{F}\, dA \approx \frac{1}{2\pi a} (\operatorname{curl}\mathbf{F})_0\, \pi a^2 = \frac{a}{2} (\operatorname{curl}\mathbf{F})_0,$$

since $\operatorname{curl}\mathbf{F}$ is nearly constant on a small disk. The angular velocity is $\omega_0 = \text{speed}/a \approx \frac{1}{2}(\operatorname{curl}\mathbf{F})_0$, with equality in the limit $a \to 0$. So the curl is twice the angular velocity of an infinitesimal paddle

wheel: it measures the VORTICITY of the flow. For $\mathbf{F} = x\,\mathbf{i} + y\,\mathbf{j}$ the curl is 0: the radial, symmetric flow favors no direction of spin. For $\omega(-y\,\mathbf{i} + x\,\mathbf{j})$ the curl is 2ω at every point, not only at the center.



C: circle of radius a

Figure 43: Top view of a paddle wheel of radius a in a plane flow (schematic). A blade tip moves with the tangential component $\mathbf{F} \cdot \mathbf{T}$ of the flow.

In space, the component of the flow along the axis of a paddle wheel has no effect on it. For a wheel in a plane with unit normal $\hat{\mathbf{n}}$, $(\operatorname{curl} \mathbf{F}) \cdot \hat{\mathbf{n}}$ is twice its angular velocity.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 28, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 29: Stokes' theorem

Stokes' theorem is the three-dimensional analogue of the work form of Green's theorem, in the same sense that the divergence theorem is the analogue of the flux form. Recall

$$\operatorname{curl}\mathbf{F} = \nabla\times\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix} = \langle R_y - Q_z,\ P_z - R_x,\ Q_x - P_y\rangle.$$

29.1 Statement and orientation

Theorem 29.1 [Stokes' theorem]. *Let C be a closed curve and S any oriented (two-sided) surface bounded by C, piecewise smooth and of finite area, with C and S compatibly oriented. If $\mathbf{F}$ is continuously differentiable on a domain containing S, then*

$$\boxed{\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_S (\nabla\times\mathbf{F})\cdot\hat{\mathbf{n}}\,dS}.$$

Compatible orientation means: walking along C in the positive direction with S on your left, your head points along $\hat{\mathbf{n}}$ (Figure 44). Equivalently, by the right-hand rule, if the thumb points along C (the 1-dimensional object) and the index finger toward S (2-dimensional), the middle finger points along $\hat{\mathbf{n}}$ (3-dimensional). Reversing the orientation changes the sign of both sides. Since S is not closed, there is no standard choice of $\hat{\mathbf{n}}$; a choice of either orientation determines the other.

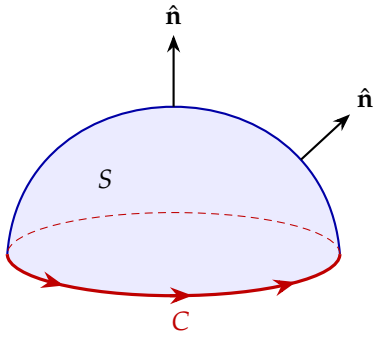


Figure 44: Compatible orientations: C runs counterclockwise seen from above, so walking along C with S on the left, the head points along the outward normal $\hat{\mathbf{n}}$ of the cap S.

Example: *Stokes and Green.* If S is a region in the xy-plane bounded by C counterclockwise, then $\hat{\mathbf{n}} = \mathbf{k}$, $dS = dx\,dy$, and for any $\mathbf{F}$, $\oint_C \mathbf{F}\cdot d\mathbf{r} = \oint_C P\,dx + Q\,dy$ (since $dz = 0$) while $(\nabla\times\mathbf{F})\cdot\mathbf{k} = Q_x - P_y$. So Stokes' theorem says $\oint_C P\,dx + Q\,dy = \iint_S (Q_x - P_y)\,dx\,dy$, which is Green's theorem.

The left side depends only on C, so the right side may be computed on *any* surface bounded by C:

if C is a circle, S could be a disk, a hemisphere, or a cone. Consistently, if $\nabla \times \mathbf{F} = \mathbf{0}$ the right side is 0, as it must be when $\mathbf{F}$ is a gradient field.

29.2 Why the theorem holds

The argument proceeds in three steps.

1. If C and S lie in the xy-plane, the statement is Green's theorem.
2. If C and S lie in any plane, it again reduces to Green's theorem in that plane. Work, flux, and curl can be defined geometrically, independently of the coordinate system, so they are invariant under rotations of space. Equivalently, choose coordinates (u, v, w) adapted to the plane; in them, the expressions for work, curl, and flux are the familiar ones with x, y, z replaced by u, v, w.
3. In general, decompose S into small, nearly flat pieces. On each, work around the boundary approximately equals the flux of the curl, by step 2. Adding up, the line integrals over the inner boundaries cancel (each is traversed twice in opposite directions), leaving the line integral over C, while the fluxes add up to the flux through S.

For a direct proof, it suffices to treat $\mathbf{F} = R(x, y, z)\,\mathbf{k}$; the cases $P\,\mathbf{i}$ and $Q\,\mathbf{j}$ are the same with the roles of the coordinates permuted, and the general case follows by adding. Write R_1, R_2, R_3 for the partial derivatives of R with respect to its first, second, and third variables, treated as independent.

Proof. Assume S is the graph of $z = f(x, y)$ over a region A of the xy-plane, with $\hat{\mathbf{n}}$ pointing upward. Let C' be the boundary of A, traversed counterclockwise, and parametrize it as $x = x(t)$, $y = y(t)$; then C is $x(t), y(t), z = f(x(t), y(t))$. The claim is $\oint_C R\,dz = \iint_S \operatorname{curl}(R\,\mathbf{k}) \cdot d\mathbf{S}$.

Line integral. By the chain rule $dz/dt = f_x\,dx/dt + f_y\,dy/dt$, so substituting $z = f(x, y)$ and $dz = f_x\,dx + f_y\,dy$,

$$\oint_C R\,dz = \oint_{C'} R(x, y, f)\,(f_x\,dx + f_y\,dy).$$

Surface integral. Here $\operatorname{curl}(R\,\mathbf{k}) = R_2\,\mathbf{i} - R_1\,\mathbf{j}$ and $d\mathbf{S} = \langle -f_x, -f_y, 1\rangle\,dx\,dy$, so

$$\iint_S \operatorname{curl}(R\,\mathbf{k}) \cdot d\mathbf{S} = \iint_A \big(R_1(x, y, f)\,f_y - R_2(x, y, f)\,f_x\big)\,dx\,dy.$$

Green's theorem. Apply $\oint_{C'} U\,dx + V\,dy = \iint_A (V_x - U_y)\,dx\,dy$ with $U = R(x, y, f)f_x$ and $V = R(x, y, f)f_y$. By the chain rule,

$$V_x = (R_1 + R_3 f_x)f_y + R\,f_{yx}, \qquad U_y = (R_2 + R_3 f_y)f_x + R\,f_{xy}.$$

Since $f_{xy} = f_{yx}$, four terms cancel and $V_x - U_y = R_1 f_y - R_2 f_x$, the integrand of the surface integral. A surface that is not a graph over each coordinate plane can usually be cut into pieces

that are; adding Stokes' theorem for the pieces, the line integrals along the cuts cancel. □

29.3 Examples

Example: *A paraboloid.* Verify Stokes' theorem for $\mathbf{F} = z\,\mathbf{i} + x\,\mathbf{j} + y\,\mathbf{k}$, C the unit circle in the xy-plane counterclockwise, and S the part of $z = 1 - x^2 - y^2$ above the xy-plane, oriented upward. Directly, with $x = \cos\theta$, $y = \sin\theta$, $z = 0$,

$$\oint_C z\,dx + x\,dy + y\,dz = \oint_C x\,dy = \int_0^{2\pi} \cos^2\theta\,d\theta = \pi.$$

By Stokes, $\nabla \times \mathbf{F} = \langle 1, 1, 1\rangle$ and $\hat{\mathbf{n}}\,dS = \langle -f_x, -f_y, 1\rangle\,dx\,dy = \langle 2x, 2y, 1\rangle\,dx\,dy$, so

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_{x^2+y^2\le 1} (2x + 2y + 1)\,dx\,dy = \iint 1\,dx\,dy = \pi,$$

since $\iint x\,dx\,dy = \iint y\,dx\,dy = 0$ by symmetry.

Example: *A hemisphere.* Let S be the half of the unit sphere with $y \ge 0$, oriented so $\hat{\mathbf{n}}$ makes an acute angle with the positive y-axis, and $\mathbf{F} = y\,\mathbf{i} + 2x\,\mathbf{j} + x\,\mathbf{k}$. The boundary C is the unit circle in the xz-plane; compatibility forces the direction $x = \cos t$, $y = 0$, $z = -\sin t$, $0 \le t \le 2\pi$, which is clockwise in the xz-plane with x and z drawn as usual. Since $y = 0$ and $dy = 0$ on C,

$$\oint_C y\,dx + 2x\,dy + x\,dz = \oint_C x\,dz = \int_0^{2\pi} -\cos^2 t\,dt = -\pi.$$

On S, $\hat{\mathbf{n}} = \langle x, y, z\rangle$ and $\nabla \times \mathbf{F} = \langle 0, -1, 1\rangle$, so $(\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}} = -y + z$. With $y = \sin\phi\sin\theta$, $z = \cos\phi$, $dS = \sin\phi\,d\phi\,d\theta$, and $0 \le \theta \le \pi$ for $y \ge 0$,

$$\iint_S (-y + z)\,dS = \int_0^{\pi}\int_0^{\pi} (-\sin\phi\sin\theta + \cos\phi)\sin\phi\,d\phi\,d\theta = \int_0^{\pi} -\frac{\pi}{2}\sin\theta\,d\theta = -\pi,$$

because $\int_0^{\pi} \sin^2\phi\,d\phi = \pi/2$ and $\int_0^{\pi} \cos\phi\sin\phi\,d\phi = 0$.

Example: *Work equals projected area.* Let $\mathbf{F} = x^2\,\mathbf{i} + x\,\mathbf{j} + z^2\,\mathbf{k}$ and S the graph of any $z = g(x, y)$ over a region A, oriented upward, with boundary C compatibly oriented. Then $\nabla \times \mathbf{F} = \mathbf{k}$, and since $dS = dA/(\hat{\mathbf{n}} \cdot \mathbf{k})$ with $\hat{\mathbf{n}} \cdot \mathbf{k} > 0$,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \mathbf{k} \cdot \hat{\mathbf{n}}\,dS = \iint_A \hat{\mathbf{n}} \cdot \mathbf{k}\,\frac{dA}{\hat{\mathbf{n}} \cdot \mathbf{k}} = \text{area}(A).$$

29.4 Curl as angular velocity

Let $\mathbf{F}$ be the velocity of a fluid flow, and place a small paddle wheel of radius a at P_0 with its axis along a unit vector $\mathbf{u}$. Its rim is a circle C bounding a disk S with normal $\mathbf{u}$. As in the plane, the wheel turns at the average tangential velocity, so by Stokes' theorem

$$\text{angular velocity} = \frac{1}{2\pi a^2}\oint_C \mathbf{F}\cdot d\mathbf{r} = \frac{1}{2\pi a^2}\iint_S (\operatorname{curl}\mathbf{F})\cdot\mathbf{u}\,dS \approx \frac{1}{2\pi a^2}(\operatorname{curl}\mathbf{F})_0\cdot\mathbf{u}\,\pi a^2,$$

and in the limit $a \to 0$ the angular velocity is $\frac{1}{2}(\operatorname{curl}\mathbf{F})_0\cdot\mathbf{u}$. This is largest when $\mathbf{u}$ points along $(\operatorname{curl}\mathbf{F})_0$. Hence the direction of the curl is the axis along which the wheel spins fastest, and its magnitude is twice that maximum angular velocity.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 29, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 30: Stokes' theorem (cont.); topology

A gradient field always has zero curl. Stokes' theorem gives the converse, but only on regions where every closed curve bounds a suitable surface.

30.1 Simply connected regions

Definition 30.1 [Simply connected]. A region D in space is simply connected if every closed curve C in D bounds some surface S lying in D. Equivalently, every closed curve in D can be shrunk to a point without leaving D.

All of space is simply connected, and so are the interior and the exterior of a sphere; in particular the complement of a point is simply connected. The complement of the z-axis is not: a loop encircling the axis bounds no surface avoiding it (Figure 45). Similarly the exterior of the cylinder $x^2 + y^2 = 1$ and the interior of a solid torus are not simply connected. Topology uses such considerations to classify surfaces: a sphere and a torus are different surfaces because every loop on the sphere bounds, while the torus has two independent loops that do not bound.

Theorem 30.2 [Curl-free fields are conservative]. *If $\mathbf{F}$ is continuously differentiable and $\nabla \times \mathbf{F} = \mathbf{0}$ in a simply connected region D, then $\mathbf{F}$ is conservative in D: its line integrals are path-independent, and $\mathbf{F} = \nabla f$ for a potential f.*

Proof. Let C_1 and C_2 be curves in D with the same endpoints. Then $C = C_1 - C_2$ (follow C_1, then C_2 backward) is a closed curve, so it bounds a two-sided surface S in D, and by Stokes' theorem

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} - \int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}\, dS = 0.$$

Path independence gives a potential by the second fundamental theorem for line integrals. □

So on a simply connected region,

$$\mathbf{F} = \nabla f \iff \nabla \times \mathbf{F} = \mathbf{0} \iff \int_P^Q \mathbf{F} \cdot d\mathbf{r} \text{ is path-independent.}$$

A potential is then found by successive integration or by a line integral along segments parallel to the axes, exactly as for fields on all of space.

30.2 Orientability

Stokes' theorem applies to any surface bounded by C, provided it has a consistent normal vector. The Möbius strip is one-sided: moving $\hat{\mathbf{n}}$ continuously once around the strip returns $-\hat{\mathbf{n}}$. No orientation is possible, so flux through it is undefined. To apply Stokes' theorem to a curve C, one must find a *two-sided* surface with boundary C.

This can be subtle. A long strip given three half-twists before its ends are joined is a one-sided surface whose boundary is the trefoil knot, so this surface is of no use for the trefoil. Two remedies:

1. Allow the surface to cross itself and to have creases. Fix a point Q not on C and join it to a point P moving around C; the segments QP sweep out a cone with boundary C (Figure 45). It is not smooth at Q and may cross itself along some lines, but it is two-sided. Stokes' theorem still applies: subdivide it into skinny triangles with a vertex at Q, including the self-intersection lines among their edges, and add the equations for the triangles.
2. In fact every closed curve bounds a smooth two-sided surface that does not cross itself.

If D is simply connected, every closed curve in D bounds a two-sided surface (possibly self-intersecting) lying in D; this is the fact used in the proof of Theorem 30.2.

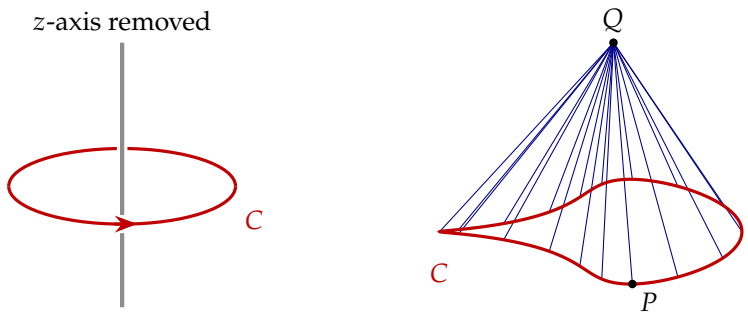


Figure 45: Left: a loop C around the z-axis bounds no surface in the complement of the axis. Right: the segments from a fixed point Q to the points P of a closed curve C sweep out a two-sided cone with boundary C (schematic curve).

30.3 Regions that are not simply connected

On a region that is not simply connected, a curl-free field need not be conservative. The standard example is

$$\mathbf{F} = \frac{-y\,\mathbf{i} + x\,\mathbf{j}}{r^2}, \qquad r = \sqrt{x^2 + y^2},$$

defined on space minus the z-axis. Here $P = -y/r^2$ and $Q = x/r^2$ do not depend on z and $R = 0$, so $\nabla \times \mathbf{F} = (Q_x - P_y)\,\mathbf{k}$, and $Q_x = P_y = (y^2 - x^2)/r^4$ gives $\nabla \times \mathbf{F} = \mathbf{0}$. Yet on a circle of radius a centered on the axis in a plane $z = z_0$, traversed counterclockwise seen from above, $x = a\cos t$, $y = a\sin t$ give $\mathbf{F} \cdot d\mathbf{r} = dt$, so $\oint \mathbf{F} \cdot d\mathbf{r} = 2\pi \neq 0$.

The same value holds for any closed curve C going once around the z-axis, oriented so that the fingers of the right hand curl along C when the thumb points along $\mathbf{k}$. Indeed C and such a circle, with its direction reversed, together bound a surface avoiding the axis. Stokes' theorem for a surface with two boundary curves and $\nabla \times \mathbf{F} = \mathbf{0}$ then shows that the two line integrals are equal. A curve going around n times gives $2n\pi$. Physically, the field outside a wire carrying a current has zero curl but is not conservative: its circulation around a loop linking the wire is nonzero.

More generally, call two closed curves C_1, C_2 in D HOMOLOGOUS, $C_1 \sim C_2$, if C_1 and C_2 reversed form the complete boundary of a surface in D. For $\nabla \times \mathbf{F} = \mathbf{0}$ in D, Stokes' theorem gives

$$\oint_{C_1} \mathbf{F} \cdot d\mathbf{r} = \oint_{C_2} \mathbf{F} \cdot d\mathbf{r} \qquad \text{whenever } C_1 \sim C_2;$$

in particular this holds when C_1 can be deformed into C_2 without leaving D. Finding the possible values of these line integrals reduces to the topological problem of finding a set of mutually non-homologous closed curves to which every closed curve in D is homologous.

30.4 Surface independence

In Stokes' theorem any surface bounded by C may be used. If C bounds two surfaces S_1 and S_2, both compatibly oriented, then $\iint_{S_1} \operatorname{curl}\mathbf{F} \cdot \hat{\mathbf{n}}\, dS = \iint_{S_2} \operatorname{curl}\mathbf{F} \cdot \hat{\mathbf{n}}\, dS$. This can be seen directly. Reverse the orientation of S_2; then $S_1 \cup (-S_2)$ is a closed surface with outward normal, enclosing a region D, and the divergence theorem gives

$$\iint_{S_1} \operatorname{curl}\mathbf{F} \cdot d\mathbf{S} - \iint_{S_2} \operatorname{curl}\mathbf{F} \cdot d\mathbf{S} = \iiint_D \operatorname{div}(\operatorname{curl}\mathbf{F})\, dV = 0,$$

since $\nabla \cdot (\nabla \times \mathbf{F}) = 0$ always, much as $\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 0$ for ordinary vectors.

30.5 The bridges between integrals

Three kinds of integrals have appeared: triple integrals $\iiint f\, dV$ (in rectangular, cylindrical, or spherical coordinates), surface integrals $\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS$ (set up from a formula for $\hat{\mathbf{n}}\, dS$), and line integrals $\int_C \mathbf{F} \cdot d\mathbf{r}$ (set up by parametrizing C). They differ in meaning and method, but three theorems connect them:

Theorem	Statement	Relates
Fundamental theorem	$\int_C \nabla f \cdot d\mathbf{r} = f(P_1) - f(P_0)$	points and curves
Stokes	$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \hat{\mathbf{n}}\, dS$	curves and surfaces
Divergence	$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS = \iiint_D \nabla \cdot \mathbf{F}\, dV$	surfaces and solids

In each, both sides are integrals of the kinds above and are evaluated by the usual methods, even when the integrand is a divergence or a curl. Each relates an integral over a region to an integral over its boundary. A generalized Stokes theorem relates an integral over an n-dimensional hypersurface to one over its $(n-1)$-dimensional boundary; Green's and Stokes' theorems are the case $n = 2$, and the divergence theorem is closely related to the case $n = 3$.

Source. Denis Auroux, MIT 18.02SC, *Multivariable Calculus*, Lecture 30, Fall 2010; lecture notes from MIT 18.02, Fall 2007. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 31: Curl and physics; Maxwell's equations

31.1 Curl of velocity and force fields

For a velocity field, the curl is twice the angular velocity vector of the rotation component of the motion; for uniform rotation about the z-axis, $\mathbf{v} = \omega(-y\,\mathbf{i} + x\,\mathbf{j})$ and $\nabla \times \mathbf{v} = 2\omega\,\mathbf{k}$. The curl singles out the rotation component of a motion, while the divergence singles out its stretching component.

Now consider a solid in a force field, or rather an acceleration field, $\mathbf{F}$: the force on a small mass Δm at (x, y, z) is $\mathbf{F}(x, y, z)\,\Delta m$. The torque of a force about the origin is $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ (for the whole solid, integrate $\mathbf{r} \times \mathbf{F}\,\delta\,dV$); it measures how the force imparts rotation. Translation and rotation obey parallel laws:

$$\frac{\text{force}}{\text{mass}} = \text{acceleration} = \frac{d}{dt}(\text{velocity}),$$
$$\frac{\text{torque}}{\text{moment of inertia}} = \text{angular acceleration} = \frac{d}{dt}(\text{angular velocity}).$$

Differentiating curl(velocity) = 2 (angular velocity) in time therefore gives

$$\operatorname{curl}\left(\frac{\text{force}}{\text{mass}}\right) = 2\,\frac{\text{torque}}{\text{moment of inertia}}.$$

In particular, if $\mathbf{F}$ derives from a potential, then $\nabla \times \mathbf{F} = \nabla \times \nabla f = \mathbf{0}$ and $\mathbf{F}$ induces no rotation. For example, gravitational attraction by itself does not affect the Earth's rotation. This is not strictly true, since the Earth is deformable; friction and tidal effects of the Earth's attraction are why the Moon's rotation is synchronized with its revolution around the Earth.

31.2 Maxwell's equations: differential and integral forms

Each of Maxwell's equations has a *differential form*, involving only partial derivatives, and an equivalent *integrated form*, involving line, surface, and volume integrals. The physical laws are empirical, but the equivalence of the two forms is mathematics: it follows from the divergence theorem or Stokes' theorem. The pattern is already familiar from the criterion for conservative fields on all of space, $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for all closed C if and only if $\nabla \times \mathbf{F} = \mathbf{0}$ everywhere, with Stokes' theorem as the bridge. Throughout, $\mathbf{E}$ is the electric field, $\mathbf{B}$ the magnetic field, and the fields are assumed continuously differentiable.

Proposition 31.1 [Gauss–Coulomb law]. *Let ρ be the charge density and ϵ_0 a physical constant. The differential form*

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

holds everywhere if and only if the integrated form

$$\iint_S \mathbf{E} \cdot \hat{\mathbf{n}}\, dS = \frac{Q}{\epsilon_0}, \qquad Q = \iiint_D \rho\, dV = \textit{total charge inside } S,$$

holds for every closed surface S bounding a region D.

Proof. If the differential form holds, the divergence theorem gives $\iint_S \mathbf{E} \cdot \hat{\mathbf{n}}\, dS = \iiint_D \nabla \cdot \mathbf{E}\, dV = \frac{1}{\epsilon_0} \iiint_D \rho\, dV = Q/\epsilon_0$.
Conversely, suppose the differential form fails at some point P_0, say $\nabla \cdot \mathbf{E} - \rho/\epsilon_0 > 0$ there (otherwise reverse the inequalities). By continuity this quantity is positive on a small ball B_0 centered at P_0, bounded by a sphere S_0. Integrating over B_0,

$$\iiint_{B_0} \nabla \cdot \mathbf{E}\, dV > \frac{1}{\epsilon_0} \iiint_{B_0} \rho\, dV = \frac{Q_0}{\epsilon_0},$$

where Q_0 is the charge inside S_0. By the divergence theorem the left side is $\iint_{S_0} \mathbf{E} \cdot \hat{\mathbf{n}}\, dS$, so the integrated form fails for S_0. □

The integrated form says how charges influence the electric field; for example, it governs the relation between the voltage across the plates of a capacitor and its charge. The flux on the left is more intuitive than div **E**, but it takes more work to compute.

Proposition 31.2 [Faraday's law]. *The differential form*

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

holds everywhere if and only if the integrated form

$$\oint_C \mathbf{E} \cdot d\mathbf{r} = -\frac{d}{dt} \iint_S \mathbf{B} \cdot \hat{\mathbf{n}}\, dS$$

holds for every closed curve C and surface S bounded by C, compatibly oriented.

In the differential form, $\partial \mathbf{B}/\partial t$ differentiates each component of $\mathbf{B}(x, y, z, t)$ with respect to t. In the integrated form, the flux of **B** is a function of t alone, so the derivative is ordinary. The line integral is the electromotive force around C, and the right side involves the magnetic flux through the loop.

Proof. If the differential form holds, then by Stokes' theorem

$$\oint_C \mathbf{E} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{E}) \cdot \hat{\mathbf{n}}\, dS = -\iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot \hat{\mathbf{n}}\, dS = -\frac{d}{dt} \iint_S \mathbf{B} \cdot \hat{\mathbf{n}}\, dS,$$

where moving d/dt outside the integral is valid when **B** has continuous derivatives and S is

smooth with finite area; this last step is subtle and belongs to advanced calculus.
Conversely, suppose $\nabla \times \mathbf{E} + \partial \mathbf{B}/\partial t \neq \mathbf{0}$ at some point P_0. Some component is nonzero there; say the $\mathbf{i}$-component is positive (other cases are the same). By continuity it stays positive on a small ball around P_0. In that ball take a small disk S_0 centered at P_0 with normal $\mathbf{i}$, and orient its boundary circle C_0 compatibly. Then

$$\iint_{S_0} \left(\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \right) \cdot \mathbf{i} \, dS > 0, \quad \text{so} \quad \oint_{C_0} \mathbf{E} \cdot d\mathbf{r} > -\frac{d}{dt} \iint_{S_0} \mathbf{B} \cdot \hat{\mathbf{n}} \, dS,$$

by Stokes' theorem on the first term and interchanging differentiation and integration on the second. The integrated form fails for C_0 and S_0. □

So in the presence of a varying magnetic field, $\mathbf{E}$ is not conservative, and a closed loop of wire carries a nonzero voltage (induction). Transformers in power adapters use this: alternating current in a wire looped around a cylinder creates an alternating magnetic field, whose flux through a second loop of wire creates the output voltage.

The other two of Maxwell's equations govern the divergence and curl of $\mathbf{B}$:

$$\nabla \cdot \mathbf{B} = 0, \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \epsilon_0 \mu_0 \frac{\partial \mathbf{E}}{\partial t},$$

where $\mathbf{J}$ is the current density. By the first, the magnetic flux through a loop C is the same for every surface spanning C: the difference of two such fluxes is the flux out of a closed surface, which is $\iiint \nabla \cdot \mathbf{B} \, dV = 0$. So one may speak of "the flux through C".

Note. In Gaussian units, used in some texts, the two laws read $\nabla \cdot \mathbf{E} = 4\pi\rho$ and $\nabla \times \mathbf{E} = -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t}$, with c the speed of light; the integrated forms change by the same constants, and the proofs are identical.

31.3 Harmonic functions

A function $f(x, y, z)$ is HARMONIC if it satisfies Laplace's equation

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0.$$

The potential of an electrostatic field is harmonic in any simply connected region free of charge. There $\nabla \cdot \mathbf{E} = 0$ by the Gauss–Coulomb law, since $\rho = 0$, and $\nabla \times \mathbf{E} = \mathbf{0}$ by Faraday's law, since static charges produce no changing magnetic field. The second equation and simple connectivity make $\mathbf{E}$ conservative, $\mathbf{E} = \nabla f$ (the physical potential is $-f$), and then the first gives $\nabla \cdot \nabla f = \nabla^2 f = 0$. Likewise the potential of a gravitational field is harmonic wherever there is no mass. These are consequences of physical laws, so they are themselves physical facts rather than mathematical ones.

Since harmonic functions represent potentials, one often prescribes the values of f on the boundary of a region D and seeks the harmonic f inside, analytically or numerically; the divergence theorem is an important tool in this.

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